Canonical labelings for graphs of maximum degree d in time $n^{\mathrm{polylog}(d)}$

A draft for expert checking

[Author names to be added]

Unrefereed draft, prepared with substantial AI assistance. Version of 5 November 2026.

Abstract

Grohe, Neuen and Schweitzer (GNS) showed that isomorphism of graphs of maximum degree d can be decided in time $n^{O((\log d)^c)}$. Whether canonical forms can be computed within the same bound was posed as open by Schweitzer and Wiebking (STOC 2019) and by Wiebking (ICALP 2020), and it is Question 9.6 of Wiebking's dissertation (2021). We give an argument that the answer is positive. For every group $G \leq \mathrm{Sym}(\Omega)$ whose composition factors all embed in S_d, we compute a canonical labeling of strings $x\colon \Omega \to \Sigma$ under G, or under a coset of G, in time $n^{O((\log d)^c)}$ with $n = |\Omega|$ (Theorem A). A canonical version of Luks's reduction, written out following Babai and Luks (1983), then gives canonical forms of n-vertex graphs of maximum degree d in time $n^{O((\log d)^c)}$ (Corollary B). The only genuinely pairwise step of GNS is replaced by a canonical-subcoset device: an ordered test tuple is moved to a reference tuple chosen from the group alone, and canonization takes place under the stabilizer of the reference tuple. The use of a canonical test set over an ordered group is due to Wiebking (dissertation, Lemma 5.34). We also give a canonical version of Neuen's hypergraph isomorphism test for such groups (Theorem H; correctness proved, modulo Neuen's constructions as listed in Section 13, which include a lemma of his dissertation that we have not seen; running time only sketched), and a proof sketch that graphs of treewidth k admit canonical forms in time $n^{(\log k)^{O(1)}}$, which needs Theorem A but not Theorem H (Theorem TW). While writing out the running time we found a step in the written potential argument of GNS (proof of Lemma 7.2) that we could not justify for the algorithm as written. We describe a choice of block system under which the step holds, and ask whether this is the intended reading; the GNS theorem itself is not affected.

This is an unrefereed draft, prepared with substantial AI assistance. Every result is conditional on the published results listed in Section 13, among them Babai's canonical form for strings (STOC 2019), whose published account is a preliminary report that sketches the proof (Theorem A needs it only for S_k, $k \leq d$, on relational structures of arity $O(\log d)$), and on the requirements D1–D4 of Section 2.4 that the group-theoretic subroutines we use be run deterministically on canonical generating sets. Each statement carries its status. The draft is offered for expert checking.

Contents

1 Introduction **3**
1.1 Canonical forms . . . 3
1.2 History . . . 3
1.3 The open problems . . . 4
1.4 Results and their status . . . 4
1.5 The new step and its provenance . . . 6
1.6 A question about the GNS analysis . . . 6

1.7 Organisation, conventions and caveats . . . 7

2 Preliminaries **7**
2.1 Conventions . . . 7
2.2 Canonizers . . . 7
2.3 The class $\widehat{\Gamma}_d$ . . . 8
2.4 Choices that are functions of the group (principle P1) . . . 8
2.5 Windowed and layered strings . . . 9
2.6 The GNS framework: the facts we use . . . 9

3 Framework lemmas **11**

4 The canonical subcoset lemma and tuple colours **13**

5 Canonical aggregation **14**
5.1 Local certificates are canonical objects . . . 14
5.2 The fullness group and the case analysis . . . 14
5.3 The canonical structure in case (v) . . . 15
5.4 Computing the colours . . . 16
5.5 Option 1: canonical find-structure . . . 17
5.6 Option 2: canonical find-symmetry . . . 18

6 The algorithm and its correctness **18**
6.1 Instances . . . 18
6.2 Algorithm CanAD . . . 18
6.3 Correctness . . . 19
6.4 The choice points of GNS . . . 19
6.5 Theorem A from CanAD . . . 21

7 Running time **21**
7.1 A recursion lemma . . . 21
7.2 A question about the written potential argument of GNS Lemma 7.2 . . . 22
7.3 The potential argument with $\mathcal{C}'$ as top partition . . . 23
7.4 One macro step and its branch counts . . . 23
7.5 Proof of the time bound . . . 24

8 Corollary B: the canonical Luks reduction **25**

9 Corollary D and relational structures **27**

10 Theorem H: canonical Neuen 2020 **28**
10.1 Neuen's algorithm . . . 28
10.2 Equivariant coset maps . . . 28
10.3 Ordered colours from a setwise reference test set . . . 29
10.4 The pairwise steps of Neu20 . . . 30
10.5 Lemmas W, F2^{set} and B . . . 30
10.6 Canonical simplification and renormalization . . . 32
10.7 Canonical local certificates . . . 33
10.8 Canonical aggregation . . . 35

10.9 The algorithm CanHyp . . . 36
10.10Theorem H . . . 38

11 Treewidth: Theorem TW (proof sketch) 40
11.1 What is known . . . 40
11.2 What Wiebking's algorithm imports . . . 40
11.3 Coset-labeled hyperedges under a hyperedge-respecting coset . . . 41
11.4 Canonical rewrites of Wiebking's isomorphism-only steps . . . 42
11.5 Theorem TW . . . 44

12 Further corollaries 44

13 Assumptions from the literature 46

14 Related work 47

15 Status and open points 49
15.1 Status of every statement . . . 49
15.2 How this draft was checked . . . 49
15.3 Questions for experts . . . 50
15.4 Known limitations . . . 50

1 Introduction

1.1 Canonical forms

A *canonical form* for a class of graphs assigns to each graph X a graph $\mathrm{CF}(X)$ on $\{1,\dots,n\}$ such that $\mathrm{CF}(X) \cong X$, and such that $X \cong Y$ implies $\mathrm{CF}(X) = \mathrm{CF}(Y)$. A *canonical labeling* is a bijection λ with $X^\lambda = \mathrm{CF}(X)$. Canonical forms solve isomorphism. They also allow hashing and database lookup, and they can be used recursively inside other canonization algorithms. That last use is why the decomposition framework of Schweitzer and Wiebking [SW19, Wie21] needs canonical forms and not just isomorphism tests.

1.2 History

- Luks [Luk82] decided isomorphism of graphs of maximum degree d in time $n^{O(d)}$, by reducing it to string isomorphism under groups in $\widehat{\Gamma}_d$ (the groups all of whose composition factors embed in S_d). Babai, Kantor and Luks [BKL83] improved this to $n^{O(d/\log d)}$.

- Babai and Luks [BL83] and Fürer, Schnyder and Specker [FSS83] computed canonical forms for graphs of bounded degree by making Luks's algorithm canonical ([BL83, Thm. 4.2]: time $n^{O(d)}$; the remark after its proof says this can be improved to $n^{O(d/\log d)}$ with the techniques of Luks).

- Babai [Bab16b, Bab16a] decided isomorphism in quasipolynomial time, and later computed canonical forms in quasipolynomial time [Bab19] ("BC" below). BC explains that the core of the 2016 algorithm, the t-ary structure built from local certificates, is only *pairwise* canonical, and its main new contribution is to repair this [Bab19, Thm. 12.1].

- Grohe, Neuen and Schweitzer [GNS18] ("GNS") decided isomorphism of graphs of maximum degree d in time $n^{O((\log d)^c)}$. GNS reuse Babai's aggregation of local certificates, so the same pairwise step appears there.

1.3 The open problems

The problem is stated as open in several places.

- Schweitzer and Wiebking [SW19, Outlook]: "Naturally, the bounded degree isomorphism algorithm of [GNS] should then be amenable to canonization. However, this remains as future work."

- Wiebking [Wie20, Outlook]: "With Babai's result, it is a natural question whether the bounded-degree isomorphism algorithm of [GNS] extends to canonization as well. The present isomorphism algorithm for graphs parameterized by treewidth should then be amenable to canonization as well." The introduction of [Wie20] says that only its final algorithm (treewidth $n^{\operatorname{polylog} k}$) depends on GNS, "for which no adequate canonization version is known".

- Wiebking's dissertation [Wie21] (printed page numbers):

  - p. 7: there are isomorphism tests with no known canonization algorithm of the same asymptotic running time, "[f]or example, ... the recent bounded-degree graph isomorphism test given by Grohe, Neuen and Schweitzer [GNS18] and also ... results building on it [Neu20a]";
  - pp. 9–10: the $2^{k \operatorname{polylog} k} n^{O(1)}$ treewidth isomorphism test of [GNSW18] "is not able to compute canonical forms as it involves subroutines [GNS18] that do not extend to canonization so far";
  - p. 179: the $n^{\operatorname{polylog} h}$ isomorphism test for graphs excluding a minor (Thm. 6.29) is only an isomorphism test "since the bounded-degree techniques [GNS18, Neu20a] are not available for canonization so far"; the treewidth result Cor. 6.30 (p. 180) is likewise stated for isomorphism only;
  - p. 181: "A natural question is whether the isomorphism test for bounded-degree graphs and finally our isomorphism test for graphs excluding a minor can be turned into a canonization algorithm";
  - p. 242: "The only isomorphism test we use for which no canonization algorithm is known so far is the isomorphism test for bounded-degree graphs given by Grohe, Neuen and Schweitzer [GNS18] and its extension to hypergraphs [Neu20a]." This is followed by **Question 9.6**: "Can canonical labelings for graphs with n vertices be computed in time $n^{\operatorname{polylog}(k)}$ where k is the maximum degree of G?", and by **Question 9.7**, the same question with k the Hadwiger number.

1.4 Results and their status

We write $\widehat{\Gamma}_d$ for the class of groups whose composition factors are all isomorphic to subgroups of S_d. Precise definitions are in Section 2. Table 1 summarizes the status of each statement. "Proof modulo §13" means that we give a complete proof, using the published results listed in Section 13 without re-proving them. None of the statements has been refereed.

Theorem A (strings under $\widehat{\Gamma}_d$-groups). *There is an algorithm that, given $G \leq \mathrm{Sym}(\Omega)$ with $G \in \widehat{\Gamma}_d$ (by generators), $\sigma \in \mathrm{Sym}(\Omega)$ and a string $x\colon \Omega \to \Sigma$, outputs a canonical form $F_{\sigma G}(x)$ and the canonical placement coset $\Lambda_{\sigma G}(x) = \{g \in \sigma G : x^g = F_{\sigma G}(x)\}$ (Definition 2.1 and the coset rule of §2.2). The output depends only on the coset σG as a set and on x, not on the generators supplied. The algorithm runs in time $n^{O((\log d)^c)}$, where $n = |\Omega|$ and c is an absolute constant.*

Corollary B (graphs of maximum degree d). *Canonical forms and canonical labelings of n-vertex graphs of maximum degree d can be computed in time $n^{O((\log d)^c)}$, with c as in Theorem A.*

Corollary D (labeling cosets). *Given a graph (or a string on V^2) and a labeling coset $\Delta\rho$ with $\Delta \in \widehat{\Gamma}_d$, a canonical labeling coset relative to $\Delta\rho$ in the sense of Schweitzer and Wiebking can be computed in time $|V|^{O((\log d)^c)}$, with c as in Theorem A.*

Corollary D is the canonical counterpart of the form in which Wiebking imports GNS [Wie20, Thm. 31].

Theorem H (canonical Neuen 2020; Section 10; running time only a proof sketch). *Given $\Gamma \in \widehat{\Gamma}_d$ on Ω ($|\Omega| = n$), a coset $\sigma\Gamma$, a partition $\mathfrak{P}$ with $\mathfrak{P}^\sigma$ Γ-invariant, and a set $\mathfrak{X}$ of m $\mathfrak{P}$-strings (in particular, a hypergraph with m edges), a canonical form of $\mathfrak{X}$ under $\sigma\Gamma$ and its canonical placement coset can be computed in time $(n+m)^{O((\log d)^{c'})}$, where c' is an absolute constant.*

Theorem TW (treewidth; Section 11; proof sketch). *Canonical forms and canonical labelings of n-vertex graphs of treewidth k can be computed in time $n^{O((\log k)^{c''})}$, where c'' is an absolute constant.*

statement	status
Theorem A	proof modulo §13 (Sections 3–7), including the running time. Depends on Babai's canonical form [Bab19, Thm. 2.2], whose published account is a preliminary report that sketches the proof, and on requirements D1–D4 (deterministic subroutines on canonical generating sets). Babai's result is needed only for S_k, $k \leq d$, on structures of arity $O(\log d)$ (Remark 5.8)
Corollary B	proof modulo Theorem A (Section 8)
Corollary D	proof modulo Theorem A (Section 9)
Theorem H	correctness: proof modulo Theorem A, the constructions of [Neu20] and the other results listed in §13, including the structure-graph lemma of [Neu19], which we have not seen (algorithm CanHyp and Proposition 10.2, §10.9); running time: **proof sketch**
Theorem TW	**proof sketch**, modulo Theorem A (via Corollary D) and canonical tools of Wiebking; it incorporates three repairs TW-1–TW-3 found in an internal review
H1 (coset-labeled hypergraphs)	proof modulo Theorem H (so its running time is only as good as the sketch for Theorem H)
H2 (t-CR-bounded graphs), H3 ($K_{3,h}$-minor-free, genus)	**proof sketches**
H4 (K_h-minor-free, K_h-topological-subgraph-free)	**conditional** on Theorem H, H1, H2, with the dynamic programming only **sketched**
Remark 7.2 (GNS Lemma 7.2)	a question about the written proof; the proposed choice is proved to make the step valid

Table 1: Statements and their status.

1.5 The new step and its provenance

GNS has exactly one step that is pairwise in an essential way. It occurs in the proof of [GNS18, Lemma 6.6] (Appendix A there), in the case where the group generated by the fullness certificates has small support. GNS define a category whose objects are test sets of *both* input strings and whose morphisms are produced by comparing local certificates [GNS18, Lemma 6.3]. The equivalence classes on ordered t-tuples of $D_1 \sqcup D_2$ then give the relations of the two t-ary structures. Since the classes are named jointly, the structure built for x_1 depends on x_2.

We instead colour each ordered t-tuple $\vec{T}$ of the non-full region $D(x)$ by the canonical form, under the pointwise stabilizer J_0 of a fixed reference tuple $\vec{T}_0$, of the certificate string of (T, x) transported by any group element that moves $\vec{T}$ to $\vec{T}_0$. The reference tuple is a function of the current group alone: the first t labels of the ideal domain. Lemma T shows that the colour is well defined and equivariant, and that it separates orbits of marked pairs. Lemma 5.4 shows that the resulting structure is strongly canonical and has relative symmetry defect at least 3/4. Lemma 5.6 shows that all colours together cost at most $k^t \cdot n$ in recursive sizes. No relation between t and n is needed.

Credit.

- The device "send a tuple to a reference tuple chosen from the group alone, then canonize under its stabilizer" is a special case of the canonical-subcoset step of the Babai–Luks framework (Lemma S; [BL83, §2], [Bab19, §5]).
- Its use with a *canonical test set* over an ordered group is due to Wiebking [Wie21, Lemma 5.34, algorithm ProdCert, pp. 101–103]: "In our framework, we define the giant representation ... for a group over a linearly ordered set V^{can}. This allows us to choose one single (canonical) test set $T^{\mathrm{can}} \subseteq W^{\mathrm{can}}$ only." There the device produces fullness certificates and partition families for multiple labeling cosets, at cost $|V|^{\mathrm{polylog}\,|V|}$. It is not used there to build a non-fullness structure in the regime of GNS.
- Babai, in [Bab19, Remark 12.5], notes that a strongly canonical choice "is obvious: we just pick an element (or a linear order) for one representative of each G-isomorphism class ... and use G to translate it to all other members of the class ... Making this selection efficient is our problem." BC's "right descent" (§16 there) transports test sets in the same way.

What we believe to be new is the observation that, at a fixed reference *ordered* tuple, this device turns the non-fullness comparison classes of GNS into canonical colours at test-set size $\Theta(\log d)$, and that the rest of GNS can be made canonical with the Babai–Luks toolkit. Together these settle Question 9.6, if the argument is correct.

Why BC's repair does not transplant. BC's replacement for the pairwise structure [Bab19, Thm. 12.1] is trivial on asymmetric test sets [Bab19, Cor. 12.3], so Babai needs [Bab19, Claim 12.4], which uses [Bab19, Cor. 13.3, Prop. 14.2]. These need test sets of size $t \geq 2\log_2 n$. In GNS, $t = \Theta(\log d)$ and $t < k/10 \leq d/10$. When $d \leq 20\log_2 n$, test sets of size $2\log_2 n$ violate the constraint $t < k/10$ of the GNS aggregation (as $k \leq d$), and when $d < 2\log_2 n$ they do not even fit into the ideal domain.

1.6 A question about the GNS analysis

While writing out the running time, we found one step in the potential argument of GNS (proof of [GNS18, Lemma 7.2]) that we could not justify for the algorithm as written. On an explicit input,

a choice of block system that the text permits gives a next giant degree $k' = k$, whereas the proof asserts $k' \le k - k/(3\log k)$. A one-line choice, which is itself a run permitted by the algorithm, makes the step valid. Remark 7.2 gives the details and phrases the matter as a question. The GNS theorem itself is not in doubt. We have not seen the SIAM journal version of GNS, which may already treat this point.

1.7 Organisation, conventions and caveats

Section 2 fixes conventions and recalls GNS. Section 3 states the framework lemmas. Section 4 contains the canonical subcoset lemma and the tuple colours. Section 5 makes the aggregation canonical. Section 6 gives the algorithm and its correctness, and Section 7 the running time. Section 8 proves Corollary B and Section 9 Corollary D. Section 10 treats Theorem H, Section 11 Theorem TW, and Section 12 the remaining corollaries. Section 13 lists what is assumed from the literature, Section 14 discusses related work, and Section 15 lists the status and the open points.

References of the form [GNS, src. l. 2126] point to a line of the arXiv LaTeX source of the cited paper (for GNS, the file `bounded-degree-polylog-d.tex` of arXiv:1802.04659v3). Theorem numbers for arXiv papers were computed from their LaTeX sources; we list the versions in Section 13.

Caveats. This draft has not been refereed. It was prepared with substantial assistance from an AI system. Three internal adversarial reviews were carried out, and their corrections are incorporated; apart from the corrections listed in Section 15, they found no gap in the main argument, but they are not a substitute for refereeing. No formal verification was attempted. Small computer checks were run only for the canonical Luks reduction (Section 8) on small graphs, for the example of Remark 7.2, and for the arithmetic in Lemma 7.4.

2 Preliminaries

2.1 Conventions

All domains Ω are finite and *linearly ordered*. $\mathrm{Sym}(\Omega)$ acts on the right: $\alpha^{gh} = (\alpha^g)^h$. A string is a map $x\colon \Omega \to \Sigma$, where Σ is a finite, linearly ordered alphabet. Groups act on strings by $x^g(\alpha) = x(\alpha^{g^{-1}})$, so $(x^g)^h = x^{gh}$. Strings on the same domain are compared lexicographically along the order of Ω. We write $\mathrm{Aut}_H(x) = \{h \in H : x^h = x\}$ and $\mathrm{Iso}_H(x,y) = \{h \in H : x^h = y\}$. For $W \subseteq \Omega$ and $K \le \mathrm{Sym}(\Omega)$ we put $\mathrm{Iso}_K^W(x,y) = \{g \in K : x(\alpha) = y(\alpha^g) \text{ for all } \alpha \in W\}$, as in [GNS18, §2], and $\mathrm{Aut}_K^W(x) = \mathrm{Iso}_K^W(x,x)$. Logarithms are to base 2.

2.2 Canonizers

Definition 2.1. Let $H \le \mathrm{Sym}(\Omega)$ and let $\mathcal{X}$ be an H-invariant set of strings. An *H-canonizer on $\mathcal{X}$* assigns to each $x \in \mathcal{X}$ a string $F_H(x)$ and a set $\Lambda_H(x) \subseteq H$ such that

(C1) $\Lambda_H(x) = \{h \in H : x^h = F_H(x)\} \neq \emptyset$;

(C2) $F_H(x^h) = F_H(x)$ for all $h \in H$.

Here H is regarded as a set: a canonizer may not depend on a generating set of H.

Lemma 2.2. *For every H-canonizer:*

1. $\Lambda_H(x^h) = h^{-1}\Lambda_H(x)$;

2. $\Lambda_H(x) = \mathrm{Aut}_H(x)\lambda = \lambda\,\mathrm{Aut}_H(F_H(x))$ *for every* $\lambda \in \Lambda_H(x)$;

3. $\mathrm{Iso}_H(x,y) = \Lambda_H(x)\Lambda_H(y)^{-1}$ *if* $F_H(x) = F_H(y)$, *and* $\mathrm{Iso}_H(x,y) = \emptyset$ *otherwise.*

Proof. Direct from (C1) and (C2). □

We call $\mathrm{Aut}_H(F_H(x))$ the *right subgroup* of the canonizer at x. A canonizer is represented by one element of $\Lambda_H(x)$ together with generators of the right subgroup. Note that $\mathrm{Aut}_H(x) = \lambda\,\mathrm{Aut}_H(F_H(x))\lambda^{-1}$ for $\lambda \in \Lambda_H(x)$; the right subgroup is $\mathrm{Aut}_H(x)$ only when $x = F_H(x)$.

Cosets. For $C = \sigma H$ put $F_C(x) := F_H(x^\sigma)$ and $\Lambda_C(x) := \sigma\Lambda_H(x^\sigma)$. By (C2) this does not depend on σ, and $\Lambda_C(x) = \{g \in C : x^g = F_C(x)\}$. We call this the *coset rule*. This is [BL83, §2.1] ("$\mathrm{CF}(X, \sigma G) := \mathrm{CF}(X^\sigma, G)$") and [Bab19, Def. 5.1(i)]. We always use *left* cosets σH. With right cosets the group visible at a child node would be conjugated by the input-dependent element.

Remark 2.3 (Labeling cosets). Schweitzer and Wiebking work with labeling cosets $\Delta\rho \subseteq \mathrm{Lab}(V)$, where $\mathrm{Lab}(V)$ is the set of bijections $V \to \{1, \ldots, |V|\}$, $\Delta \leq \mathrm{Sym}(V)$ and $\rho \in \mathrm{Lab}(V)$. Put $H := \rho^{-1}\Delta\rho \leq \mathrm{Sym}([n])$; it depends only on $\Delta\rho$. Then $\mathrm{CL}(x, \Delta\rho) := \rho\Lambda_H(x^\rho)$ is a canonical labeling relative to $\Delta\rho$ in their sense, and conversely a canonical labeling function in their sense yields canonizers. The canonical labelings of [BL83] are the same notion. All three formalisms are interchangeable, and we use Definition 2.1.

2.3 The class $\widehat{\Gamma}_d$

$\widehat{\Gamma}_d$ is the class of groups all of whose composition factors are isomorphic to subgroups of S_d [GNS18, Def. 2.3]. It is closed under subgroups and homomorphic images [GNS18, Lemma 2.4], and, by the Jordan–Hölder theorem, under extensions. A group of degree n lies in $\widehat{\Gamma}_n$, so we may assume $d \leq n$.

2.4 Choices that are functions of the group (principle P1)

The output $F_H(x)$ must depend on H only as a *set*. All auxiliary choices are therefore made by *deterministic* procedures whose input is a *canonical generating set* of the current group, together with the ordered domain. A canonical generating set depends only on $H \leq \mathrm{Sym}(\Omega)$ for ordered Ω, and is computable in polynomial time [AGvM+18, Lemma 6.2]; see also [GNSW18, Lemma 21] (arXiv numbering) and the corresponding lemma of [SW19] (`lem:canGen`, Lemma 15 in the arXiv numbering). Concretely, one may take the lexicographically least coset representatives along the point-stabilizer chain of the ordered domain.

We write **(P1)** for a choice made this way. Choices of this kind include block systems, Babai's giant representation [GNS18, Lemma 7.1] and the labeling of its ideal domain $[k]$, reference tuples, lexicographically least elements of cosets, and the linear order of every derived domain (Ω^*, $[k]^t$, slot domains). Groups computed along different routes, for example as right subgroups of canonizer calls on different inputs, are re-normalized to canonical generators before any (P1) choice.

Determinism requirement. This requires that every subroutine feeding a (P1) choice be deterministic and be run on the canonical generating set. Concretely, we require deterministic polynomial-time versions of:

(D1) the permutation-group toolkit of [GNS18, Thm. 2.1] (orders, membership, orbits, pointwise stabilizers, block systems, kernels, preimages). Seress's book [Ser03] contains randomized

algorithms as well as deterministic ones; the deterministic polynomial-time versions are the ones meant here;

(D2) the algorithm of [GNS18, Lemma 7.1] (= [Bab16a, Thm. 3.2.1]);

(D3) the constructions in the proofs of [GNS18, Thms. 4.3, 4.6], and, for Theorem H, the constructions of [Neu20, Thms. 4.8, 4.9, 4.17] and the structure-graph lemma they rely on [Neu19, Lemma 6.1.17];

(D4) Babai's canonical form of strings [Bab19, Thm. 2.2]. BC's canonical form takes "(a set of generators of) the group G" as input, and its invariance property is stated for a fixed input group. Theorem A calls it only for the group S_k ($k \le d$) in its action on $[k] \sqcup [k]^2 \sqcup [k]^t$, on the fixed generating set determined by (k,t) (Proposition 5.7 and Remark 5.8), so its output is a function of (k,t) and the string, whatever BC does on other generating sets. We also require the computation to be deterministic.

2.5 Windowed and layered strings

For $W \subseteq \Omega$ the *windowed string* $x^{[W]}$ carries an explicit membership tag: $x^{[W]}(\alpha) = (1, x(\alpha))$ for $\alpha \in W$, and $x^{[W]}(\alpha) = (0,*)$ otherwise. Thus $x^{[W]}$ determines W, whatever Σ is. For a chain $\emptyset = U_0 \subset U_1 \subset \cdots \subset U_r$ the *layered string* $x^{\langle\langle U_r \rangle\rangle}$ records the layer index: $x^{\langle\langle U_r \rangle\rangle}(\alpha) = (i, x(\alpha))$ for $\alpha \in U_i \setminus U_{i-1}$, and $(0,*)$ for $\alpha \notin U_r$. All tags live in one fixed totally ordered set $\{(0,*)\} \cup (\mathbb{N} \times \Sigma)$.

The tags matter for three reasons.

- $\mathrm{Aut}_H(x^{[W]}) = \{h \in H : W^h = W,\ x(\alpha^h) = x(\alpha) \text{ for } \alpha \in W\}$. With an untagged blank symbol that also occurs in Σ, h need not stabilize W.
- The tags are needed for the identity $G(T,x) = \mathrm{Aut}_{H_T}(x^{\langle\langle W(T,x) \rangle\rangle})$ used in Lemma 5.4(2).
- They are needed so that Lemma T(c), applied to windowed strings, means "the same window and the same values on it".

Recursive calls never receive windowed strings. They receive restrictions $x|_V$ to invariant sets (Lemma F1), so alphabets do not grow along the recursion. Tagged strings occur only as the colours of Section 5.3 and as the encodings passed to BC.

2.6 The GNS framework: the facts we use

Result numbers refer to arXiv:1802.04659v3.

Almost d-ary sequences [GNS18, Def. 4.1, Obs. 4.2]. A G-invariant sequence of partitions $\{\Omega\} = \mathfrak{B}_0 \succ \mathfrak{B}_1 \succ \cdots \succ \mathfrak{B}_m$ (discrete) is *almost d-ary* if for every i and every $B \in \mathfrak{B}_{i-1}$, either G_B acts semi-regularly on $\mathfrak{B}_i[B]$, or $|\mathfrak{B}_i[B]| \le d$. Such a sequence is inherited by subgroups and by restrictions to invariant subsets. Inserting a G-invariant partition $\mathfrak{P}$ between $\mathfrak{B}_{i-1}$ and $\mathfrak{B}_i$ keeps the sequence almost d-ary provided each class of $\mathfrak{B}_{i-1}$ contains at most d classes of $\mathfrak{P}$ and each class of $\mathfrak{P}$ contains at most d classes of $\mathfrak{B}_i$ (or the corresponding actions are semi-regular); this is immediate from the definition.

Change of action [GNS18, Thms. 4.3, 4.6]. For transitive $G \in \widehat{\Gamma}_d$ there are a set Ω^* with $|\Omega^*| \le n^{O(\log^2 d)}$, a homomorphism $\iota\colon G \to \mathrm{Sym}(\Omega^*)$, $g \mapsto g^*$, whose image carries an almost d-ary sequence, and a map $x \mapsto x^*$ on strings. The proofs give [GNS, src. l. 1131–1178, 1207–1213, 1452–1530]:

- $(x^g)^* = (x^*)^{g^*}$, because $x^*((Nh, \alpha)) = x(\alpha)$, respectively $x^*((\alpha, \bar{v})) = x(\alpha)$;
- $x \mapsto x^*$ is injective, also after restriction to one G^*-orbit A^*, because every $\alpha \in \Omega$ occurs as a second coordinate (G is transitive).

Giant representations and affected points [GNS18, Def. 5.6]. A homomorphism $\varphi\colon H \to S_k$ is a *giant representation* if $\varphi(H) \ge A_k$. A point α is *affected* if $\varphi(H_\alpha) \not\ge A_k$; $\mathrm{Aff}(H, \varphi)$ denotes the set of affected points. For $B \le H$ with $\varphi(B) \ge A_k$, $\mathrm{Aff}(B, \varphi)$ is B-invariant, and $\mathrm{Aff}(B^h, \varphi^h) = \mathrm{Aff}(B, \varphi)^h$, where $\varphi^h\colon B^h \to S_k$, $c \mapsto \varphi(hch^{-1})$. (For $h \in H$, φ^h and $\varphi|_{B^h}$ differ by conjugation with $\varphi(h)$, so also $\mathrm{Aff}(B^h, \varphi) = \mathrm{Aff}(B, \varphi)^h$; the same holds for ψ_T and $\psi_{T^{\varphi(h)}}$ below.) Aff is antimonotone: if $B' \le B$ and both images are giants, then $\mathrm{Aff}(B, \varphi) \subseteq \mathrm{Aff}(B', \varphi)$.

Unaffected stabilizers [GNS18, Thm. 5.8]. If H has an almost d-ary sequence, $k > \max\{8, 2 + \log d\}$, and D is the set of unaffected points, then $\varphi(H_{(D)}) \ge A_k$.

Affected orbits [GNS18, Thm. 5.9] (= [Bab16b, Thm. 6(b)]). For $k \ge 5$, each orbit of $\ker\varphi$ inside an affected H-orbit Δ has length at most $|\Delta|/k$.

Local certificates [GNS18, Def. 6.1, Alg. 2, Lemma 6.2]. Fix a node group H with a giant representation φ and a t-set $T \subseteq [k]$. Put $H_T := \varphi^{-1}(\varphi(H)_T)$ and let $\psi_T\colon H_T \to \mathrm{Sym}(T)$ be the induced action. The *window chain* is: $G_0 := H_T$, $W_0 := \emptyset$; while $\psi_T(G_i) \ge \mathrm{Alt}(T)$ and $W_i \ne \mathrm{Aff}(G_i, \psi_T)$, put $W_{i+1} := \mathrm{Aff}(G_i, \psi_T)$ and $G_{i+1} := \mathrm{Aut}_{G_i}^{W_{i+1} \setminus W_i}(x)$. Let i^* be the final index, $W(T, x) := W_{i^*}$ and $G(T, x) := G_{i^*}$. We use:

(L1) $G_i = \mathrm{Aut}_{H_T}^{W_i}(x)$ (proof of [GNS18, Lemma 6.2]).

(L2) G_i stabilizes $W_1, \dots, W_{i+1}$ setwise, because $W_{j+1} = \mathrm{Aff}(G_j, \psi_T)$ is G_j-invariant and $G_i \le G_j$. Hence $G_i = \mathrm{Aut}_{H_T}(x^{\langle\langle W_i \rangle\rangle}) = \mathrm{Aut}_{H_T}(x^{[W_i]})$, using the tags of §2.5.

(L3) T is *full* if $\psi_T(\mathrm{Aut}_{H_T}(x)) \ge \mathrm{Alt}(T)$. If T is full, every G_i contains $\mathrm{Aut}_{H_T}(x)$, the loop ends with $W_{i^*} = \mathrm{Aff}(G_{i^*}, \psi_T)$, and $K(T, x) := (G_{i^*})_{(\Omega \setminus W)} \le \mathrm{Aut}_{H_T}(x)$ is a fullness certificate (by [GNS18, Thm. 5.8]). If T is not full, the window grows strictly until $\psi_T(G_i)$ is not a giant, since otherwise [GNS18, Thm. 5.8] would make T full [GNS, src. l. 1931–1934]. So T *is non-full if and only if* $\psi_T(G(T, x))$ *is not a giant.*

(L4) (Equivariance.) For $h \in H$: $W_i(T^{\varphi(h)}, x^h) = W_i(T, x)^h$ and $G_i(T^{\varphi(h)}, x^h) = G_i(T, x)^h$, by induction on i, using $H_{T^{\varphi(h)}} = (H_T)^h$ and equivariance of Aff. Hence W, G and K are equivariant, and fullness is invariant.

(L5) (Cost, [GNS18, Lemma 6.2].) For one T: recursive calls on domains of size $\le n/2$ whose sizes sum to $\le t! \cdot n$, plus $O(t!\, n^c)$ further work. The large-window case uses [GNS18, Thm. 5.9].

Symmetry defect [GNS18, §6.3]. The symmetry defect of $A \le \mathrm{Sym}(\Gamma)$ is $|\Gamma| - \max\{|\Delta| : \mathrm{Alt}(\Delta) \le A\}$; the relative defect divides this by $|\Gamma|$. [GNS18, Cor. 6.5] (from [DM96, Thm. 5.2A,B]): if $A_k \le S \le S_k$, $k \ge 24$, $A \le S$, and the relative symmetry defect of A is at least $1/4$, then $|S : A| \ge (4/3)^k$.

Giant representation of a large primitive group [GNS18, Lemma 7.1] (= [Bab16a, Thm. 3.2.1]). For primitive $P \le S_b$ with $|P| \ge b^{1+\log b}$ (b larger than an absolute constant), one computes in polynomial time a normal subgroup $N \trianglelefteq P$ with $|P : N| \le b$, an N-invariant equipartition $\mathcal{C}$ of $[b]$, and a giant representation $\psi\colon N \to S_k$ with $k \ge \log b$ and $\ker\psi = N_{(\mathcal{C})}$. We use two consequences:

- $|\mathcal{C}| \ge 2$, since $\ker\psi \ne N$ (as ψ is a giant with $k \ge 3$); as $\mathcal{C}$ is an equipartition, each class of its lift to Ω has size $n/|\mathcal{C}| \le n/2$;
- $k \le d$ when $b = |\mathfrak{B}| \le d$, since $N \le P \le \mathrm{Sym}(\mathfrak{B})$ [GNS, src. l. 2092]; we use this for the height of the recursion and for the size of the call tree.

The main recursion [GNS18, Lemma 7.2, Alg. 4]. For transitive G with almost d-ary sequence:

- if $G^{\mathfrak{B}_1}$ is semi-regular, Luks reduction;
- otherwise $|\mathfrak{B}_1| \le d$: coarsen $\mathfrak{B}_1$ to a block system $\mathfrak{B}$ on which $P = G^{\mathfrak{B}}$ is primitive; if $|P|$ is small, Luks reduction; otherwise apply [GNS18, Lemma 7.1], put $G' := \{g : g^{\mathfrak{B}} \in N\}$ (index $\le d$) and $\varphi := \psi \circ (g \mapsto g^{\mathfrak{B}})$, so that $\ker\varphi = G'_{(\mathcal{C}')}$, where $\mathcal{C}'$ is the lift of $\mathcal{C}$ to Ω;
- put $t := \max\{9, 3 + \log d\}$; if $k \le 10t$, Luks reduction to $\ker\varphi$;
- otherwise aggregate local certificates and recurse on at most k^6 groups $H \le G'$ with $|\varphi(G') : \varphi(H)| \ge (4/3)^k$.

3 Framework lemmas

All of these are standard ([BL83, §2]; [Bab19, §§5, 8]). We restate and prove them in the conventions of Section 2, because the left/right conventions matter.

Lemma F1 (restriction). *Let $U \subseteq \Omega$ be H-invariant and let $\rho_U\colon H \to H^U$ be restriction. Let (F', Λ') be an H^U-canonizer for $x|_U$. Define $F(\alpha) := (1, F'(\alpha))$ for $\alpha \in U$, $F(\alpha) := (0, *)$ otherwise, and $\Lambda := \rho_U^{-1}(\Lambda')$. Then (F, Λ) is an H-canonizer for $x^{[U]}$. Its right subgroup is $\mathrm{Aut}_H(F) = \rho_U^{-1}(\mathrm{Aut}_{H^U}(F'))$, and $\mathrm{Aut}_H(x^{[U]}) = \lambda\,\mathrm{Aut}_H(F)\lambda^{-1}$ for every $\lambda \in \Lambda$. Everything is computable in polynomial time from (F', Λ') (preimage of a coset, [GNS18, Thm. 2.1]).*

Proof. Since U is H-invariant, $h \in H$ maps $x^{[U]}$ to F iff $\rho_U(h)$ maps $x|_U$ to F'. (C2) holds because F' is constant on H^U-orbits. The last statement is Lemma 2.2(2). □

Lemma F2 (sequential chain rule with input-dependent windows). *Suppose windows V_{i+1} are given as functions of the layered string $x^{\langle\langle U_i\rangle\rangle}$, with $V_{i+1} \subseteq \Omega \setminus U_i$, that are H-equivariant ($V_{i+1}(y^h) = V_{i+1}(y)^h$), and put $U_{i+1} := U_i \sqcup V_{i+1}$; suppose the number of layers $r(x)$ is H-invariant. Given an H-canonizer (F_i, Λ_i) for $x^{\langle\langle U_i\rangle\rangle}$:*

- *pick $\tau \in \Lambda_i$ and put $w := x^\tau$;*
- *put $A_i := \mathrm{Aut}_H(F_i)$ and $V' := V_{i+1}(F_i)$, which is A_i-invariant;*
- *let (F', Λ') be an A_i-canonizer of $w^{[V']}$;*
- *let F_{i+1} be F_i overwritten on V' by F' with layer tag $i+1$, and $\Lambda_{i+1} := \tau\Lambda'$.*

Then (F_{i+1}, Λ_{i+1}) is an H-canonizer for $x^{\langle\langle U_{i+1}\rangle\rangle}$.

Proof. Independence of τ. Replacing τ by τa ($a \in A_i$) replaces w by w^a. Since $V'^a = V'$, F' is unchanged and Λ' becomes $a^{-1}\Lambda'$, so Λ_{i+1} is unchanged. *(C2).* $\Lambda_i(x^h) = h^{-1}\Lambda_i(x)$, so for x^h we may use $\tau' = h^{-1}\tau$, which gives the same w. *(C1), $\subseteq$.* Let $a \in \Lambda'$. Then $(x^{\langle\langle U_i\rangle\rangle})^{\tau a} = F_i^a = F_i$ and $V_{i+1}(x^{\langle\langle U_i\rangle\rangle})^\tau = V_{i+1}(F_i) = V'$, so $(x^{[V_{i+1}]})^{\tau a} = (w^{[V']})^a = F'$. *(C1), $\supseteq$.* If h maps $x^{\langle\langle U_{i+1}\rangle\rangle}$ to F_{i+1}, the layer tags force $(x^{\langle\langle U_i\rangle\rangle})^h = F_i$, so $h = \tau a$ with $a \in A_i$, and $(w^{[V']})^a = F'$, so $a \in \Lambda'$. □

Definition 3.1 (The chain rule used in this paper)**.** The *chain rule* is the special case of Lemma F2 in which the windows are orbits of the current right subgroup. At stage i, let V_{i+1}^* be the first A_i-orbit, in the order (size, least point), among the positions where F_i carries the tag $(0, *)$ (that is, in $\Omega \setminus U_i^\tau$ for $\tau \in \Lambda_i$), and for y in the H-orbit of $x^{\langle\langle U_i\rangle\rangle}$ put $V_{i+1}(y) := (V_{i+1}^*)^{\lambda^{-1}}$ for any $\lambda \in \Lambda_i(y)$. This is well defined because V_{i+1}^* is A_i-invariant and $\Lambda_i(y) = \lambda A_i$. It is H-equivariant by Lemma 2.2(1), and $V_{i+1}(y)$ is an orbit of $\mathrm{Aut}_H(y)$. The canonizer of $w^{[V']}$ under A_i is obtained by Lemma F1 from one recursive call on the *transitive* instance $(V', A_i^{V'}, w|_{V'})$.

Applied to an intransitive H, the chain rule processes "orbit by orbit" as in [GNS18, Alg. 1]. Its windows are orbits of the shrinking right subgroups, so they are finer than the fixed orbits used by GNS and never cost more. We use only this chain rule. In particular, every recursive call it makes is on a transitive instance, which is used in Sections 7.3 and 7.5. (Note that "least point" is not H-invariant on the input side; this is why the window is defined in the canonical frame and pulled back.)

Lemma F3 (descent). *Let $K \le H$ with $H = \bigsqcup_{\rho \in Q} \rho K$. Put $m := \min_\rho F_K(x^\rho)$ and $\Lambda_H(x) := \bigcup_{\rho : F_K(x^\rho) = m} \rho \Lambda_K(x^\rho)$. This is an H-canonizer with $F_H(x) = m$, and $\Lambda_H(x) = \{h \in H : x^h = m\}$.*

Proof. (C2). For x^h the branch values are $F_K(x^{h\rho})$. The cosets $h\rho K$ run over all left cosets of K, and F_K is constant on K-orbits, so the set of branch values is unchanged. *(C1).* Every $\rho\lambda$ in the union maps x to m. Conversely, let $x^{\rho k} = m$. Then m lies in the K-orbit of x^ρ, and $m = F_K(x^{\rho_0})$ for a minimizer ρ_0, so $F_K(m) = m$. Hence $F_K(x^\rho) = m$ and $k \in \Lambda_K(x^\rho)$. □

The output depends only on the branch values, so branches whose values are known to coincide need not be evaluated separately (Lemma 5.9 uses this). The right subgroup is generated by $\mathrm{Aut}_K(m)$ and one element $\lambda_1^{-1}\lambda_j$ per further minimizing branch, where $\lambda_j \in \rho_j \Lambda_K(x^{\rho_j})$.

Lemma F4 (equivariant branch families). *Let $x \mapsto \mathfrak{F}(x)$ assign to each x a finite set of objects with $\mathfrak{F}(x^h) = \mathfrak{F}(x)^h$, and suppose an H-canonizer for pairs (x, b), $b \in \mathfrak{F}(x)$, is given. Let $F := \min_{b \in \mathfrak{F}(x)} F_H(x, b)$ (pairs compared lexicographically), let F^x be its first component, and let $\Lambda_H(x)$ be the union of $\Lambda_H(x, b)$ over the minimizing b. Then $\Lambda_H(x) = \{h : x^h = F^x\}$, so this is an H-canonizer for x with $F_H(x) = F^x$.*

Proof. (C2) is clear. Every element of the union maps x to F^x. Conversely let $x^h = F^x$. Write $F = (F^x, F^b) = (x, b_0)^{\lambda_0}$ for a minimizing b_0. Then $F^b \in \mathfrak{F}(x)^{\lambda_0} = \mathfrak{F}(F^x) = \mathfrak{F}(x)^h$, so $b := (F^b)^{h^{-1}} \in \mathfrak{F}(x)$ and $(x, b)^h = F$. Since $F_H(F) = F$, we get $F_H(x, b) = F$, so b minimizes and $h \in \Lambda_H(x, b)$. □

Lemma R (change of action). *Let $\iota : G \to \mathrm{Sym}(\Omega^*)$, $g \mapsto g^*$, be a homomorphism and $x \mapsto x^*$ an injective map with $(x^g)^* = (x^*)^{g^*}$. Given a G^*-canonizer (F_{G^*}, Λ_{G^*}), $G^* := \iota(G)$, on the G^*-invariant set $\{x^*\}$ of encoded strings, put $\Lambda_G(x) := \iota^{-1}(\Lambda_{G^*}(x^*))$, and let $F_G(x)$ be the unique y with $y^* = F_{G^*}(x^*)$. This is a G-canonizer.*

Proof. y exists: $F_{G^*}(x^*) = (x^*)^{g^*} = (x^g)^*$ for some $g \in G$. (C1): $g \in \Lambda_G(x)$ iff $(x^g)^* = F_{G^*}(x^*) = (F_G(x))^*$ iff $x^g = F_G(x)$, by injectivity. (C2) follows from (C2) for G^* and equivariance. □

In the application, GNS's Ω^*, G^*, the almost d-ary sequence and the order on Ω^* are computed from G alone (P1; rows 1–5 of Table 2).

4 The canonical subcoset lemma and tuple colours

Lemma S (canonical subcoset). *Let $H \leq \mathrm{Sym}(\Omega)$ and let $\varphi\colon H \to \mathrm{Sym}(\Gamma)$ be a homomorphism. Let $\mathcal{S}$ be a $\varphi(H)$-invariant set of structures on Γ with a $\varphi(H)$-canonizer (F', Λ'). Let $x \mapsto S(x) \in \mathcal{S}$ satisfy $S(x^h) = S(x)^{\varphi(h)}$, and write $S_{\mathrm{can}} = S_{\mathrm{can}}(x) := F'(S(x))$ and $R = R(x) := \varphi^{-1}(\mathrm{Aut}_{\varphi(H)}(S_{\mathrm{can}}))$; both depend only on the H-orbit of x. Suppose that for each value of S_{can} an R-canonizer is available on the R-invariant set of strings x' with $S(x') = S_{\mathrm{can}}$. Define $E(x) := \varphi^{-1}(\Lambda'(S(x)))$; this is a left coset σR, and we put $\Lambda_H(x) := \Lambda_{\sigma R}(x)$, i.e. $F_H(x) := F_R(x^\sigma)$. Then this is an H-canonizer, with right subgroup $\mathrm{Aut}_R(F_H(x)) = \mathrm{Aut}_H(F_H(x))$.*

Proof. Shape of $E(x)$. $\Lambda'(S) = \pi\,\mathrm{Aut}_{\varphi(H)}(S_{\mathrm{can}})$, so $E(x) = \sigma R$. *Equivariance.* $\Lambda'(S^\pi) = \pi^{-1}\Lambda'(S)$ gives $E(x^h) = h^{-1}E(x)$. *(C2).* For x^h, the representative $h^{-1}\sigma$ gives $F_R(x^{hh^{-1}\sigma}) = F_R(x^\sigma)$. *(C1), $\subseteq$.* Every $e \in \Lambda_{E(x)}(x)$ satisfies $x^e = F_H(x)$. *(C1), $\supseteq$.* Suppose $x^h = F := F_H(x)$. We have $F = x^e$ for some $e \in E(x)$, so $S(F) = S(x)^{\varphi(e)} = S_{\mathrm{can}}$. Then $S(x)^{\varphi(h)} = S(F) = S_{\mathrm{can}}$, so $\varphi(h) \in \Lambda'(S(x))$ and $h \in E(x)$. Hence $h \in \Lambda_{E(x)}(x)$. □

Isomorphic inputs x and x^h lead to the children σR and $h^{-1}\sigma R$. The coset rule maps both to the same group node (R, x^σ): the same group as a set, on the same ordered domain, with inputs in one R-orbit. Every (P1) choice made at that node is therefore the same for both inputs.

Lemma T (tuple colours). *Let H and $\varphi\colon H \to \mathrm{Sym}(\Gamma)$ be as above, let $\vec{T}_0$ be an injective t-tuple of Γ chosen by (P1), let $\mathcal{O} := \vec{T}_0^{\varphi(H)}$, and let $J_0 := \{h \in H : \vec{T}_0^{\varphi(h)} = \vec{T}_0\}$. Let $\mathcal{P}$ be an H-invariant set of pairs $(y, \vec{T})$ with $\vec{T} \in \mathcal{O}$, let $\mathcal{Y}_0 := \{y : (y, \vec{T}_0) \in \mathcal{P}\}$ (a J_0-invariant set), and let F_{J_0} be a J_0-canonizer on $\mathcal{Y}_0$. For $(y, \vec{T}) \in \mathcal{P}$ put*

$$c(y, \vec{T}) := F_{J_0}(y^\sigma) \quad \text{for any } \sigma \in H \text{ with } \vec{T}^{\varphi(\sigma)} = \vec{T}_0.$$

Then:

- *(a) c is independent of σ;*
- *(b) $c(y^h, \vec{T}^{\varphi(h)}) = c(y, \vec{T})$ for all $h \in H$;*
- *(c) $c(y, \vec{T}) = c(y', \vec{T}')$ if and only if there is $h \in H$ with $y^h = y'$ and $\vec{T}^{\varphi(h)} = \vec{T}'$.*

Proof. Note $(y, \vec{T})^\sigma = (y^\sigma, \vec{T}_0) \in \mathcal{P}$, so $y^\sigma \in \mathcal{Y}_0$. (a) Two valid σ, σ' differ by $\sigma^{-1}\sigma' \in J_0$, and F_{J_0} is constant on J_0-orbits. (b) $h^{-1}\sigma$ is valid for $(y^h, \vec{T}^{\varphi(h)})$. (c) "If" is (b). For "only if", equal canonical forms give $j \in J_0$ with $y^{\sigma j} = y'^{\sigma'}$. Put $h := \sigma j \sigma'^{-1}$. Then $y^h = y'$ and $\vec{T}^{\varphi(h)} = \vec{T}_0^{\varphi(j\sigma'^{-1})} = \vec{T}'$. □

Lemma T is Lemma S with $S(y, \vec{T}) := \vec{T}$ and $\Lambda'(\vec{T}) := \{\pi \in \varphi(H) : \vec{T}^\pi = \vec{T}_0\}$, so that $R = J_0$.

Remark 4.1 (Two naive variants fail). Take $H = S_4$ on [4], $\varphi = \mathrm{id}$, $t = 1$, and let $F_K(z)$ be the lexicographically least string in the K-orbit of z, with $a < b$.

- *No transport*, $c(x, \vec{T}) := F_{H_{(T)}}(x)$. For $x = (a, b, b, b)$ and $T = (1)$ this gives (a, b, b, b). For $h = (1\,2)$, $x^h = (b, a, b, b)$ and $T^h = (2)$, it gives (b, a, b, b), since position 2 is pinned to a. Two isomorphic marked pairs get different colours, so (C2) fails. With transport, $\sigma = (1\,2)$ maps $2 \mapsto 1$ and $(x^h)^\sigma = x$, which gives the same colour.
- *Reference tuple read off a coset representative* (the first t labels pulled back by the representative of the node coset). Isomorphic inputs then use conjugate groups $J_0^{\varphi(h)}$, whose canonical forms are different strings; the ranking of colours is permuted, and strong canonicity fails.

Our $\vec{T}_0$ depends only on the group of the node (P1).

5 Canonical aggregation

Throughout this section the node group is H (in the algorithm, $H = G'$), it carries an almost d-ary sequence, $\varphi\colon H \to S_k$ satisfies $\varphi(H) \in \{A_k, S_k\}$, $t := \max\{9, 3 + \lceil \log d \rceil\}$ (GNS write $3 + \log d$; we round up so that t is an integer) and $k > 10t$ (so $k \geq 91$). The ideal domain $[k]$ is ordered (P1).

5.1 Local certificates are canonical objects

Lemma 5.1. *For every t-set $T \subseteq [k]$, the chain $(W_i(T, x), G_i(T, x))_i$, the fullness of T, and $K(T, x)$ for full T can be computed within the recursive budget of L5. All of these are set-theoretic functions of (H, φ, T, x) and are equivariant by L4. In particular, the way the groups are computed does not affect canonicity.*

Proof. Put $W^* := W_{i+1} \setminus W_i$; it is G_i-invariant by L2, and $G_{i+1} = \mathrm{Aut}_{G_i}(x^{[W^*]})$. Compute a G_i-canonizer (F, Λ) for $x^{[W^*]}$ by Lemma F1 and output $G_{i+1} = \lambda\,\mathrm{Aut}_{G_i}(F)\lambda^{-1}$ for $\lambda \in \Lambda$. If $|W^*| \leq n/2$, this is one recursive call on W^* under $G_i^{W^*}$; the almost d-ary sequence is inherited. Otherwise, W^* is a union of affected G_i-orbits (with respect to $\psi_T|_{G_i}$, a giant by the loop condition). Descend (Lemma F3) from G_i to $N := \ker(\psi_T|_{G_i})$ over at most $t!$ cosets; in each branch apply the chain rule (on W^*, via Lemma F1). Its windows lie inside N-orbits, which have size $\leq |W^*|/t$ by [GNS18, Thm. 5.9] ($t \geq 9$). These are GNS's counts. □

5.2 The fullness group and the case analysis

Put $\Phi(x) := \langle K(T, x) : T \text{ full} \rangle \leq \mathrm{Aut}_H(x)$ (the *fullness group*) and $\mathrm{supp}_x := \mathrm{supp}(\varphi(\Phi(x))) \subseteq [k]$ (its *support*). Both are equivariant: $\Phi(x^h) = \Phi(x)^h$ and $\mathrm{supp}_{x^h} = (\mathrm{supp}_x)^{\varphi(h)}$ by L4. We follow [GNS18, App. A], canonically.

case	GNS (App. A)	canonical object	count
(i) $k/4 \le \lvert\operatorname{supp}_x\rvert \le 3k/4$	colouring by supp_x	$\mathfrak{A} = ([k]; \operatorname{supp}_x)$	$r = 1$
(ii) $\lvert\operatorname{supp}_x\rvert > 3k/4$, no $\varphi(\Phi)$-orbit $\ge 3k/4$	orbit partition	$\mathfrak{A} = ([k]; \text{orbit partition})$	$r = 1$
(iii) $\lvert\operatorname{supp}_x\rvert > 3k/4$, orbit C with $\lvert C\rvert \ge 3k/4$ and $\varphi(\Phi)\lvert_C \ge \mathrm{Alt}(C)$	"Option 2"	$\Delta(x) := C$, $K(x) := \Phi(x)$	—
(iv) as (iii), $\varphi(\Phi)\lvert_C$ not a giant	individualize I, an orbital R, possibly a vertex v	the family of *all* structures over all choices of (I, R, v), each on $[k]$ with $C' = C \setminus I$ marked	$r \le k^6$
(v) $\lvert\operatorname{supp}_x\rvert < k/4$	joint comparison classes (pairwise)	$\mathfrak{A}(x)$ of Definition 5.3	$r = 1$

In case (iv), $|I|$ is the degree of transitivity of $\varphi(\Phi)|_C$ minus one, which is at most 4 by [GNS18, Thm. A.2] (a consequence of the classification of finite simple groups). In every structure, the relevant domain $D \subseteq [k]$ is marked by a unary relation.

Proposition 5.2. *1. The case is an invariant of the H-orbit of x.*

2. In cases (i), (ii), (iv), (v) the family $\mathfrak{F}(x)$ satisfies $\mathfrak{F}(x^h) = \mathfrak{F}(x)^{\varphi(h)}$, and every member, viewed as a structure on $[k]$, has relative symmetry defect at least $1/4$.

3. In case (iii), $\Delta(x^h) = \Delta(x)^{\varphi(h)}$, $K(x^h) = K(x)^h$, $K(x) \le \mathrm{Aut}_{H_{\Delta(x)}}(x)$ and $\varphi(K(x))|_\Delta \ge \mathrm{Alt}(\Delta)$.

Proof. Equivariance. All ingredients are equivariant: Φ, supp_x, the orbits, the degree of transitivity, the orbitals, and in case (v) Lemma 5.4(1). The family in case (iv) is the set of all choices, so it is equivariant as a set. GNS's "without loss of generality R_1 is the diagonal" [GNS, src. l. 2357] is harmless since the diagonal is an invariant. GNS's "it can be assumed that $|D_1| = |D_2|$" [GNS, src. l. 2368] is not needed, since $|D(x)|$ is an invariant.

Defects. Let $\mathrm{Alt}(\Delta) \le \mathrm{Aut}(\mathfrak{A})$ with $|\Delta| \ge 3$. Since D is a unary relation and $\mathrm{Alt}(\Delta)$ is transitive on Δ, either $\Delta \subseteq D$ or $\Delta \cap D = \emptyset$. (i) $|\Delta| \le \max(|\operatorname{supp}_x|, k - |\operatorname{supp}_x|) \le 3k/4$. (ii) $\mathrm{Alt}(\Delta)$ preserves the orbit partition and is primitive on Δ. If Δ lies in one class, $|\Delta| < 3k/4$. Otherwise Δ meets each class in at most one point, and a class P meeting Δ with $|P| \ge 2$ would contain a point fixed by $\mathrm{Alt}(\Delta)$, forcing $P^a = P$ for all $a \in \mathrm{Alt}(\Delta)$, which is impossible; so every class meeting Δ is a singleton, i.e. a point of $[k] \setminus \operatorname{supp}_x$, and $|\Delta| \le |[k] \setminus \operatorname{supp}_x| < k/4$. (iv) If $\Delta \cap C' = \emptyset$ then $|\Delta| \le k - |C'| \le k/4 + 4$. If $\Delta \subseteq C'$, then either $\mathrm{Alt}(\Delta)$ preserves a non-trivial regular graph on C', whose relative symmetry defect is at least $1/2$ [GNS18, Lemma A.3] (= [Bab16a, Cor. 2.4.13]), or it preserves the colouring by in- and out-neighbours of an individualized vertex, whose classes have size $\le |C'|/2$. So $|\Delta| \le k/2$. (v) Lemma 5.4(3). This also covers a small slip in GNS: [GNS, src. l. 2324] asserts $|D_{i,j}| \ge 3k/4$, whereas $C' = C \setminus I$ only has at least $3k/4 - 4$ elements; the defect bound is unaffected.

Case (iii). This is GNS's Option 2: $\Phi(x) \le \mathrm{Aut}_H(x)$, and C is a $\varphi(\Phi)$-orbit, so $\Phi(x) \le H_C$. □

5.3 The canonical structure in case (v)

Let $D(x) := [k] \setminus \operatorname{supp}_x$, so $|D(x)| > 3k/4$. Every t-set $T \subseteq D(x)$ is non-full: a full T has a certificate in $\Phi(x)$ acting on T as at least $\mathrm{Alt}(T)$, with $t \ge 3$, so $T \subseteq \operatorname{supp}_x$. Let $\vec{T}_0 := (1, \dots, t)$

be the first t labels of $[k]$ (P1). Since $\varphi(H) \geq A_k$ is $(k-2)$-transitive, it is transitive on injective t-tuples. Put $J_0 := H_{(\vec{T}_0)} = \ker(\psi_{T_0} \colon H_{T_0} \to \mathrm{Sym}(T_0))$.

Definition 5.3. Let $\mathcal{P}$ be the set of all pairs $(z^{\langle\langle W(T',z)\rangle\rangle}, \vec{T}')$, where z is a string and $\vec{T}'$ an injective t-tuple with underlying set T'. For $\vec{T} \in D(x)^{\langle t\rangle}$ with underlying set T, put $c_x(\vec{T}) := c(x^{\langle\langle W(T,x)\rangle\rangle}, \vec{T})$, as in Lemma T for this $\mathcal{P}$ and the J_0-canonizer of Lemma 5.6. The structure is

$$\mathfrak{A}(x) := ([k];\ \text{unary } D(x);\ t\text{-ary colouring } \kappa_x), \qquad \kappa_x(\vec{T}) := \text{rank of } c_x(\vec{T}) \text{ among all values of } c_x,$$

encoded as a string on $[k] \sqcup [k]^t$, with value 0 on tuples outside $D(x)^{\langle t\rangle}$.

The set $\mathcal{P}$ is H-invariant by L4. The set of values of c_x is an invariant of the H-orbit of x by Lemma T(b) and L4, so the ranking is canonical.

Lemma 5.4. *1. (Strong canonicity.)* $\mathfrak{A}(x^h) = \mathfrak{A}(x)^{\varphi(h)}$ *for* $h \in H$.

2. (Key identity.) For non-full T: $G(T,x) = \mathrm{Aut}_{H_T}(x^{\langle\langle W(T,x)\rangle\rangle})$, *and* $\psi_T(G(T,x))$ *is not a giant.*

3. (Symmetry defect.) If $\mathrm{Alt}(\Delta) \leq \mathrm{Aut}(\mathfrak{A}(x))$ *with* $|\Delta| \geq 3$, *then* $\Delta \subseteq [k] \setminus D(x)$ *or* $|\Delta| \leq t+1$. *Hence the symmetry defect of* $\mathfrak{A}(x)$ *is at least* $k - \max(k/4, t+1) \geq 3k/4$.

Proof. (1) $W(T^{\varphi(h)}, x^h) = W(T,x)^h$ by L4, D is equivariant, and Lemma T(b) applies. (2) This is L2 together with L3. (3) Either $\Delta \subseteq D$ or $\Delta \cap D = \emptyset$. Suppose $\Delta \subseteq D$ and $|\Delta| \geq t+2$, and pick $T \subseteq \Delta$ with $|T| = t$ and an ordering $\vec{T}$. Every $\pi \in \mathrm{Sym}(T)$ is induced by an element of $\mathrm{Alt}(\Delta)$ (correct the parity on $\Delta \setminus T$, which has at least two points). So all orderings of T have the same colour. The window $W(T,x)$ depends only on the set T, so Lemma T(c), applied to $y = x^{\langle\langle W(T,x)\rangle\rangle}$, gives for every ordering $\vec{T}'$ of T an $h \in H$ with $y^h = y$ and $\vec{T}^{\varphi(h)} = \vec{T}'$. Such h lie in H_T, and $y^h = y$ means $h \in \mathrm{Aut}_{H_T}(y) = G(T,x)$ by (2). So $\psi_T(G(T,x)) = \mathrm{Sym}(T)$, contradicting (2). □

Remark 5.5. Running GNS's comparison [GNS18, Alg. 3] for (T_1, x_1) and (T_2, x_2), the final coset is $\{h : T_1^{\varphi(h)} = T_2,\ (x_1^{[W(T_1,x_1)]})^h = x_2^{[W(T_2,x_2)]}\}$. So the colour classes of Definition 5.3, taken over $D(x_1)^{\langle t\rangle} \sqcup D(x_2)^{\langle t\rangle}$, coincide with GNS's comparison classes, under the reading that an empty comparison coset means "no morphism" (GNS's Alg. 3 does not specify what happens when the coset becomes empty). The algorithm never uses this fact; only the symmetry defect is used. Our bound $t+1$ differs from GNS's $t-1$ [GNS, src. l. 2383]; both suffice.

5.4 Computing the colours

Lemma 5.6. *For* $\vec{T} \in D(x)^{\langle t\rangle}$, $c_x(\vec{T})$ *can be computed in polynomial time plus recursive canonizer calls on pairwise disjoint subsets of* $W(T,x)$, *each of size at most* $n/2$, *with groups that are restrictions of subgroups of* H. *Over all at most* k^t *tuples, the recursive sizes add up to at most* $k^t \cdot n$.

Proof. Transport. Compute $\sigma \in H$ with $\vec{T}^{\varphi(\sigma)} = \vec{T}_0$ (a preimage) and put $z := x^\sigma$. By L4, $(x^{\langle\langle W(T,x)\rangle\rangle})^\sigma = z^{\langle\langle W(T_0,z)\rangle\rangle}$ and $G_i(T_0,z) = G_i(T,x)^\sigma$.

Definition of F_{J_0}. Apply Lemma F2 with group J_0 on the J_0-invariant set $\mathcal{Y}_0 := \{z^{\langle\langle W(T_0,z)\rangle\rangle}\}$, with windows $V_{i+1}(y) := \mathrm{Aff}(\mathrm{Aut}_{H_{T_0}}(y), \psi_{T_0}) \setminus U_i$ for as long as $\psi_{T_0}(\mathrm{Aut}_{H_{T_0}}(y)) \geq \mathrm{Alt}(T_0)$ and this set is non-empty. These windows are H_{T_0}-equivariant, hence J_0-equivariant, and by L2 they reproduce the chain of (T_0, z). So F_{J_0} is a fixed deterministic function on $\mathcal{Y}_0$, which is what Lemma T needs.

Stage i. Let F_i be the current canonical form and $\tau_i \in \Lambda_i$. Then $B := \mathrm{Aut}_{H_{T_0}}(F_i) = G_i(T,x)^{\sigma\tau_i}$, which is known from Lemma 5.1, and $A_i = \mathrm{Aut}_{J_0}(F_i) = \ker(\psi_{T_0}|_B)$. The new window V' is a union of B-orbits that are affected with respect to the giant representation $\psi_{T_0}|_B$ of degree $t \geq 9$. By [GNS18, Thm. 5.9], every A_i-orbit in V' has length at most n/t. Canonize $w^{[V']}$ under A_i by the chain rule (after restricting to V' by Lemma F1); its windows lie in A_i-orbits. (If $|V'| \leq n/2$ one may instead make one call on V'; whichever rule is used must be a fixed function of $|V'|$, which is J_0-invariant, so that F_{J_0} remains a single function on $\mathcal{Y}_0$.)

Sizes. The sizes for one tuple sum to at most $|W(T,x)| \leq n$. No descent over $t!$ cosets is needed, because the right subgroup A_i is already the kernel. GNS spend $\binom{k}{t}^2 t!\, n$ on comparisons; we spend at most $k^t n$. □

5.5 Option 1: canonical find-structure

Proposition 5.7. *In cases (i), (ii), (iv), (v), an H-canonizer for x is obtained as follows. For each $\mathfrak{A} \in \mathfrak{F}(x)$, let $s_{\mathfrak{A}}$ be the string encoding $\mathfrak{A}$ on $[k] \sqcup [k]^2 \sqcup [k]^t$. Let F^{BC} be Babai's canonical form [Bab19, Thm. 2.2], called only for the group S_k in its induced action on this domain, on the generating set determined by (k,t) (requirement D4), and put $\Lambda^{S_k}(\mathfrak{A}) := \mathrm{Iso}_{S_k}(s_{\mathfrak{A}}, F^{\mathrm{BC}}(s_{\mathfrak{A}}))$, computed by Babai's string isomorphism test [Bab19, Thm. 2.1]. If $\varphi(H) = S_k$, put $\Lambda'(\mathfrak{A}) := \Lambda^{S_k}(\mathfrak{A})$. If $\varphi(H) = A_k$, then*

- *if $\mathrm{Aut}_{S_k}(s_{\mathfrak{A}})$ contains an odd permutation, put $\Lambda'(\mathfrak{A}) := \Lambda^{S_k}(\mathfrak{A}) \cap A_k$, with form $F^{\mathrm{BC}}(s_{\mathfrak{A}})$;*
- *otherwise the elements of $\Lambda^{S_k}(\mathfrak{A})$ all have one parity; if it is even, put $\Lambda'(\mathfrak{A}) := \Lambda^{S_k}(\mathfrak{A})$, with form $F^{\mathrm{BC}}(s_{\mathfrak{A}})$, and if it is odd, put $\Lambda'(\mathfrak{A}) := \Lambda^{S_k}(\mathfrak{A})\,(1\,2)$, with form $F^{\mathrm{BC}}(s_{\mathfrak{A}})^{(1\,2)}$.*

Then $\varphi^{-1}(\Lambda'(\mathfrak{A})) = \sigma_{\mathfrak{A}} R_{\mathfrak{A}}$ with $R_{\mathfrak{A}} = \varphi^{-1}(\mathrm{Aut}_{\varphi(H)}(\mathfrak{A}_{\mathrm{can}}))$. Recursively canonize $x^{\sigma_{\mathfrak{A}}}$ under $R_{\mathfrak{A}}$, and combine over $\mathfrak{A} \in \mathfrak{F}(x)$ by Lemma S (for pairs $(x,\mathfrak{A})$) and Lemma F4.

Proof. Λ' is a $\varphi(H)$-canonizer on structures: for $\varphi(H) = S_k$ because F^{BC}, on a generating set determined by (k,t), is constant on S_k-orbits and takes values in them; for $\varphi(H) = A_k$ because $\Lambda^{S_k}(s^a) = a^{-1}\Lambda^{S_k}(s)$ for $a \in A_k$, so the case distinction and the parity are constant on A_k-orbits, and in each case the displayed coset is exactly the set of $g \in A_k$ mapping $s_{\mathfrak{A}}$ to the stated form (in the odd case, $s^g = F^{(1\,2)}$ iff $g(1\,2) \in \Lambda^{S_k}(s)$). Lemma S with $S(x,\mathfrak{A}) = \mathfrak{A}$ is equivariant on pairs; then apply Lemma F4. □

Each $R_{\mathfrak{A}}$ satisfies $|\varphi(H) : \varphi(R_{\mathfrak{A}})| \geq (4/3)^k$, by Proposition 5.2(2) and [GNS18, Cor. 6.5] ($k \geq 24$). The number of branches is $r \leq k^6$, as in [GNS18, Lemma 6.7], where $\mathfrak{A}_{1,1}$ is paired with each $\mathfrak{A}_{2,j}$. The calls of BC are on domains of size at most $2k^t = d^{O(\log d)}$, so each costs $\exp(O((\log d)^{2c_{\mathrm{BC}}}))$, where $\exp(O((\log n)^{c_{\mathrm{BC}}}))$ bounds the running time of BC and of the isomorphism test [Bab19, Thm. 2.1] on domains of size n.

Remark 5.8 (What Theorem A needs from BC). BC is called only in Proposition 5.7, only at Option 1 nodes, where $\varphi(H) \in \{A_k, S_k\}$ and $k \leq d$ (§2.6), and the case A_k is obtained from S_k by the parity argument above. So Theorem A needs exactly: a canonical form, under S_k with $k \leq d$, of relational structures on $[k]$ of arity at most $t = O(\log d)$ (arity at most 2 in cases (i), (ii), (iv), and t in case (v)), together with Babai's string isomorphism test [Bab19, Thm. 2.1] (the main result of [Bab16a]) for Λ^{S_k}. Through the standard encoding of a t-ary structure by a graph on $k^{O(t)}$ vertices (cf. [Neu20, proof of Lemma 4.33]), this is a special case of [Bab19, Cor. 2.3] on graphs with $k^{O(t)} = d^{O(\log d)}$ vertices. It is *not* a reduction to graphs on at most d vertices: for $t \geq 3$ there are more t-ary structures on $[k]$ than graphs on $[k]$, so case (v) cannot be encoded on

k vertices. Since [Bab19, Cor. 2.3] is deduced from [Bab19, Thm. 2.2], the dependency is also no lighter in status: the published account of both is the preliminary report [Bab19], which says of its main theorem, "we sketch the proof". This is why the dependency is flagged in the abstract and in the status of Theorem A.

5.6 Option 2: canonical find-symmetry

Lemma 5.9 (canonical find-symmetry)**.** *In case (iii), an H-canonizer is obtained with two recursive canonizer calls under $L := H_{(\Delta_0)} := \varphi^{-1}(\varphi(H)_{(\Delta_0)})$, where $\Delta_0 := \{1, \dots, |\Delta|\}$ (P1). Moreover $|\varphi(H) : \varphi(L)| \geq |\operatorname{Alt}(\Delta_0)| \geq (4/3)^k$.*

Proof. Move Δ to Δ_0. Apply Lemma S with $S(x) = \Delta(x)$ and $\Lambda'(\Delta) = \{\pi \in \varphi(H) : \Delta^\pi = \Delta_0\}$; this is well defined because A_k is transitive on $|\Delta|$-sets. We get $E(x) = \sigma H_{\Delta_0}$. Put $z := x^\sigma$, so $\Delta(z) = \Delta_0$ and $K(z) \leq \operatorname{Aut}(z) \cap H_{\Delta_0}$.

Collapse to two cosets. $L \trianglelefteq H_{\Delta_0}$, and the image of $K(z)$ on Δ_0 contains $\operatorname{Alt}(\Delta_0)$, so $K(z)L$ has index at most 2 in H_{Δ_0} and is normal. Hence $H_{\Delta_0} = K(z)L \cup K(z)L\tau$, where, if $K(z)L \neq H_{\Delta_0}$, τ is any element of $H_{\Delta_0} \setminus K(z)L$ (one inducing an odd permutation on Δ_0); if $K(z)L = H_{\Delta_0}$, omit the second part (this also covers the case $\varphi(H) = A_k$, which GNS do not discuss).

Branch values. In the descent (Lemma F3) from H_{Δ_0} to L, a coset ρL with $\rho = \kappa l \in K(z)L$ has branch value $F_L(z^{\kappa l}) = F_L(z)$. A coset in $K(z)L\tau$ has a representative $\kappa\tau$, and branch value $F_L(z^{\kappa\tau}) = F_L(z^\tau)$.

Output. Two L-calls suffice. With $\lambda_1 \in \Lambda_L(z)$ and $\lambda_\tau \in \Lambda_L(z^\tau)$: $\operatorname{Aut}(z) \cap K(z)L = K(z) \operatorname{Aut}_L(z)$; $\operatorname{Aut}(z)$ meets $K(z)L\tau$ iff $F_L(z^\tau) = F_L(z)$, and then it contains $g_\tau := \tau\lambda_\tau\lambda_1^{-1}$; so $\operatorname{Aut}_{H_{\Delta_0}}(z) = \langle K(z), \operatorname{Aut}_L(z)[, g_\tau]\rangle$. Finally $\Lambda = \operatorname{Aut}(z)\lambda_1$ if $F_L(z) \leq F_L(z^\tau)$, and $\Lambda = \operatorname{Aut}(z)\tau\lambda_\tau$ otherwise. By Lemma F3 this is exactly $\{g : z^g = m\}$. □

This matches [GNS18, Lemma 6.9]: $r \in \{1, 2\}$ and $H = G_{(\Delta_1)}$.

6 The algorithm and its correctness

6.1 Instances

An *instance* is a tuple $(\Omega, H, \vec{\mathfrak{B}}, x)$, where Ω is ordered, $H \leq \operatorname{Sym}(\Omega)$ is given by its canonical generating set, $\vec{\mathfrak{B}}$ is an H-invariant almost d-ary sequence, and d and t are fixed globally. The sequence $\vec{\mathfrak{B}}$ is auxiliary data. It is always computed from groups higher in the recursion and never from x, so isomorphic strings lead to instances that differ only in the string.

6.2 Algorithm CanAD

Step 1. If $|\Omega| = 1$, return (x, H).

Step 2. Intransitive H. Apply the chain rule (Definition 3.1). Each window V' is handled by CanAD on the transitive instance $(V', A_i^{V'}, \vec{\mathfrak{B}}[V'], w|_{V'})$.

Step 3. Transitive H, $H^{\mathfrak{B}_1}$ semi-regular. Then $H^{\mathfrak{B}_1}$ is regular, with $b := |\mathfrak{B}_1| \geq 2$ elements. Descend (Lemma F3) to $H_{(\mathfrak{B}_1)}$ over b cosets. In each branch apply the chain rule. Its windows are orbits of subgroups of $H_{(\mathfrak{B}_1)}$, so each lies in a block of $\mathfrak{B}_1$ and has size at most n/b; per branch the window sizes sum to n (there may be more than b windows).

Step 4. Transitive H, $|\mathfrak{B}_1| \leq d$. Choose (P1) a block system $\mathfrak{B} \succeq \mathfrak{B}_1$ of H on which $P := H^{\mathfrak{B}}$ is primitive, and put $b := |\mathfrak{B}| \leq d$.

(a) If $|P| \le b^{1+\log b}$, or b is below the constant of [GNS18, Lemma 7.1]: descend to $H_{(\mathfrak{B})}$ over $|P|$ cosets and apply the chain rule; its windows have size $\le n/b \le n/2$.

(b) Otherwise apply [GNS18, Lemma 7.1] to P (P1), obtaining N, $\mathcal{C}$, ψ and k. Put $G' := \{g \in H : g^{\mathfrak{B}} \in N\}$, $\varphi := \psi \circ (g \mapsto g^{\mathfrak{B}})$, and let $\mathcal{C}'$ be the lift of $\mathcal{C}$ to Ω, so $\ker\varphi = G'_{(\mathcal{C}')}$. Descend (Lemma F3) from H to G' over at most $b \le d$ cosets, and in each branch ρ call $\mathrm{Agg}(G', \varphi, \mathcal{C}', \vec{\mathfrak{B}}, x^\rho)$.

Step 5. $\mathrm{Agg}(G', \varphi, \mathcal{C}', \vec{\mathfrak{B}}, y)$**.**

- If $k \le 10t$: descend to $\ker\varphi = G'_{(\mathcal{C}')}$ over at most $k!$ cosets, then apply the chain rule. The windows lie inside $\mathcal{C}'$-classes, of size $\le n/2$.
- Otherwise compute the local certificates (Lemma 5.1), Φ, supp_x, and the case (Proposition 5.2). In cases (i), (ii), (iv), (v) apply Proposition 5.7 (for (v) the colours come from Lemma 5.6). In case (iii) apply Lemma 5.9.
- Every recursive call *on the same domain*, under some $R_{\mathfrak{A}}$ or L, is made with the sequence $\vec{\mathfrak{B}} \oplus \mathcal{C}'$ of Step 6.

Step 6. The top partition of same-domain children. Put $\vec{\mathfrak{B}} \oplus \mathcal{C}' := (\{\Omega\} \succ \mathcal{C}' \succ \mathfrak{B}_1 \succ \mathfrak{B}_2 \succ \cdots)$, with duplicates removed. Here $\mathcal{C}' \succeq \mathfrak{B} \succeq \mathfrak{B}_1$. The sequence is invariant under $R_{\mathfrak{A}}, L \le G'$, because $\mathcal{C}'$ is G'-invariant. It is almost d-ary: $|\mathcal{C}'| = |\mathcal{C}| \le b \le d$, and each $\mathcal{C}'$-class contains at most $|\mathfrak{B}_1| \le d$ blocks of $\mathfrak{B}_1$; the lower levels are inherited.

Step 6 is the choice discussed in Remark 7.2. Since $\mathcal{C}' \succeq \mathfrak{B} \succeq \mathfrak{B}_1$, whenever Step 4 applies at a same-domain child, the algorithm picks a block system above $\mathcal{C}'$ restricted to the relevant orbit, and such a block system is also one above $\mathfrak{B}_1$, so it is a choice that the GNS algorithm also permits there. (CanAD tests semi-regularity on $\mathcal{C}'[\Omega']$ rather than on $\mathfrak{B}_1[\Omega']$, so it is a variant of the GNS algorithm rather than literally one of its runs; Section 7 analyses CanAD directly.)

6.3 Correctness

Proposition 6.1. *CanAD returns an H-canonizer for x.*

Proof. Termination. Induction on $(|\Omega|, |H|)$, ordered lexicographically. Every same-domain recursive call is on a proper subgroup, since $R_{\mathfrak{A}}$ and L have index at least $(4/3)^k > 1$ in $G' \le H$. All other calls are on smaller domains. *Each step preserves canonizers.* Each step is one of Lemmas F1–F4, S, 5.9, Proposition 5.7, or a (P1) choice, and each of these maps canonizers of the sub-instances to a canonizer of the instance. *Canonicity of auxiliary data.* Every auxiliary object is either a function of the node's group (P1) or an H-equivariant function of x (Lemma 5.1, Proposition 5.2, Lemma 5.4). The colours use one fixed J_0-canonizer (Lemma 5.6). □

6.4 The choice points of GNS

Table 2 lists every place where GNS make an arbitrary choice or compare two inputs, and how CanAD treats it. Line numbers refer to the arXiv v3 source.

No other pairwise step exists in GNS. The places where two strings occur are [GNS, src. l. 1866–1940], [GNS, src. l. 1980–1990], [GNS, src. l. 1999–2057] and [GNS, src. l. 2370–2385]. The phrase "for $i = 1, 2$" in [GNS18, Lemma 6.6] is presentational everywhere except row 17.

#	GNS place (src. line)	choice	canonical treatment
1	Thm. 4.3 proof; 1103	maximal block of G_B^B, "B arbitrary"	P1 (block of the least point)
2	Thm. 4.3; 1120–1121	numbering $B_1, \dots, B_s$; transporters $\sigma_{1\to i}$	P1 (blocks by least point; lex-least transporters)
3	Thm. 4.3; 1207, 1254	one G^*-orbit A^*	P1 (orbit of the least point of Ω^*)
4	Thm. 4.3; 1216	$\mathcal{P}$-sequence of Thm. 3.13 for N^{B_1}	P1 (deterministic computation)
5	Thm. 4.6; 1315	Johnson identifications $\rho_{i,B}$ ([BLS87]; Lemma 4.4)	P1
6	Lemmas 6.2/6.3; 1802, 1908	windows $\mathrm{Aff}(G_i, \psi)$	equivariant L4; no choice
7	Lemmas 6.2/6.3; 1810, 1917	coset representatives of N	descent over all cosets (F3)
8	Lemma 6.3; 1902	σ_0 with $T_1 \mapsto T_2$	disappears (Lemma T(a))
9	Lemma 7.2; 2087	coarsening to primitive P	P1
10	Lemma 7.2; 2090–2093	$N, \mathcal{C}, \psi$ (Lemma 7.1); G'; labels of $[k]$	P1, then descent over $\le d$ cosets
11	Lemma 7.2; 2101	$k \le 10t$: Luks reduction to $\ker\varphi$	descent over $\le k!$ cosets
12	Lemma 7.2, Alg. 1; 2121–2122	order of orbits	chain rule (Definition 3.1)
13	App. A; 2337–2349	Φ, supp_x, case split	equivariant (Proposition 5.2)
14	App. A; 2352, 2359, 2364	I, orbital R_i, vertex	equivariant family and F4
15	App. A; 2357–2358	"WLOG R_1 is the diagonal"	the diagonal is an invariant
16	App. A; 2368	"WLOG $\lvert D_1\rvert = \lvert D_2\rvert$"	pairwise only; $\lvert D(x)\rvert$ is an invariant
17	App. A; 2370–2383	joint classes on $D_1' \sqcup D_2'$	**Lemma T: the only essential change**
18	Lemma 7.2; 2118–2126 (implicit)	block system of the next round	$\mathcal{C}'$ as top partition (Step 6; Remark 7.2)
19	Lemma 6.7; 2017–2019	pair $\mathfrak{A}_{1,1}$ with each $\mathfrak{A}_{2,j}$; isomorphism test of [Bab16b] on structures	Lemma S with BC per member of the family, then F4 (Proposition 5.7)
20	Lemma 6.9; 2049–2051	g with $\Delta_1 \mapsto \Delta_2$; a transposition τ on Δ_1	Lemma S to Δ_0 (P1), then Lemma 5.9

Table 2: Choice points and pairwise steps of GNS.

6.5 Theorem A from CanAD

Proof of Theorem A (correctness). Cosets are reduced to groups by the coset rule. The first step computes canonical generators, so the output depends only on the coset as a set. For intransitive G, apply the chain rule over orbits; each constituent is a recursive call of the $\widehat{\Gamma}_d$-canonizer, and each restricted group lies in $\widehat{\Gamma}_d$. For transitive G, compute Ω^*, ι, $x \mapsto x^*$ and the almost d-ary sequence by [GNS18, Thms. 4.3, 4.6] (P1; rows 1–5 of Table 2), run CanAD on $(\Omega^*, G^*, \vec{\mathfrak{B}}^*, x^*)$, and pull back by Lemma R. The almost d-ary recursion never re-applies the change of action [GNS18, Obs. 4.2]. Correctness follows from Proposition 6.1 and Lemma R. The time bound is proved in Section 7.5. □

Status. Proof modulo the results listed in Section 13, in particular Babai's canonical form [Bab19, Thm. 2.2], whose published account is a preliminary report that sketches the proof (used only for S_k, $k \le d$, on structures of arity $O(\log d)$; Remark 5.8), the internal constructions of [GNS18, Thms. 4.3, 4.6], and requirements D1–D4 (deterministic subroutines on canonical generating sets); see Questions 15.2, 15.3.

7 Running time

7.1 A recursion lemma

Lemma REC (a recursion lemma). *Let $\tau\colon \mathbb{N} \to \mathbb{N}$ with $\tau(1) = 1$ and let $K \ge 1$. Suppose every $n \ge 2$ satisfies $\tau(n) \le \sum_{i=1}^{\ell} \tau(n_i)$ with $1 \le n_i < n$, where either*

(A) $n_i \le n/2$ for all i and $\sum_i (n_i/n)^{K+1} \le 1$, or

(B) $\sum_i n_i \le n$.

Then $\tau(n) \le n^{K+1}$.

Proof. By induction: $\sum n_i^{K+1} = n^{K+1} \sum (n_i/n)^{K+1} \le n^{K+1}$ in case (A), and $\sum n_i^{K+1} \le (\sum n_i)^{K+1} \le n^{K+1}$ in case (B). □

We apply the lemma to $\tau(n) :=$ the maximum, over all instances with domain of size n, of the number of leaves of the recursion tree of the algorithm on that instance. The hypothesis is a statement about one instance at a time, and each instance of size n has children of sizes n_i as in (A) or (B).

The hypothesis of (A) holds in two situations.

- $n_i \le n/2$ and $\sum n_i \le 2^K n$ [GNS18, Lemma 2.6].
- The semi-regular step (Step 3): $b \ge 2$ branches, each with pieces of size $\le n/b$ that sum to n. Per branch $\sum (n_i/n)^{K+1} \le b^{-K} \sum n_i/n = b^{-K}$, so over all branches the sum is at most $b^{1-K} \le 1$.

Remark 7.1. [GNS18, Lemma 2.8] requires $\sum n_i \le 2^K n$. The semi-regular step of [GNS18, Alg. 4] has recursive sizes summing to $b \cdot n$ with b unbounded (e.g. C_n acting regularly), so the appeal to Lemma 2.8 in the proof of [GNS18, Thm. 7.3] does not literally cover it. Lemma REC covers it. This is a point of presentation only.

7.2 A question about the written potential argument of GNS Lemma 7.2

Remark 7.2. In the proof of [GNS18, Lemma 7.2], the height of the recursion is bounded by showing that the degree of the giant representation drops from k to $k' \le k - k/(3\log k)$ from one round to the next. The proof reads [GNS, src. l. 2126]: "Observe that $|(H')^{\mathfrak{C}'}| \ge \frac{(k')!}{2}$ because $(H')^{\varphi'} \ge A_{k'}$ and $H'_{(\mathfrak{C}')} \le \ker(\varphi')$." Combined with $|H^{\mathcal{C}'}| \le k!/(4/3)^k$, this gives the inequality and a recursion of height $O(\log^2 d)$.

As far as we can see, the algorithm as written does not ensure $H'_{(\mathcal{C}')} \le \ker\varphi'$. In the next round the algorithm starts again from $\mathfrak{B}_1$, restricted to the current orbit, and replaces it by "a block system of smaller size" on which the group acts primitively [GNS, src. l. 2087]. Neuen's version of the same proof says "Let $\mathfrak{B} \succ \mathfrak{B}_1$ be a minimal block system" [Neu20, src. l. 2683]. Nothing ties this block system to $\mathcal{C}'$. The partition $\mathcal{C}'$ is used only in the analysis. The remark "Recall that $\mathfrak{C}'$ is G'-invariant and thus, it is also H-invariant" [GNS, src. l. 2120] is needed in any case, so that $H^{\mathcal{C}'}$ is defined.

An example. Let m be large, $d := m^2$, $q \ge 1$, $\Omega := [m]^2 \times [q]$, and $G := (S_m \wr S_2) \times C_q$, with the product action on $[m]^2$ and C_q regular on $[q]$. Take the sequence $\{\Omega\} \succ \mathfrak{B}_1 := \{\{(a,b)\} \times [q]\} \succ$ discrete, which is almost d-ary. Let $U \subset [m]$ with $|U| \approx 0.3m$, and $x((a,b),c) := [a \in U]$.

- *First round.* $P = G^{\mathfrak{B}_1} = S_m \wr S_2$ is primitive. Every output of [GNS18, Lemma 7.1] has $N \le S_m \times S_m$, $k = m$, and $\mathcal{C}$ the rows (up to swapping coordinates), so $G' = S_m \times S_m \times C_q$ and $\ker\varphi = 1 \times S_m \times C_q$.
- *Aggregation.* A test set $T \subseteq [k]$ (a set of rows) is full iff $T \subseteq U$ or $T \subseteq U^c$. So $\varphi(\Phi(x))$ has orbits U and U^c, $\mathrm{supp}_x = [k]$, and no orbit has $\ge 3k/4$ points: case (ii), with structure $\{U, U^c\}$ and $H = \varphi^{-1}(\mathrm{Sym}(U) \times \mathrm{Sym}(U^c)) \cap G'$.
- *Next round.* The large H-orbit is $\Omega' = (U^c \times [m]) \times [q]$, of size about $0.7n$. The group H' induced on the cells of Ω' is $\mathrm{Sym}(U^c) \times S_m$ in product action, which has exactly two minimal block systems: rows and columns. If the columns are chosen, $P' \cong S_m$ in its natural action, so $k' = m = k$, while $H'_{(\mathcal{C}')}$ (which fixes all rows) acts as S_m on the columns and is not contained in $\ker\varphi'$.

For all sufficiently large m the numerical hypotheses of [GNS18, Lemma 7.2] hold (for instance $m \ge 200$ with $t = \max\{9, 3 + \lceil \log d \rceil\}$, apart from the unspecified constant of [GNS18, Lemma 7.1]). At $m = 200$ the bound $k - k/(3\log k)$ is about 191.3, the choice of columns gives $k' = 200$, and the choice of rows gives $k' = 140$. A small computation confirmed the block systems and groups above for $q = 2$, $|U| = 2$ and $m = 5, \dots, 8$.

The "fewest blocks" reading. If "a block system of smaller size" is read as "a block system with the fewest blocks", this example picks the rows ($0.7m$ blocks rather than m) and is avoided. We do not know whether that reading suffices in general. The text does not state it, and Neuen's wording does not suggest it; in any case the proof would need an argument for it.

A choice that makes the step valid. Choose the next block system above $\mathcal{C}'[\Omega']$. This is always possible: $\mathcal{C}'$ is G'-invariant, hence invariant under every same-domain descendant, and it has at least two classes on the large orbit. Since $\mathcal{C}' \succeq \mathfrak{B} \succeq \mathfrak{B}_1$, it is a run permitted by the algorithm as written. With this choice the inequality $k' \le k - k/(3\log k)$ holds (Section 7.3), and the rest of the analysis is unchanged. Neuen's proof [Neu20, Lemma 4.36] contains the same step [Neu20, src. l. 2716], in the notation $\Lambda'_{(\mathfrak{C}')} \le \ker(\varphi')$, and the same choice applies there.

What is and is not claimed. The theorem of GNS is not in doubt: the proposed choice is a legal run of their algorithm, and with it their own inequality holds. We have not shown that the

running time fails under other choices; in the example the chain ends one round later. The same passage appears in all arXiv versions of GNS and of [Neu20]. We found no erratum, but we could not consult the SIAM journal version of GNS.

Question 7.3. Is "choose the next block system above $\mathcal{C}'$" the intended reading of the proof of [GNS18, Lemma 7.2] (and of [Neu20, Lemma 4.36])? Is the point addressed in the SIAM journal version? Does any other part of the GNS analysis use the original $\mathfrak{B}_1$ in a way this choice disturbs? (We checked that local certificates need only some almost d-ary sequence, and that [GNS18, Thm. 5.9, Cor. 6.5] and the case analysis do not use the sequence at all.)

7.3 The potential argument with $\mathcal{C}'$ as top partition

Consider a same-domain chain: an aggregation node (G', φ, k); a child $(R, \cdot)$ with $R \in \{R_{\mathfrak{A}}, L\}$ and $|\varphi(G') : \varphi(R)| \geq (4/3)^k$; the chain rule on R; and the window Ω' of size $> n/2$, if any, processed under the current right subgroup $R^\circ \leq R$. By the choice of chain rule, the next node $H' := (R^\circ)^{\Omega'}$ is transitive, with top partition $\mathcal{C}'[\Omega']$. Note $|\mathcal{C}'[\Omega']| \geq 2$, since $\mathcal{C}'$-classes have size $\leq n/2 < |\Omega'|$.

At H', Step 3 or Step 4 is applied with $\mathfrak{B}_1 = \mathcal{C}'[\Omega']$. Let $\mathfrak{D} \succeq \mathcal{C}'[\Omega']$ be the block system chosen there. If Step 4(b) applies, it produces $G'' \leq H'$ (index $\leq d$) and $\varphi' = \psi' \circ$ (action on $\mathfrak{D}$) of degree k'.

Lemma 7.4. *In this situation, $k'!/2 \leq k!/(4/3)^k$, and hence $k' \leq k - k/(3 \log k)$.*

Proof.

1. An element of G'' fixing every class of $\mathcal{C}'[\Omega']$ fixes every class of the coarser partition $\mathfrak{D}$, so it lies in $\ker \varphi'$. Hence $A_{k'} \leq \varphi'(G'')$ is a quotient of $(G'')^{\mathcal{C}'[\Omega']}$.

2. The $\mathcal{C}'$-classes meeting Ω' form an R°-invariant subset of $\mathcal{C}'$, and $C \mapsto C \cap \Omega'$ is a bijection onto $\mathcal{C}'[\Omega']$. So $(H')^{\mathcal{C}'[\Omega']}$ is a quotient of $(R^\circ)^{\mathcal{C}'}$, and $|(G'')^{\mathcal{C}'[\Omega']}| \leq |(R^\circ)^{\mathcal{C}'}| \leq |R^{\mathcal{C}'}|$.

3. $R^{\mathcal{C}'} \cong R/R_{(\mathcal{C}')} = R/(R \cap \ker \varphi) \cong \varphi(R)$, using $\ker \varphi = G'_{(\mathcal{C}')}$, and $|\varphi(R)| \leq |\varphi(G')|/(4/3)^k \leq k!/(4/3)^k$.

4. So $k!/k'! \geq (4/3)^k/2$. Since $k!/k'! \leq k^{k-k'}$, we get $k \log(4/3) \leq 1 + (k - k') \log k$, hence $k - k' \geq (0.415k - 1)/\log k \geq k/(3 \log k)$ for $k \geq 13$.

□

(The arithmetic in step 4 was also checked numerically, by computing the largest k' with $k'!/2 \leq k!/(4/3)^k$ for every $13 \leq k \leq 5000$.)

Height. Every giant degree is at most d, and aggregation stops once $k \leq 10t$. By Lemma 7.4, $k_{j+1} \leq k_j(1 - 1/(3 \log d))$ along a root-to-leaf path. So a path contains at most $h := 3 \log d \cdot \ln(d/(10t)) + 1 = O(\log^2 d)$ aggregation nodes.

7.4 One macro step and its branch counts

A *macro step* is the computation for one transitive instance on n points to which Step 4 applies. It explores a tree $\mathcal{T}$ of same-domain nodes as in Section 7.3: descents to G', aggregation nodes, their children $R_{\mathfrak{A}}$ or L, the chain rule on these children, and the transitive node H' on the window of size $> n/2$. Calls on domains of size $\leq n/2$ are the *exits* of the macro step. If Step 3 (semi-regular) applies at an inner node H', that node is a leaf of $\mathcal{T}$; its children have size $\leq |\Omega'|/b \leq n/2$ and total size $\leq b|\Omega'| \leq d\,n$, since $b = |\mathcal{C}'[\Omega']| \leq |\mathcal{C}| \leq d$, and they are exits. (A semi-regular step at the *root* of an instance, with b unbounded, is a separate (A)-node of Lemma REC.)

quantity	GNS	canonical	where
descent $H \to G'$	$\le d$	$\le d$	F3
Luks reduction, small primitive P	$\lvert P\rvert \le d^{1+\log d}$	same	F3
$k \le 10t$, to $\ker\varphi$	$\le k!$	same	F3
semi-regular inner leaf	$b \le d$ cosets	same	F3, chain rule
local certificates	$\binom{k}{t} t!\, n$	same	Lemma 5.1
comparisons / colours	$\binom{k}{t}^2 t!\, n$	$\le k^t n$	Lemma 5.6
Option 1 branches	$r \le k^6$	$r \le k^6$	Prop. 5.7
Option 2 branches	$r \le 2$	2 calls under L	Lemma 5.9
index of children	$\ge (4/3)^k$	$\ge (4/3)^k$	[GNS18, Cor. 6.5]
black box on structures	$\le k^6$ isomorphism tests	$\le k^6$ BC and isomorphism calls	Prop. 5.7
extra non-recursive work	—	canonical generators, lex-least transporters: poly(n) per node	§2.4

Canonization explores no branch that the isomorphism test does not. GNS never prune by "one isomorphism suffices": Luks reduction, the descent to G' and the descent to $\ker\varphi$ all enumerate every coset [GNS, src. l. 459–470, 2097, 2101]. Their only pruning by automorphisms is find-symmetry, which Lemma 5.9 reproduces. GNS pair $\mathfrak{A}_{1,1}$ with each of the r structures of x_2, which is also r branches. Early exits of GNS (the two strings fall into different cases) correspond to computing one side only, which costs the same.

Bounds for one macro step.

- *Nodes of $\mathcal{T}$.* Each level branches by at most $d \cdot k^6 \le d^7$, so $\mathcal{T}$ has at most $d^{7(h+2)} = 2^{O(\log^3 d)}$ nodes.
- *Exits per node.* At most $2k^t n$ (certificates and colours), plus $k^6 n$ (small windows of the children), plus, at leaves, $\max(d^{1+\log d}, k!, d) \cdot n$. In total $2^{O(\log^2 d)} n$.
- *Total exits.* $\sum n_i \le 2^{K_0 \log^3 d}\, n$ with every $n_i \le n/2$, for an absolute constant K_0.
- *Non-recursive work.* $2^{O(\log^3 d)} \cdot \left(n^{O(1)} + \exp(O((\log d)^{2c_{\mathrm{BC}}}))\right) = d^{O((\log d)^{c_1})} n^{O(1)}$ with $c_1 := \max(2, 2c_{\mathrm{BC}} - 1)$.

7.5 Proof of the time bound

Proof of Theorem A (running time). Almost d-ary instances with domain size N (and $d \le N$). A macro step satisfies Lemma REC(A) with $K := K_0 \log^3 d$, by [GNS18, Lemma 2.6]. The semi-regular step satisfies (A). The intransitive step satisfies (B), with $\ell \ge 2$. So the recursion tree has at most N^{K+1} leaves. Its depth is at most $2\log N + 1$: by the choice of chain rule, every child of a (B)-node is transitive, and a transitive node is an (A)-node, so sizes halve at least every two levels. Hence there are at most $(2\log N + 2)N^{K+1}$ macro nodes, each costing $d^{O((\log d)^{c_1})} N^{O(1)}$. The total is $N^{O((\log d)^{c_2})}$ with $c_2 := \max(3, c_1)$, since $d \le N$.

$\widehat{\Gamma}_d$ *instances.* A transitive constituent of size n_i becomes an almost d-ary instance on $N_i := |\Omega_i^*| \le n_i^{O(\log^2 d)}$ points [GNS18, Thm. 4.6], costing $n_i^{O((\log d)^{c_2+2})}$. The constituents are disjoint and the bound is superadditive, so the total is $n^{O((\log d)^c)}$ with $c := c_2 + 2$. The coset rule and Lemma R add polynomial overhead (preimages, [GNS18, Thm. 2.1]). □

The only changes to GNS's analysis are the choice of Remark 7.2 and the recursion lemma (Lemma REC).

Remark 7.5 (The exponent). We have not tried to optimize c, and we make no claim about its value here. A separate analysis of the exponent in GNS and in the canonical version is work in progress.

8 Corollary B: the canonical Luks reduction

We follow [BL83, §4.3, Thm. 4.2] and the "fourth trick, due to Babai" of [BL83, §4.2], a half-edge encoding that keeps the string domain of polynomial size independently of d. We deviate in two cosmetic ways: we pad every vertex to $d-1$ slots instead of first canonizing the valence string, and we fold the edges inside the new layer into the same string.

Setting. Let X be a graph on V, $|V| = n$, of maximum degree $d \geq 3$ (for $d \leq 2$ everything is trivial), and put $q := d - 1$.

Step 0 (components). Canonize each connected component C, obtaining $\mathrm{CF}(C)$ on $[|C|]$ and its coset $\Lambda(C)$. $\mathrm{CF}(X)$ places the component forms on consecutive intervals of $[n]$, sorted by (size, form). $\Lambda(X)$ consists of all bijections that map each component, by a member of its own $\Lambda(C)$, onto an interval carrying an equal form. It is represented by one element together with generators of $\prod \mathrm{Aut}(\mathrm{CF}(C))$ and the swaps of intervals with equal forms. This is the first paragraph of [BL83, §4.3].

Step 1 (edge rooting). For connected X with at least two vertices and each *ordered* edge (u, v) (at most nd of them), compute $\mathrm{CF}(X, u, v)$ on $[n]$ with $u \mapsto 1$, $v \mapsto 2$, together with $\Lambda(X, u, v) = \{\mu : X^\mu = \mathrm{CF}(X, u, v), u^\mu = 1, v^\mu = 2\}$. Then put $\mathrm{CF}(X) := \min \mathrm{CF}(X, u, v)$ over all ordered edges, and let $\Lambda(X)$ be the union of the $\Lambda(X, u, v)$ attaining the minimum. This is Lemma F4 with the equivariant family of ordered edges. The roots can be forgotten because they always go to 1 and 2.

Layers. Put $V_0 := \{u, v\}$ and $V_r := \{w : \mathrm{dist}(w, \{u, v\}) = r\}$ for $r \geq 1$; let $X_r := X[V_0 \cup \cdots \cup V_r]$, $N_r := |V_{\leq r}|$, and $I_r := (N_{r-1}, N_r]$ (the labels of layer r). Every $y \in V_r$ has at most q neighbours in V_{r+1} (its *down-degree* $\deg_\downarrow y$): for $r \geq 1$ because y has a neighbour in V_{r-1}, and for $r = 0$ because u and v are adjacent. This is the "outvalence $\leq d - 1$" of [BL83, Lemma 4.4].

Invariant (I_r)**.** CF_r is a graph on $[N_r]$ that depends only on the isomorphism type of (X, u, v), and $\Lambda_r := \{\lambda \colon V_{\leq r} \to [N_r] : u \mapsto 1, v \mapsto 2, X_r^\lambda = \mathrm{CF}_r\}$ is non-empty. Every such λ maps V_i onto I_i, because isomorphisms fixing the roots preserve the distance to the roots, and distances in X_r equal those in X. The group $\hat{A}_r := \mathrm{Aut}(\mathrm{CF}_r; 1, 2)$ (the *doubly rooted* automorphism group) satisfies $\Lambda_r = \lambda \hat{A}_r$. The base case is $\mathrm{CF}_0 =$ the edge $\{1, 2\}$ on $[2]$, $\Lambda_0 = \{u \mapsto 1, v \mapsto 2\}$.

Step $r \to r+1$: the group. Let $L := I_r$ and let $\hat{H} := S_q \wr_L \hat{A}_r$ act on $\Xi_r := [N_r] \sqcup (L \times [q]) \sqcup (L \times [q])^2$: an element $(\hat{a}, s)$ acts by $\hat{a}$ on $[N_r]$, by $(\ell, i) \mapsto (\ell^{\hat{a}}, i^{s_\ell})$ on slots, and coordinatewise on pairs of slots. $\hat{H}$ is generated by the generators of $\hat{A}_r$ and two generators of S_q per $\ell \in L$, and it acts faithfully (this needs the summand $[N_r]$, since $\hat{A}_r$ need not act faithfully on L). It lies in $\widehat{\Gamma}_q$ if $\hat{A}_r$ does, being an extension of S_q^L by $\hat{A}_r$. The domain has $|\Xi_r| \leq n + nq + (nq)^2 \leq 3n^4$ points

and is ordered with $[N_r]$ first, then slots lexicographically, then pairs lexicographically. $\hat{H}$ and the order of Ξ_r are functions of CF_r (as a set) and of (N_r, L, q).

The string. Given $\lambda \in \Lambda_r$ and a slot assignment β (bijections $\beta_y \colon N_\downarrow(y) \to [\deg_\downarrow y]$ for $y \in V_r$), put $\mathrm{slot}(y, w) := (y^\lambda, \beta_y(w))$ for each down-edge yw, and define $x_{\lambda,\beta}$ on Ξ_r by: $*$ on $[N_r]$; on a slot (ℓ, i), 1 if it is occupied ($i \le \deg_\downarrow(\ell^{\lambda^{-1}})$) and 0 otherwise; on a pair of occupied slots with endpoints $w, w' \in V_{r+1}$, the value "=" if $w = w'$, "E" if $ww' \in E(X)$, and "$\neq$" otherwise; 0 if either slot is unoccupied.

Claim 1 (one orbit)**.** *All $x_{\lambda,\beta}$, for $\lambda \in \Lambda_r$ and all β, lie in one $\hat{H}$-orbit.*

Proof. Let $\lambda' = \lambda\hat{a}$ and let β' be another slot assignment. For $\ell \in L$ put $y := \ell^{\lambda^{-1}}$ and let s_ℓ map $\beta_y(w)$ to $\beta'_y(w)$, extended arbitrarily on unoccupied slots. Then $(\hat{a}, s)$ maps $\mathrm{slot}_{\lambda,\beta}(y, w) = (y^\lambda, \beta_y(w))$ to $(y^{\lambda\hat{a}}, \beta'_y(w)) = \mathrm{slot}_{\lambda',\beta'}(y, w)$, so it preserves occupancy and the pair values. □

Claim 2 (invariance)**.** *Let $\pi \colon (X_{r+1}, u, v) \to (X'_{r+1}, u', v')$ be an isomorphism of rooted graphs (here X_{r+1} is the subgraph induced on the first $r+1$ layers). Then $x_{\pi^{-1}\lambda,\beta\circ\pi^{-1}}(X') = x_{\lambda,\beta}(X)$, and $\hat{H}$ depends only on CF_r. In particular the orbit $\mathcal{O}_r$ is an isomorphism invariant of (X, u, v); in particular the first statement applies to automorphisms of (X_{r+1}, u, v).*

Proof. $x_{\lambda,\beta}$ depends only on (X_{r+1}, u, v), λ and β: the layer V_{r+1}, the down-edges and the edges inside V_{r+1} are read off X_{r+1}. Transporting everything along π gives the claim. □

Canonization. Let $\hat{F} := F_{\hat{H}}(x_{\lambda,\beta})$ and $\hat{\Lambda} := \Lambda_{\hat{H}}(x_{\lambda,\beta})$, computed by *one call of Theorem A* with group $\hat{H} \in \widehat{\Gamma}_{d-1}$ on at most $3n^4$ points. $\hat{F}$ depends only on $\mathcal{O}_r$.

Reading off CF_{r+1}. In $\hat{F}$, "=" is an equivalence relation on occupied slots (it is one in $x_{\lambda,\beta}$, and $\hat{F} = x^{\hat{h}}_{\lambda,\beta}$). Order its classes by least slot and label them $N_r + 1, \dots, N_{r+1}$. Let CF_{r+1} be CF_r together with an edge $\{\ell, N_r + j\}$ for each occupied slot (ℓ, i) in class j, and an edge $\{N_r + j, N_r + j'\}$ whenever $\hat{F}$ has "E" on a pair of slots from classes j and j' (this value does not depend on the choice of slots, since it depends only on the endpoints). So CF_{r+1} is a function of $(\mathrm{CF}_r, \hat{F})$, hence an invariant.

The labelings. For $\hat{h} = (\hat{a}, s) \in \hat{\Lambda}$ define $\mu = \mu(\lambda, \beta, \hat{h})$ by $y \mapsto y^{\lambda\hat{a}}$ on $V_{\le r}$, and on V_{r+1} by $w \mapsto N_r + j$, where j is the class of $\hat{F}$ containing $\hat{h}(\mathrm{slot}(y, w))$ for any up-neighbour y of w. This is well defined, because the slots of w are "="-related and every w has an up-neighbour, and it is a bijection onto I_{r+1}.

Claim 3. $X^\mu_{r+1} = \mathrm{CF}_{r+1}$.

Proof. $X^{\lambda\hat{a}}_r = \mathrm{CF}^{\hat{a}}_r = \mathrm{CF}_r$. For $y \in V_r$, $w \in V_{r+1}$: $yw \in E$ iff $\mathrm{slot}(y, w)$ is occupied with endpoint w, iff its $\hat{h}$-image (a slot of y^μ) lies in class w^μ. This uses that $x_{\lambda,\beta}$ and $\hat{F}$ agree along $\hat{h}$, and that the slots of a fixed y have distinct endpoints (β_y is a bijection). No edge joins V_{r+1} to $V_{\le r-1}$. Edges inside V_{r+1} are read off the "E" values in the same way. □

Claim 4. $\{\mu(\lambda, \beta, \hat{h})\} = \Lambda_{r+1} := \{\mu \text{ rooted} \colon X^\mu_{r+1} = \mathrm{CF}_{r+1}\}$.

Proof. "$\subseteq$" is Claim 3. For "$\supseteq$": by Claim 2 applied to an automorphism g of (X_{r+1}, u, v), the choice $(g^{-1}\lambda, \beta \circ g^{-1}, \hat{h})$ is valid, satisfies $x_{g^{-1}\lambda,\beta\circ g^{-1}} = x_{\lambda,\beta}$, and yields $g^{-1}\mu$. So the set of outputs is closed under $\mathrm{Aut}(X_{r+1}, u, v)$; Λ_{r+1} is a single coset of that group, and it contains an output. □

Claim 5 (generators)**.** *Let* $\theta\colon \mathrm{Aut}_{\hat{H}}(\hat{F}) \to \mathrm{Sym}([N_{r+1}])$ *act by* $\hat{a}$ *on* $[N_r]$ *and by the induced permutation of "=" -classes on the new labels. Then* $\theta(\mathrm{Aut}_{\hat{H}}(\hat{F})) = \hat{A}_{r+1}$.

Proof. "$\subseteq$" is clear. For "$\supseteq$", let $b \in \hat{A}_{r+1}$. It fixes $1, 2$ and preserves layers, so it preserves $[N_r]$; put $\hat{a} := b|_{[N_r]} \in \hat{A}_r$. For $\ell \in L$, the classes adjacent to ℓ correspond bijectively to the occupied slots of ℓ, and b maps them to the classes adjacent to ℓ^b. Let s_ℓ follow this bijection, extended arbitrarily on unoccupied slots (b preserves down-degrees). Then $(\hat{a}, s)$ fixes $\hat{F}$ and $\theta(\hat{a}, s) = b$. □

So $\Lambda_{r+1} = \mu_0 \hat{A}_{r+1}$ is obtained in polynomial time from $\hat{\Lambda}$, and by induction (I_r) holds for all r; at $r = R$, the eccentricity of $\{u, v\}$, we have $X_R = X$. By induction $\hat{A}_r \in \widehat{\Gamma}_{d-1}$: $\hat{A}_0 = 1$, and $\hat{A}_{r+1}$ is a quotient of a subgroup of $\hat{H} \in \widehat{\Gamma}_{d-1}$. This reproves [BL83, Lemma 4.3] (due to Luks) inside the construction.

Proof of Corollary B. There are at most n^2 rooted edges and at most n layers, so at most n^3 calls of Theorem A on domains of size at most $3n^4$ with groups in $\widehat{\Gamma}_{d-1}$, each costing $(3n^4)^{O((\log d)^c)}$, plus polynomial overhead. The total is $n^{O((\log d)^c)}$. As in [GNS18, Thm. 8.1], the dependence on d comes only from Theorem A. □

Status. Proof modulo Theorem A. In addition, an independent implementation of this construction, with brute-force canonization in place of Theorem A, was checked on 47 small connected graphs ($n \le 8$, $d \le 4$, every ordered edge and every layer: Claims 1, 3, 4, 5 and the minimum over roots) and on six disconnected graphs (component rule). This is a sanity check only.

Comparison with [BL83]. [BL83, §4.3] alternates two steps: canonize Y_r (the edges inside V_r) relative to the current coset, then extend through the bipartite graph Z_r between $V(X_r)$ and V_{r+1} by the algorithm of [BL83, §4.2]. Trick 4 there canonizes the valence string, extends the coset to the edges "allowing arbitrary permutations of edges with common origin", canonizes $y(e_1, e_2) = [e_1, e_2 \text{ end at the same vertex}]$ on $E \times E$, and orders the new vertices by their least incident edge. Our padded slots replace the valence step (the group S_q^L is larger than $\prod S_{\deg_\downarrow}$, still in $\widehat{\Gamma}_{d-1}$, and Claim 1 shows the extra freedom is harmless), and our pair values "E" replace Y_{r+1}.

9 Corollary D and relational structures

Proof of Corollary D. Let x be a string on V^2 (e.g. the adjacency string of a graph on V) and $\Delta\rho$ a labeling coset with $\Delta \in \widehat{\Gamma}_d$. By Remark 2.3, $\mathrm{CL}(x, \Delta\rho) := \rho\Lambda_H(x^\rho)$ with $H := \rho^{-1}\Delta\rho \le \mathrm{Sym}([n])$. The induced action of H on $[n]^2$ is faithful and isomorphic to H, so it lies in $\widehat{\Gamma}_d$. Theorem A on n^2 points gives the bound $|V|^{O((\log d)^c)}$. □

Status. Proof modulo Theorem A.

Corollary D is the canonical counterpart of [Wie20, Thm. 31], which computes $\mathrm{Iso}(G_1; G_2) \cap \Delta\varphi$ in time $|V|^{\mathrm{polylog}(\mathrm{cw}\,\Delta)}$. Theorem A with $\mathrm{Sym}(D)$ acting on D^t also gives canonical forms of t-ary structures and hypergraphs in time $n^{O(t(\log n)^c)}$, the canonical counterparts of [GNS18, Thm. 8.3, Cor. 8.4]. These are weaker than known canonization results [Wie21, Thms. 5.2, 5.4], so we claim nothing new there.

10 Theorem H: canonical Neuen 2020

Throughout this section, "Neu20" refers to arXiv:2002.06997v2, and result numbers were computed from its source. Neuen's results are used as listed in Section 13.

10.1 Neuen's algorithm

Neuen solves *generalized string isomorphism*: a group Γ, a Γ-invariant partition $\mathfrak{P}$, and sets $\mathfrak{X}, \mathfrak{Y}$ of $\mathfrak{P}$-strings, i.e. pairs (P, x) with $P \in \mathfrak{P}$ and $x\colon P \to \Sigma$. Hypergraphs and sets of strings reduce to it equivariantly [Neu20, src. l. 987–1005]. Progress is measured by the *d-virtual size* $s = \sum_{P\in\mathfrak{P}} |P| \cdot m_{\mathfrak{X}}(P)^{f(d)+1}$ with $f(d) = O(\log d)$, which is at most $(n+m)^{O(\log d)}$ [Neu20, src. l. 1570–1578]. An instance is normalized once by [Neu20, Thm. 4.8], which produces an equivalent instance with an almost d-ary sequence, built from structure graphs. The main algorithm [Neu20, Lemma 4.36, Thm. 4.37] proceeds as follows: intransitive groups are processed orbit by orbit and combined by [Neu20, Lemma 4.15] (CombineWindows); a semi-regular top is handled by Luks reduction with BalanceOrbits [Neu20, Lemma 4.3]; otherwise a giant representation [Neu20, Lemma 4.35], the local certificates routine [Neu20, Thm. 4.24], aggregation [Neu20, Lemma 4.29], find-structure and find-symmetry [Neu20, Lemmas 4.33, 4.34], and recursion on the potential k.

The difference from GNS lies inside the local certificates loop. There, the window automorphisms $\Gamma_{(\Omega\setminus W)}$ are automorphisms of a set of strings only if $\mathfrak{X}[W]$ is *simple* [Neu20, Lemma 4.16]. So after each window step the instance is simplified [Neu20, Thm. 4.17] (copies of each block, one per restriction to W) and renormalized [Neu20, Thm. 4.9] (implanted structure graphs). Neuen notes that renormalization "is not isomorphism-invariant" [Neu20, src. l. 1930], and therefore runs the local certificates routine on all pairs of test objects simultaneously [Neu20, src. l. 270–275].

Sets of strings as objects. For $U \subseteq \Omega$ put $\mathfrak{P}[U] := \{P \cap U : P \in \mathfrak{P},\ P \cap U \neq \emptyset\}$, and for a set Z of $\mathfrak{P}$-strings let $Z[U] := \{(P \cap U, x[P \cap U]) : (P, x) \in Z,\ P \cap U \neq \emptyset\}$, a set of $\mathfrak{P}[U]$-strings. A permutation acts by $Z^g := \{(P^g, x^g) : (P, x) \in Z\}$. We call Z *completely occupied* if $m_Z(P) \geq 1$ for all blocks P of its partition; this is Neuen's standing assumption [Neu20, src. l. 1036], it is inherited by every $Z[U]$, and we need it for [Neu20, Obs. 4.12]: if Z is completely occupied and $U \subsetneq U'$, then the d-virtual size of $Z[U]$ is strictly smaller than that of $Z[U']$. Sets of strings on a fixed Ω are compared in a fixed total order (the sorted list of the strings, compared along the order of Ω). The notions of canonizer (Definition 2.1), Lemma 2.2, the coset rule, and the statements and proofs of Lemmas F3, F4, R, S, Lemma 5.9 and Proposition 5.7 use only a right group action and this total order on the objects, so they apply verbatim to sets of strings in place of strings.

10.2 Equivariant coset maps

Definition 10.1. Let $H \leq \mathrm{Sym}(\Omega)$ and let $\mathcal{Z}$ be an H-invariant set of objects. An *H-equivariant coset map* (ECM) assigns to each $Z \in \mathcal{Z}$ a left coset $C(Z) = \pi R(Z) \subseteq H$ of a subgroup $R(Z) \leq H$ such that $C(Z^h) = h^{-1}C(Z)$ for all $h \in H$. We call $R(Z)$ the *right group* of C at Z (the analogue of the right subgroup of a canonizer).

The Λ_H of a canonizer is an ECM with $R = \mathrm{Aut}_H(F_H(x))$. ECMs are the weak notion of canonical labeling used by Schweitzer and Wiebking.

Lemma E1. *For an H-ECM, $R(Z^h) = R(Z)$, and $\mathrm{Aut}_H(Z) \leq C(Z)C(Z)^{-1} = \pi R(Z)\pi^{-1}$ for any $\pi \in C(Z)$.*

Proof. $h^{-1}C(Z)$ has right group $R(Z)$. If $a \in \mathrm{Aut}_H(Z)$ then $C(Z) = C(Z^a) = a^{-1}C(Z)$, so $a \in C(Z)C(Z)^{-1}$. □

Lemma E2 (composition at a prefix). *Let C_1 be an H-ECM on $\mathcal{Z}$ with right group $R_1(Z)$. A* prefix *is a value $\bar{p}(Z)$ that is computed from Z^π and is the same for every $\pi \in C_1(Z)$, satisfies $\bar{p}(Z^h) = \bar{p}(Z)$, and determines $R_1(Z)$. For each prefix value p with right group R, the set $\mathcal{Z}_p := \{Z^\pi : \pi \in C_1(Z),\ \bar{p}(Z) = p\}$ is R-invariant; let C_2^p be an R-ECM on $\mathcal{Z}_p$. Put $C(Z) := \pi \cdot C_2^{\bar{p}(Z)}(Z^\pi)$ for any $\pi \in C_1(Z)$. Then C is a well-defined H-ECM with $C(Z) \subseteq C_1(Z)$.*

Proof. For $\pi' = \pi r$ ($r \in R$): $\pi r \cdot C_2(Z^{\pi r}) = \pi r \cdot r^{-1}C_2(Z^\pi)$. For Z^h use $\pi' = h^{-1}\pi \in C_1(Z^h)$; then $Z^{h\pi'} = Z^\pi$ and $\bar{p}$ is unchanged, so $C(Z^h) = h^{-1}C(Z)$. □

When C_1 and C_2 come from canonizers, this is Lemma F2. The point of E2 is that *anything computed from the prefix by a deterministic procedure, for example a renormalization, is identical for every object in the H-orbit.* This replaces Neuen's synchronization of pairs.

Lemma E3 (change of action at a prefix). *In the setting of E2, suppose the prefix p determines an injective homomorphism $\iota_p \colon R \to \mathrm{Sym}(\Omega_p)$ and an injective map $Z' \mapsto Z'_*$ on $\mathcal{Z}_p$ with $(Z'^r)_* = (Z'_*)^{\iota_p(r)}$. If C^* is an $\iota_p(R)$-ECM on $\{Z'_* : Z' \in \mathcal{Z}_p\}$, then $C_2(Z') := \iota_p^{-1}(C^*(Z'_*))$ is an R-ECM on $\mathcal{Z}_p$.*

Proof. $\iota_p^{-1}(C^*((Z'^r)_*)) = \iota_p^{-1}(\iota_p(r)^{-1}C^*(Z'_*)) = r^{-1}C_2(Z')$. □

Lemma E4 (from an ECM to a canonizer). *Let C be an H-ECM on an H-invariant set $\mathcal{Z}$ with right group $R(Z)$. Suppose that for every value R of the right group an R-canonizer (F_R, Λ_R) is given on the R-invariant set $\{Z^\pi : \pi \in C(Z),\ R(Z) = R\}$. Then*

$$F_H(Z) := F_{R(Z)}(Z^\pi), \qquad \Lambda_H(Z) := \pi\, \Lambda_{R(Z)}(Z^\pi) \qquad (\pi \in C(Z))$$

defines an H-canonizer on $\mathcal{Z}$. Its right subgroup is $\mathrm{Aut}_{R(Z)}(F_H(Z))$, and $\Lambda_H(Z) \subseteq C(Z)$.

Proof. Independence of π. Let $\pi' = \pi r$ with $r \in R$. Then $F_R(Z^{\pi r}) = F_R(Z^\pi)$, and $\pi r\, \Lambda_R(Z^{\pi r}) = \pi r\, r^{-1}\Lambda_R(Z^\pi)$. *(C2).* For Z^h use $h^{-1}\pi \in C(Z^h)$; then $Z^{hh^{-1}\pi} = Z^\pi$, and $R(Z^h) = R(Z)$ by Lemma E1. *(C1).* Every element of $\Lambda_H(Z)$ maps Z to $F := F_H(Z)$. Conversely, let $Z^h = F$. Write $F = Z^{\pi\lambda}$ with $\lambda \in \Lambda_R(Z^\pi)$. Then $C(F) = (\pi\lambda)^{-1}C(Z) = \lambda^{-1}R = R$, so $1 \in C(F)$; on the other hand $C(F) = C(Z^h) = h^{-1}C(Z)$, so $h \in C(Z)$, say $h = \pi r$ with $r \in R$. Then $(Z^\pi)^r = F$, so $r \in \Lambda_R(Z^\pi)$. *Right subgroup.* If $h \in \mathrm{Aut}_H(F)$, the same computation gives $h \in C(F) = R$. □

This is Lemma S in the language of ECMs: an ECM combined with canonizers at the right group gives a canonizer.

10.3 Ordered colours from a setwise reference test set

This is Lemma T moved from the pointwise stabilizer of a reference *tuple* to the setwise stabilizer of a reference *set*, which is the form of Wiebking's canonical test set [Wie21, pp. 102–103] ("$T^{\mathrm{can}} := \{1, \dots, m\} \subseteq W^{\mathrm{can}}$"; the group "stabilize[s] the test set T^{can} setwise").

Lemma T$^{\mathbf{set}}$. *Let $H \leq \mathrm{Sym}(\Omega)$ and $\varphi \colon H \to \mathrm{Sym}([k])$ with $\varphi(H) \geq A_k$ and $k \geq t + 2$. Let $T_0 := \{1, \dots, t\}$ (P1), $H_0 := \varphi^{-1}(\mathrm{Stab}(T_0))$, and $\psi \colon H_0 \to \mathrm{Sym}(T_0)$ the induced action. Let C be an H_0-ECM with right group R on an H-invariant set $\mathcal{Z}$. For $Z \in \mathcal{Z}$ and an ordered tuple $\vec{T}$ of*

distinct points with underlying set T, choose $\sigma \in H$ with $T^{\varphi(\sigma)} = T_0$ and put $E(Z,T) := \sigma C(Z^\sigma)$, $R(Z,T) := R(Z^\sigma)$, and

$$c(Z,\vec{T}) := \big(R(Z^\sigma),\ \textit{the } \psi(R(Z^\sigma))\textit{-orbit of } \vec{T}^{\varphi(e)}\big) \quad \textit{for any } e \in E(Z,T).$$

Then:

(a) E, R and c are well defined;

(b) $c(Z^h, \vec{T}^{\varphi(h)}) = c(Z,\vec{T})$ for $h \in H$;

(c) if $c(Z,\vec{T}^\pi) = c(Z,\vec{T})$ for all $\pi \in \mathrm{Sym}(T)$, then $\psi(R(Z^\sigma)) = \mathrm{Sym}(T_0)$;

(d) $\psi_T(\mathrm{Aut}_{H_T}(Z))$ is conjugate to a subgroup of $\psi(R(Z^\sigma))$, where $H_T := \varphi^{-1}(\mathrm{Stab}(T))$.

Proof. (a) If $\sigma' = \sigma g$ with $g \in H_0$, then $\sigma g \cdot C(Z^{\sigma g}) = \sigma g \cdot g^{-1} C(Z^\sigma)$, and R is unchanged by Lemma E1. Two elements of E differ by $r \in R$ on the right, and $\vec{T}^{\varphi(er)} = (\vec{T}^{\varphi(e)})^{\psi(r)}$. (b) Use $h^{-1}\sigma$ for $(Z^h, T^{\varphi(h)})$: the transported object is again Z^σ, E becomes $h^{-1}E$, and $(\vec{T}^{\varphi(h)})^{\varphi(h^{-1}e)} = \vec{T}^{\varphi(e)}$. (c) With σ and e fixed (they depend on the set T only), $(\vec{T}^\pi)^{\varphi(e)}$ runs through all $t!$ orderings of T_0, which then lie in one $\psi(R)$-orbit. (d) Lemma E1 for H_0 and Z^σ, conjugated by σ. □

Colours are compared through canonical generating sets of R and the lexicographically least ordering in the orbit. Only (b) and (c) are used below. In particular, the colours need *not* reproduce Neuen's comparison classes.

10.4 The pairwise steps of Neu20

An internal reading of Neu20 identified 21 choice points or pairwise steps. Four are pairwise in an essential way:

(a) the local certificates routine, run "on all pairs in parallel" with an arbitrary test object and a realignment [Neu20, src. l. 2201–2258];

(b) the representative group Γ_1'' in Simplify [Neu20, src. l. 2053–2061];

(c) the joint category on $D_1' \sqcup D_2'$ in the aggregation [Neu20, src. l. 2589–2596];

(d) find-structure and find-symmetry [Neu20, src. l. 2603–2653].

(c) and (d) are as in GNS and are handled by Lemma $\mathrm{T}^{\mathrm{set}}$, Lemma S with BC, and Lemma 5.9. (a) and (b) are new. The others are handled as in Section 6: (P1) choices (structure graphs and their transporters, the choice of v_P, the minimal block system, the giant representation), descents (F3), equivariant branch families (F4), equivariant encodings (Lemma R), and the two routine pairwise subroutines CombineWindows and BalanceOrbits (Lemmas W and B below). All group-only constructions are subject to requirement D3: they must be deterministic functions of the group and the ordered domain, computed from canonical generating sets.

10.5 Lemmas W, $\mathrm{F2}^{\mathrm{set}}$ and B

Theorem A is a canonizer for strings under a group. Below it is also applied to strings on a set Γ on which the current group A acts through a homomorphism $\theta\colon A \to \mathrm{Sym}(\Gamma)$. This is Lemma S with $S(x)$ the string and Λ' the Theorem A canonizer under $\theta(A) \in \widehat{\Gamma}_d$ (quotients stay in $\widehat{\Gamma}_d$).

Lemma W (canonical CombineWindows). *Let $A \le \mathrm{Sym}(\Omega)$ be in $\widehat{\Gamma}_d$, let $\mathfrak{P}$ be A-invariant, let $W_1, W_2 \subseteq \Omega$ be disjoint and A-invariant, let F_j be a set of $\mathfrak{P}[W_j]$-strings with $A[W_j] \le \mathrm{Aut}(F_j)$ $(j = 1, 2)$, and put $W := W_1 \cup W_2$. Let $\mathcal{Z}$ be the A-invariant set of sets Z of $\mathfrak{P}$-strings with $Z[W_j] = F_j$ for $j = 1, 2$. Let $V := F_1 \sqcup F_2$, let $\theta\colon A \to \mathrm{Sym}(V)$ be the induced action, and for $Z \in \mathcal{Z}$ let the* linking graph $G(Z)$ *be the graph on V with edges $\{(P \cap W_1, x[W_1]), (P \cap W_2, x[W_2])\}$ for $(P, x) \in Z$ with $P \cap W_1 \neq \emptyset \neq P \cap W_2$ [Neu20, src. l. 1648–1651]. Let (F_G, Λ_G) be the $\theta(A)$-canonizer on these graphs given by Corollary D under $\theta(A) \in \widehat{\Gamma}_d$. Then there is a set $F_W(Z)$ of $\mathfrak{P}[W]$-strings, depending only on V and $F_G(G(Z))$, such that for $Z \in \mathcal{Z}$ and $E(Z) := \theta^{-1}(\Lambda_G(G(Z)))$, a left coset σR with $R = \theta^{-1}(\mathrm{Aut}_{\theta(A)}(F_G(G(Z))))$,*

$$E(Z) = \{h \in A : Z^h[W] = F_W(Z)\}, \qquad F_W(Z^a) = F_W(Z) \quad \text{for } a \in A.$$

Thus (F_W, E) is an A-canonizer of $Z \mapsto Z[W]$ on $\mathcal{Z}$. Cost: one call of Theorem A on a graph with at most $2|Z[W]|$ vertices, i.e. on a string on at most $(2|Z[W]|)^2$ points (through θ).

Proof. For $h \in A$ we have $F_j^h = F_j$, so $\theta(h)$ permutes V, $Z^h \in \mathcal{Z}$, and $G(Z^h) = G(Z)^{\theta(h)}$. A set $Z \in \mathcal{Z}$ is recovered on W from its linking graph: $Z[W]$ consists of the strings $(Q_1 \cup Q_2, z_1 \cup z_2)$ over the edges $\{(Q_1, z_1), (Q_2, z_2)\}$, together with the strings of $F_1 \cup F_2$ on blocks that meet only one of W_1, W_2; and $G(Z)$ is determined by $Z[W]$ and V. So $Z[W] \leftrightarrow G(Z)$ is a bijection; let $F_W(Z)$ be the set of strings corresponding to the graph $F_G(G(Z))$. For $h \in A$ this gives $Z^h[W] = F_W(Z)$ iff $G(Z)^{\theta(h)} = F_G(G(Z))$ iff $\theta(h) \in \Lambda_G(G(Z))$ iff $h \in E(Z)$. The remaining statements follow from (C1), (C2) for (F_G, Λ_G). (This is the forward and backward direction of the proof of [Neu20, Lemma 4.15].) □

Lemma F2^{set} (one step of the chain rule for sets of strings). *Let $A \le \mathrm{Sym}(\Omega)$ be in $\widehat{\Gamma}_d$, let $\mathfrak{P}$ be A-invariant, and let $\mathcal{Z}$ be an A-invariant set of sets of $\mathfrak{P}$-strings. Let C be an A-ECM on $\mathcal{Z}$ with right group R, and suppose that there is a prefix (U, F, R), constant on A-orbits, where U is R-invariant and $Z^\pi[U] = F$ for all $\pi \in C(Z)$ (so $R[U] \le \mathrm{Aut}(F)$). Let $V \subseteq \Omega \setminus U$ be R-invariant, given by a function of the prefix, and let (F_V, Λ_V) be an R-canonizer on the R-invariant set $\{Z^\pi[V] : \pi \in C(Z)\}$. For $\pi \in C(Z)$ put $w := Z^\pi$, take $\lambda \in \Lambda_V(w[V])$, put $y := w^\lambda$ and $R_2 := \mathrm{Aut}_R(F_V(w[V]))$, and apply Lemma W to R_2, $W_1 := U$, $W_2 := V$, $F_1 := F$, $F_2 := F_V(w[V])$ and the object y, obtaining $E(y) = \sigma R_3$ and $F_+ := F_W(y)$. Then $C_+(Z) := \pi\lambda E(y)$ is an A-ECM on $\mathcal{Z}$ with right group R_3 and $C_+(Z) \subseteq C(Z)$; and $(U \cup V, F_+, R_3)$ is a prefix of the same kind: it is constant on A-orbits, and $Z^{\pi'}[U \cup V] = F_+$ for all $\pi' \in C_+(Z)$.*

Proof. Since $V \cap U = \emptyset$, we have $y[U] = F$ and $y[V] = F_V(w[V])$, and R_2 preserves both F and $F_V(w[V])$, so Lemma W applies. By Lemma E2 with $C_1 := C$ (prefix (U, F, R)) it suffices that $w \mapsto \lambda E(w^\lambda)$ is an R-ECM on $\{Z^\pi\}$. This is again Lemma E2, now with $C_1 := (w \mapsto \Lambda_V(w[V]))$, an R-ECM by Lemma 2.2(1) and $(w^r)[V] = (w[V])^r$, with prefix $(F_V(w[V]), R_2)$, and with the R_2-ECM $y \mapsto E(y)$ of Lemma W. For $\pi' = \pi\lambda e$ with $e \in E(y)$ we get $Z^{\pi'}[U \cup V] = y^e[U \cup V] = F_+$ by Lemma W. Replacing (π, λ) by another choice replaces y by an element of y^{R_2}, and Z by Z^h leaves y unchanged, so $F_+ = F_W(y)$ and R_3 are constant on A-orbits by Lemma W. □

This is the analogue of Lemma F2 for sets of strings; Lemma W replaces the concatenation of canonical forms. The chain rule itself is the iteration of this step in the subroutine Chain of §10.9.

Lemma B (canonical BalanceOrbits). *Let $A \le \mathrm{Sym}(\Omega)$ be in $\widehat{\Gamma}_d$, let $\mathfrak{P}$ be A-invariant, and let $\mathcal{Z}$ be an A-invariant set of sets of $\mathfrak{P}$-strings. There is an A-ECM C_{B} on $\mathcal{Z}$, with right group R, such that for every $Z \in \mathcal{Z}$, every $\pi \in C_{\mathrm{B}}(Z)$ and every R-orbit O, the set $Z^\pi[O]$ is balanced. It is computed by at most $n \log n$ calls of Theorem A on at most n points, with $n = |\Omega|$. (This is [Neu20, Lemma 4.3] made canonical.)*

Proof. Start with $C_0(Z) := A$, $R_0 := A$. Suppose C_i is an A-ECM with right group R_i; let $\pi \in C_i(Z)$ and $z := Z^\pi$. For $r \in R_i$ and an R_i-orbit O we have $z^r[O] = (z[O])^r$, so whether $z[O]$ is balanced depends only on the R_i-orbit of z. If all R_i-orbits are balanced, stop. Otherwise let O be the first R_i-orbit in the order (size, least point) with $z[O]$ unbalanced; it depends only on R_i and the R_i-orbit of z. Let $\mu\colon \mathfrak{P}[O] \to \mathbb{N}$, $Q \mapsto m_{z[O]}(Q)$, and let $\rho\colon R_i \to R_i[\mathfrak{P}[O]] =: B \in \widehat{\Gamma}_d$ be the natural homomorphism. Then B is transitive on $\mathfrak{P}[O]$ and μ is not constant. Let (F_μ, Λ_μ) be the Theorem A canonizer of μ under B and $E(z) := \rho^{-1}(\Lambda_\mu(\mu))$, a coset σR_{i+1} with $R_{i+1} := \rho^{-1}(\mathrm{Aut}_B(F_\mu))$. Since $\mu_{z^r} = \mu_z^{\rho(r)}$, we have $E(z^r) = r^{-1}E(z)$, so E is an R_i-ECM and Lemma E2 (with prefix R_i) makes $C_{i+1}(Z) := \pi E(z)$ an A-ECM with right group R_{i+1}. Moreover R_{i+1} is a proper subgroup of R_i, since F_μ is not constant and B is transitive. A subgroup chain in $\mathrm{Sym}(\Omega)$ has length at most $\log_2 n! \le n \log n$, which bounds the number of rounds. □

10.6 Canonical simplification and renormalization

For a set Y of $\mathfrak{Q}$-strings and $U \subseteq \bigcup \mathfrak{Q}$ put $\widehat{s}(Y,U) := \sum_{Q\in\mathfrak{Q}} |Q \setminus U| \cdot m_Y(Q)^{f(d)+1}$. Thus $\widehat{s}(Y,\emptyset)$ is the d-virtual size of Y. If $Y[U]$ is simple, then $\widehat{s}(Y,U)$ is the d-virtual size of $Y[\bigcup \mathfrak{Q} \setminus U]$, since the strings on a block then differ only outside U; in general it can be larger. For U' disjoint from U we have $\widehat{s}(Y,U) - \widehat{s}(Y,U\cup U') = \sum_Q |Q \cap U'|\, m_Y(Q)^{f(d)+1}$, which is at least the d-virtual size of $Y[U']$, and at least $|U'|$ if Y is completely occupied.

Lemma SIMP. *Let $A \le \mathrm{Sym}(\Omega)$ be in $\widehat{\Gamma}_d$, with an almost d-ary A-invariant sequence $\vec{\mathfrak{B}}$ having $\mathfrak{P}$ as a level; let $W \subseteq \Omega$ be A-invariant, let F be a set of $\mathfrak{P}[W]$-strings with $A[W] \le \mathrm{Aut}(F)$, and let $\mathcal{Z}_F$ be the A-invariant set of completely occupied sets Z of $\mathfrak{P}$-strings with $Z[W] = F$. From these data one computes an A-ECM C on $\mathcal{Z}_F$ with right group $A'' \le A$ (by one call of Theorem A) and, from $(A, \mathfrak{P}, \vec{\mathfrak{B}}, W, F)$ and the canonical string of step 3 below, a domain Ω^*, a partition $\mathfrak{P}^*$ with a sequence $\vec{\mathfrak{B}}^*$, a window W^*, a monomorphism $\iota\colon A'' \to \mathrm{Sym}(\Omega^*)$ with image Λ^*, and, for each $\pi \in C(Z)$, a completely occupied set $(Z^\pi)_*$ of $\mathfrak{P}^*$-strings, such that*

(S1) $\vec{\mathfrak{B}}^*$ is an almost d-ary sequence of Λ^*-invariant partitions having $\mathfrak{P}^*$ as a level;

(S2) W^* is Λ^*-invariant, $\Lambda^*[W^*] \le \mathrm{Aut}(F^*)$, where $F^* := (Z^\pi)_*[W^*]$ is simple and depends only on the computed data, not on Z or π;

(S3) $Z^\pi \mapsto (Z^\pi)_*$ is injective and $(Z^{\pi r})_* = ((Z^\pi)_*)^{\iota(r)}$ for $r \in A''$;

(S4) the d-virtual size of $(Z^\pi)_*$ is at most that of Z, and $\widehat{s}((Z^\pi)_*, W^*) \le \widehat{s}(Z, W)$.

If F is simple, put $C := A$, $\Omega^* := \Omega$, $(Z^\pi)_* := Z^\pi$ (nothing is done).

Proof. We follow the proof of [Neu20, Thm. 4.17] for a single object.

Step 1 (copy blocks). $\Omega' := \bigcup_P P \times \{z : (P \cap W, z) \in F\}$, where a block P with $P \cap W = \emptyset$ gets the single copy $P \times \{\emptyset\}$ [Neu20, src. l. 1974–1985]; since Z is completely occupied, Ω' depends only on $\mathfrak{P}$ and F. Since $A[W] \le \mathrm{Aut}(F)$, the rule $(\alpha, z)^\gamma := (\alpha^\gamma, z^\gamma)$ defines an action $\psi'\colon A \to \mathrm{Sym}(\Omega')$, faithful because every point of Ω occurs as a first coordinate. The partition $\mathfrak{P}'$, the copies $Z \mapsto Z'$, and $W' := W \times (\cdot)$ are as in [Neu20, src. l. 1974–2003]: $(P, x) \mapsto (P \times \{x[W \cap P]\}, (\alpha, x[W \cap P]) \mapsto x(\alpha))$, so Z' determines Z and $(Z^\gamma)' = (Z')^{\psi'(\gamma)}$ (claim `preserve-isomorphisms`, [Neu20, src. l. 1987–2002]). Moreover Z' is completely occupied, $Z'[W']$ depends only on F, $\psi'(A)[W'] \le \mathrm{Aut}(Z'[W'])$ and $Z'[W']$ is simple [Neu20, src. l. 2006–2029].

Step 2 (lift the sequence) to $\vec{\mathfrak{B}}'$ [Neu20, src. l. 2031–2040] (blocks with $P \cap W = \emptyset$ handled as in Step 1, a detail omitted in the text there), a function of $(\vec{\mathfrak{B}}, F)$; only the level $\mathfrak{P}'$ can violate almost d-arity [Neu20, src. l. 2041–2046].

Step 3 (balance). Apply Lemma S to the string $x(Z') : \mathfrak{P}' \to \mathbb{N}^2$, $P' \mapsto (m_{Z'}(P'), m_{Z'[\Omega'\setminus W']}(P'\setminus W'))$ [Neu20, src. l. 2048], canonized by Theorem A under $\psi'(A)[\mathfrak{P}']$. This yields a coset $E(Z') = \sigma A'''$, with $A''' \le \psi'(A)$ the preimage of the stabilizer of the canonical string, and we put $C(Z) := (\psi')^{-1}(E(Z'))$, $A'' := (\psi')^{-1}(A''')$. For $\pi \in C(Z)$ put $Z'' := (Z^\pi)' = Z'^{\psi'(\pi)}$. Then $x(Z'')$ is the canonical string, so the claim `respect-block-sizes` [Neu20, src. l. 2069–2074] holds for A''' and Z''.

Step 4 (renormalize). Apply [Neu20, Thm. 4.9] to $(A''', \mathfrak{P}', \vec{\mathfrak{B}}')$. The construction in its proof [Neu20, src. l. 1455–1554] uses these data only; the strings enter only through the definition of $\mathfrak{X}^*$ [Neu20, src. l. 1521–1525]. So $\Omega^*, \mathfrak{P}^*, \vec{\mathfrak{B}}^*, f, f'$ and $\iota_0 \colon A''' \to \mathrm{Sym}(\Omega^*)$ are functions of the prefix, and $\iota := \iota_0 \circ \psi'|_{A''}$. Define $(Z^\pi)_* := \{(P^*, \alpha^* \mapsto x(f(\alpha^*))) : (f'(P^*), x) \in Z''\}$ (property (VII)) and $W^* := f^{-1}(W')$.

Verification. (S1) is (I), (VI) of [Neu20, Thm. 4.9], whose hypothesis is met by Step 2. (S2): invariance of W^* follows from (II); $\Lambda^*[W^*] \le \mathrm{Aut}(F^*)$ from (IX), applied to the relation $\psi'(A)[W'] \le \mathrm{Aut}(Z'[W'])$ of Step 1. Since $Z''[W']$ is simple and equals the object $Z'[W']$ of Step 1 (a function of F), and the strings of $(Z^\pi)_*$ on a block P^* are $x \circ f$ with $(f'(P^*), x) \in Z''$, the set F^* is simple and determined by the computed data. Complete occupancy of $(Z^\pi)_*$ is immediate from (VII) and the complete occupancy of Z''. (S3): f' is onto $\mathfrak{P}'$ (otherwise (VIII) fails for two sets differing on a block outside the image); each P^* is mapped bijectively onto $f'(P^*)$ by f ((III), (V)), so $x \circ f$ determines x; equivariance is (VIII) applied to $\mathfrak{X} := Z''$, $\mathfrak{Y} := Z''^r$, together with Step 1. (S4): the first bound is the estimate in the proof of property (F) of [Neu20, Thm. 4.17] [Neu20, src. l. 2115–2141], which uses only (I)–(V) and the first half of `respect-block-sizes`; it holds with $\mathfrak{X}_1 := Z^\pi$, $\mathfrak{X}_1' = \mathfrak{X}_1'' := Z''$ and $\Gamma'' := A'''$. For the second bound, the same computation with $|P'|$ replaced by $|P'\setminus W'|$ applies: by (III), (V), (VII), $\widehat{s}((Z^\pi)_*, W^*) = \sum_{P'} |P'\setminus W'|\, m_{Z''}(P')^{f(d)+1}\, |(f')^{-1}(P')|$. For a block P of $\mathfrak{P}$ put $B_P := \bigcup_z P \times \{z\}$. Every copy P' of P has $|P'\setminus W'| = |P\setminus W|$, and on an orbit A of $(A''')_{B_P}$ on the copies of P the value $m_{Z''}(P') =: m(A)$ is constant. So by (IV) the contribution of P is at most $|P\setminus W| \sum_A (|A|\, m(A))^{f(d)+1} \le |P\setminus W|\, (\sum_A |A|\, m(A))^{f(d)+1} = |P\setminus W|\, m_{Z^\pi}(P)^{f(d)+1}$, and the sum over P is $\widehat{s}(Z^\pi, W) = \widehat{s}(Z, W)$, since $\pi \in A$ and W is A-invariant. (Neuen's property (G) asks for the same bound for the virtual size of the restriction $(Z^\pi)_*[\Omega^* \setminus W^*]$; we could not verify it for this construction, see Remark 10.3(3). It is not used.)

In the language of ECMs: C is an A-ECM because $x(Z'^{\psi'(a)}) = x(Z')^{\psi'(a)}$ (action on $\mathfrak{P}'$) and the Theorem A canonizer of $x(Z')$ is equivariant; and $Z \mapsto Z'$ and $Z'' \mapsto (Z^\pi)_*$ are the changes of action of Lemma E3 (using the injectivity of Step 1 and (S3)) that transport canonizers back along ψ' and ι_0. □

Neuen runs steps 3 and 4 once per equivalence class of pairs, with a representative group Γ_1''. Here step 3 is a canonizer and step 4 is computed from its canonical output.

10.7 Canonical local certificates

Lemma LC. *Assume the hypotheses of [Neu20, Thm. 4.24]:* $\Gamma \in \widehat{\Gamma}_d$ *on* Ω *with an almost* d*-ary sequence containing* $\mathfrak{P}$*, a giant representation* $\varphi \colon \Gamma \to S_k$ *with* $\max\{8, 2+\log d\} < t \le k-2$*, and* T_0*,* $H_0 = \Gamma_{T_0}$*,* ψ *as in Lemma* T^{set}*. Let* $\mathcal{Z}$ *be the set of completely occupied sets of* $\mathfrak{P}$*-strings of* d*-virtual size at most* s*. Algorithm CanLC below computes an* H_0*-ECM* C *on* $\mathcal{Z}$ *with right group* R*, and an* H_0*-invariant flag, such that:*

(LC1) $C(Z^g) = g^{-1}C(Z)$ *for* $g \in H_0$*;*

(LC2) if the flag is nonfull*, then* $\psi(R(Z)) \not\ge \mathrm{Alt}(T_0)$*;*

(LC3) if the flag is full*, CanLC also outputs* $\Delta(Z) \le \mathrm{Aut}_{H_0}(Z)$ *with* $\psi(\Delta(Z)) \ge \mathrm{Alt}(T_0)$ *and* $\Delta(Z^g) = g^{-1}\Delta(Z)g$*.*

Cost: polynomial work in s*, polynomially many calls of Theorem A on domains of size polynomial in* s*, and recursive calls of the main canonizer on instances of virtual size at most* $s/2$*, of total virtual size at most* $t!\,s$*.*

State at round i (for an object Z, with $\pi \in C_i(Z)$). C_i is an H_0-ECM with right group R_i. Its prefix $\bar{p}_i(Z)$ consists of a domain Ω_i, a partition $\mathfrak{P}_i$ and a sequence $\mathfrak{B}^i$; a window W_i; a group $\Lambda_i \le \mathrm{Sym}(\Omega_i)$ with an injective $\tau_i\colon \Lambda_i \to H_0$ such that $\tau_i(\Lambda_i) = R_i$; the object map $Z' \mapsto Z'_*$; and the window part $F_i := (Z^\pi)_*[W_i]$, the same for all $\pi \in C_i(Z)$. Invariants:

(I1) $\mathfrak{B}^i$ is an almost d-ary sequence of Λ_i-invariant partitions containing $\mathfrak{P}_i$;

(I2) W_i is Λ_i-invariant, $\Lambda_i[W_i] \le \mathrm{Aut}(F_i)$, and F_i is simple;

(I3) for $\lambda \in \Lambda_i$: $\lambda \in \mathrm{Aut}((Z^\pi)_*)$ iff $\tau_i(\lambda) \in \mathrm{Aut}(Z^\pi)$;

(I4) the virtual size of $(Z^\pi)_*$ is at most s, and the virtual size s'_i of $(Z^\pi)_*[\Omega_i \setminus W_i]$ strictly decreases with i;

(I5) $(Z^\pi)_*$ is completely occupied.

Start: $C_0 = H_0$, $\Omega_0 = \Omega$, $W_0 = \emptyset$, $\Lambda_0 = H_0$, $\tau_0 = \mathrm{id}$.

Round i (each step is Lemma E2 applied with the named ECM):

(a) If $\psi(R_i) \not\ge \mathrm{Alt}(T_0)$, stop with *nonfull*.

(b) Put $W' := \mathrm{Aff}(\Lambda_i, \psi\tau_i) \setminus W_i$, a function of the prefix. If $W' = \emptyset$, stop with *full* and $\Delta(Z) := \pi\,\tau_i\big((\Lambda_i)_{(\Omega_i \setminus W_i)}\big)\,\pi^{-1}$.

(c) *Balance*: Lemma B on $(Z^\pi)_*$ under Λ_i, giving the group Λ'. If $\psi\tau_i(\Lambda')$ is not $\ge \mathrm{Alt}(T_0)$, stop with *nonfull*.

(d) *Window*: canonize $(Z^\pi)_*[W']$ under Λ', joining it to the window part F_i on W_i, by iterating Lemma F2^{set} from the prefix (W_i, F_i, Λ') over the Λ'-orbits O of W' in (P1) order. For an orbit O, processed under the current right group $R \le \Lambda'$, the required R-canonizer of the object on O is obtained by descent (F3) from R to $R \cap N'$, where $N' := \ker(\psi\tau_i|_{\Lambda'})$ (index at most $t!$ in R), and by the subroutine Chain of §10.9 under $(R \cap N')^O$ (intransitive on O) on the object in each branch; its windows are recursive calls of the main canonizer (CanHyp, at smaller virtual size).

(e) *Simplify and renormalize*: Lemma SIMP at the new prefix (window $W_i \cup W'$). Lemma E3 yields Λ_{i+1}, τ_{i+1}, Ω_{i+1}, W_{i+1}.

Proof of Lemma LC. (LC1). By induction on rounds. Each step composes (Lemma E2) an ECM that comes from a canonizer (Theorem A, recursive calls, Lemmas B and W), a descent (F3), or a change of action computed from the prefix (Lemma E3, Lemma SIMP). The stopping tests depend only on R_i and the prefix, which are H_0-invariant.

Invariants. (I1) and (I2) are (S1), (S2) of Lemma SIMP. (I3) follows from (S3), the claim `preserve-isomorphisms` and property (VIII) of [Neu20, Thm. 4.9], and the fact that balancing and windowing only pass to subgroups containing the relevant automorphisms (Lemma E1). (I5) holds at the start by hypothesis and is preserved by Lemma SIMP, whose objects are completely

occupied; it is needed for strict monotonicity of the virtual size. (I4): the bound on the virtual size of $(Z^\pi)_*$ is (S4). For the second statement let $Y := (Z^\pi)_*$ at round i. Since F_i is simple, $s'_i = \widehat{s}(Y, W_i)$. Extending the window gives $\widehat{s}(Y, W_i \cup W') \le s'_i - |W'|$ by (I5) (the object changes only by elements of the right group during the round, which does not change $\widehat{s}$), and Simplify does not raise $\widehat{s}$ (S4), so $s'_{i+1} = \widehat{s}(Y_{i+1}, W_{i+1}) \le \widehat{s}(Y, W_i \cup W') < s'_i$. This replaces the appeal to property (G) in Neuen's argument [Neu20, src. l. 2400–2412] (see the question in Remark 10.3(3)). Hence there are at most s rounds.

(LC2) holds by the tests at (a) and (c).

(LC3). At (b), $\mathrm{Aff}(\Lambda_i, \psi\tau_i) \subseteq W_i$, Λ_i is d-normalized by (I1), and $\psi\tau_i(\Lambda_i)$ is a giant by test (a). So [Neu20, Thm. 4.6] (= [GNS18, Thm. 5.8]) gives $\psi\tau_i((\Lambda_i)_{(\Omega_i \setminus W_i)}) \ge \mathrm{Alt}(T_0)$. [Neu20, Lemma 4.16] with (I2) gives $(\Lambda_i)_{(\Omega_i \setminus W_i)} \le \mathrm{Aut}((Z^\pi)_*)$, and (I3) transfers this to $\tau_i(\dots) \le \mathrm{Aut}_{H_0}(Z^\pi)$. Conjugating by π gives $\Delta(Z) \le \mathrm{Aut}_{H_0}(Z)$ with a giant image. $\Delta(Z)$ does not depend on π, because $(\Lambda_i)_{(\Omega_i \setminus W_i)} \trianglelefteq \Lambda_i$ (W_i is Λ_i-invariant), so its τ_i-image is normalized by R_i. Equivariance follows from (LC1).

Cost. Step (d) is Neuen's recursion for one pair [Neu20, src. l. 2436–2452], with one change: [Neu20, Lemma 4.7] (= [Bab16b, Thm. 6(b)]) is applied to Λ' itself, which is a giant by test (c); the orbits in W' are affected for Λ' because Aff is antimonotone. So each N'-orbit in a Λ'-orbit O has size at most $|O|/t$, and the $(R \cap N')$-orbits lie inside N'-orbits. Since $(Z^\pi)_*[O]$ is balanced (Lemma B), each piece has virtual size at most $s/2$ [Neu20, Obs. 4.13]. Since the virtual size of $Y[W']$ is at most $\widehat{s}(Y, W_i) - \widehat{s}(Y, W_i \cup W') \le s'_i - s'_{i+1}$, the window virtual sizes sum to at most s over the rounds (this is the accounting of [Neu20, src. l. 2425–2434] with $\widehat{s}$ in place of t_i); with the $t!$ cosets the total is at most $t!\, s$. The calls of Theorem A (Lemmas B, W and SIMP) are polynomially many, on domains of polynomial size. □

Comparison with Neu20. Neuen computes, for every pair of test objects, the coset of isomorphisms, and runs all $4k^{2t}$ pairs in one synchronized loop so that renormalizations are comparable [Neu20, src. l. 270–275, 1930–1933]. CanLC runs $\binom{k}{t}$ independent single-object loops, one per t-set: transport the t-set to T_0, then run the loop on Z^σ. Consistency between test sets is automatic, because every auxiliary construction is a function of the canonical prefix, and Lemma $\mathrm{T}^{\mathrm{set}}$(a) makes the result independent of the transporter. The recursive cost drops from $4k^{2t}t!\, s$ to $k^t t!\, s$.

Why the setwise stabilizer. In Section 5 we canonize under $J_0 = \ker\psi$ and take the windows from the GNS groups $G_i(T, x) = \mathrm{Aut}_{H_T}(x^{[W_i]})$, which are intrinsic to (T, x). In Neu20 the corresponding groups live on renormalized domains, and they depend on structure-graph choices made at earlier rounds. They are intrinsic only once those choices are made canonically, which requires canonizing at the level H_0 at which the windows are defined.

10.8 Canonical aggregation

Lemma AGG. *In the setting of [Neu20, Lemma 4.29], for a single $\mathfrak{X}$ (and Γ', φ, $t < k/10$ as there), one computes one of:*

(i) a family $\{\mathfrak{A}_j(\mathfrak{X})\}_{j \le k^6}$ of t-ary structures on subsets $V_j \subseteq [k]$ with $|V_j| \ge 3k/4 - 4$ (the 4 being the individualized points, as in Proposition 5.2), each, viewed as a structure on $[k]$ with V_j marked, of relative symmetry defect at least $1/4$, equivariant as a family: $\{\mathfrak{A}_j(\mathfrak{X}^h)\} = \{\mathfrak{A}_j(\mathfrak{X})\}^{\varphi(h)}$;

(ii) a pair $(M(\mathfrak{X}), \Delta(\mathfrak{X}))$ with $|M| \ge 3k/4$, $\Delta \le \mathrm{Aut}_{\Gamma'_M}(\mathfrak{X})$ and $\varphi(\Delta)[M] \ge \mathrm{Alt}(M)$, both equivariant.

In the case $|\mathrm{supp}_{\mathfrak{X}}| < k/4$ the structure has relative symmetry defect at least $3/4$.

Proof. Run CanLC on $\mathfrak{X}^\sigma$ for each t-set T. This gives the canonical fullness certificates $\Delta(T,\mathfrak{X}) := \sigma\Delta(\mathfrak{X}^\sigma)\sigma^{-1}$ for routine-full T (equivariant, since $\Delta(\mathfrak{X}^{\sigma g}) = g^{-1}\Delta(\mathfrak{X}^\sigma)g$), and the colours $c(\mathfrak{X},\vec{T})$ of Lemma $\mathrm{T}^{\mathrm{set}}$ for all orderings of routine-nonfull T. By (LC2), (LC3) and Lemma E1, "routine-full" coincides with Neuen's "full". The fullness group $\Lambda_{\mathfrak{X}} := \langle \Delta(T,\mathfrak{X}) : T \text{ routine-full}\rangle$ and its support $\mathrm{supp}_{\mathfrak{X}}$ are equivariant. The cases $|\mathrm{supp}_{\mathfrak{X}}| \in [k/4, 3k/4]$ and $|\mathrm{supp}_{\mathfrak{X}}| > 3k/4$ [Neu20, src. l. 2554–2582] produce GNS's structures from $\Lambda_{\mathfrak{X}}$ alone, and each individualization is a member of an equivariant branch family of size at most k^6 (Proposition 5.2). In the case $|\mathrm{supp}_{\mathfrak{X}}| < k/4$, put $D := [k] \setminus \mathrm{supp}_{\mathfrak{X}}$ and $\mathfrak{A}(\mathfrak{X}) := ([k]; \text{unary } D;\ \vec{T} \mapsto \text{rank of } c(\mathfrak{X},\vec{T}) \text{ for } \vec{T} \in D^{\langle t\rangle})$. Every $T \subseteq D$ is routine-nonfull. $\mathfrak{A}$ is strongly canonical by Lemma $\mathrm{T}^{\mathrm{set}}$(b), and the ranking is canonical because the set of colour values is an invariant. If $\mathrm{Alt}(M) \le \mathrm{Aut}(\mathfrak{A})$ with $|M| \ge t+2$ and $M \subseteq D$, then for a t-set $T \subseteq M$ all orderings are induced by $\mathrm{Alt}(M)$ and have equal colour, so Lemma $\mathrm{T}^{\mathrm{set}}$(c) gives $\psi(R) = \mathrm{Sym}(T_0)$, contradicting (LC2). Hence every such M lies in $\mathrm{supp}_{\mathfrak{X}}$ or has at most $t+1$ points, and the relative defect is at least $1 - \max(1/4, 1/10 + 1/k) = 3/4$ for $k \ge 7$. □

Neuen uses the comparison classes only to build his structures [Neu20, src. l. 2583–2597], and find-structure uses only the invariance of the family and the defect bound [Neu20, src. l. 2610–2630]; exactness of the colour classes is not needed.

10.9 The algorithm CanHyp

This is the counterpart of Section 6 for sets of strings. *Instances* are tuples $(\Omega,\Gamma,\mathfrak{P},\vec{\mathfrak{B}},\mathfrak{X})$, where Ω is ordered, $\Gamma \le \mathrm{Sym}(\Omega)$ is in $\widehat{\Gamma}_d$ and given by its canonical generating set, $\vec{\mathfrak{B}}$ is a Γ-invariant almost d-ary sequence with $\mathfrak{P}$ as a level, and $\mathfrak{X}$ is a completely occupied set of $\mathfrak{P}$-strings. As in Section 6, d and $t := \max\{9, 3 + \lceil \log d \rceil\}$ are fixed globally, and $(\mathfrak{P},\vec{\mathfrak{B}})$ are auxiliary data: they are functions of the group data above the node, or of canonical prefixes (Lemma E2), and never depend on $\mathfrak{X}$ directly. The *measure* of an instance is

$$\mu(\Omega,\Gamma,\mathfrak{P},\vec{\mathfrak{B}},\mathfrak{X}) := (s, |\Gamma|) \in \mathbb{N}^2, \qquad s := \text{the } d\text{-virtual size of } \mathfrak{X},$$

ordered lexicographically; this order is a well-order. All recursive calls of CanHyp below are made on instances of strictly smaller measure (Proposition 10.2).

Chain. For $A \le \mathrm{Sym}(\Omega)$ intransitive and Z a completely occupied set of $\mathfrak{P}$-strings, $\mathrm{Chain}(A,Z)$ (with $\Omega,\mathfrak{P},\vec{\mathfrak{B}}$ understood from the context) is the iteration of Lemma $\mathrm{F2}^{\mathrm{set}}$ over orbits of the right groups: put $C_0(Z) := A$, $R_0 := A$, $U_0 := \emptyset$, $F_0 := \emptyset$. At stage i, if $U_i = \Omega$, stop and return $(F_i, C_i(Z))$. Otherwise let V be the first R_i-orbit in $\Omega \setminus U_i$ in the order (size, least point), let $\pi \in C_i(Z)$, and call CanHyp on the transitive instance $(V, R_i^V, \mathfrak{P}[V], \vec{\mathfrak{B}}[V], Z^\pi[V])$ (a valid instance, by [GNS18, Obs. 4.2], and the restriction of a completely occupied set is completely occupied). Its output, lifted through the restriction $R_i \to R_i^V$ to an R_i-canonizer of $Z^\pi[V]$, is the input of Lemma $\mathrm{F2}^{\mathrm{set}}$; this gives $C_{i+1}, R_{i+1}, F_{i+1}$ and $U_{i+1} := U_i \cup V$.

Luks. For an A-invariant partition $\mathfrak{D}$ with at least two classes, $\mathrm{Luks}(A,\mathfrak{D},Z)$ is: apply Lemma B to $(A,\mathfrak{P},Z)$, obtaining C_{B} with right group B; let $\pi \in C_{\mathrm{B}}(Z)$ and $y := Z^\pi$. If B is intransitive, let $(F,\Lambda) := \mathrm{Chain}(B,y)$. Otherwise descend (Lemma F3) from B to $B_{(\mathfrak{D})}$ over at most $|B^{\mathfrak{D}}|$ cosets and run $\mathrm{Chain}(B_{(\mathfrak{D})}, y^\rho)$ on each branch ($B_{(\mathfrak{D})}$ is intransitive, as its orbits lie in classes of $\mathfrak{D}$), obtaining a B-canonizer (F,Λ) of y. Return $(F,\pi\Lambda)$, an A-canonizer by Lemma E4.

Algorithm CanHyp. On an instance $(\Omega, \Gamma, \mathfrak{P}, \vec{\mathfrak{B}}, \mathfrak{X})$, with $\vec{\mathfrak{B}} = (\{\Omega\} = \mathfrak{B}_0 \succ \mathfrak{B}_1 \succ \dots)$:

Step 0. If $|\Omega| = 1$, return $(\mathfrak{X}, \Gamma)$.

Step 1. Intransitive Γ. Return Chain$(\Gamma, \mathfrak{X})$.

Step 2. Transitive Γ, $\Gamma^{\mathfrak{B}_1}$ semi-regular. Return Luks$(\Gamma, \mathfrak{B}_1, \mathfrak{X})$.

Step 3. Transitive Γ, $|\mathfrak{B}_1| \le d$. Choose (P1) a block system $\mathfrak{B} \succeq \mathfrak{B}_1$ of Γ with $|\mathfrak{B}| \ge 2$ on which $P := \Gamma^{\mathfrak{B}}$ is primitive, and put $b := |\mathfrak{B}| \le d$.

(a) If $|P| \le b^{1+\log b}$, or b is below the constant of [Neu20, Lemma 4.35]: return Luks$(\Gamma, \mathfrak{B}, \mathfrak{X})$.

(b) Otherwise apply [Neu20, Lemma 4.35] (= [GNS18, Lemma 7.1]) to P (P1), obtaining N, $\mathcal{C}$, ψ and k. Put $G' := \{g \in \Gamma : g^{\mathfrak{B}} \in N\}$, $\varphi := \psi \circ (g \mapsto g^{\mathfrak{B}})$, and let $\mathcal{C}'$ be the lift of $\mathcal{C}$, so that $\ker\varphi = G'_{(\mathcal{C}')}$, $|\mathcal{C}'| \ge 2$ and $\mathcal{C}' \succeq \mathfrak{B} \succeq \mathfrak{B}_1$. Descend (Lemma F3) from Γ to G' over at most b cosets, and in each branch ρ run Agg$(G', \mathfrak{X}^\rho)$.

Step 4. Agg(G', y) (with $\varphi, \mathcal{C}', \vec{\mathfrak{B}}$ from Step 3) returns a G'-canonizer of y.

- If $k \le 10t$: return Luks$(G', \mathcal{C}', y)$ (its descent is to $B_{(\mathcal{C}')} \le \ker\varphi$, of index at most $k!$).
- If $k > 10t$: run CanLC for every t-set of $[k]$ (with recursive calls of CanHyp, Lemma LC) and apply Lemma AGG to y. In case (i) apply Proposition 5.7 with the family of Lemma AGG, each structure encoded on $[k]$ with its domain marked by a unary relation; for each member $\mathfrak{A}$ of the family, call CanHyp on $(\Omega, R_{\mathfrak{A}}, \mathfrak{P}, \vec{\mathfrak{B}} \oplus \mathcal{C}', y^{\sigma_{\mathfrak{A}}})$, and combine by Lemma S and Lemma F4. In case (ii) apply Lemma 5.9 with $\Delta(y) := M(y)$ and $K(y) := \Delta(y)$ and call CanHyp on its two children $(\Omega, L, \mathfrak{P}, \vec{\mathfrak{B}} \oplus \mathcal{C}', \cdot)$.

Step 5. Top partition of same-domain children. Put $\vec{\mathfrak{B}} \oplus \mathcal{C}' := (\{\Omega\} \succ \mathcal{C}' \succ \mathfrak{B}_1 \succ \mathfrak{B}_2 \succ \cdots)$, with duplicates removed. As in Step 6 of §6.2, this is an almost d-ary sequence for every subgroup of G', since $|\mathcal{C}'| \le b \le d$ and each $\mathcal{C}'$-class contains at most $|\mathfrak{B}_1| \le d$ blocks of $\mathfrak{B}_1$; and $\mathfrak{P}$ is still a level.

Proposition 10.2. *CanHyp terminates on every instance $(\Omega, \Gamma, \mathfrak{P}, \vec{\mathfrak{B}}, \mathfrak{X})$, and it returns a Γ-canonizer for $\mathfrak{X}$ (on the completely occupied sets of $\mathfrak{P}$-strings): its output (F, Λ) satisfies $\Lambda = \{g \in \Gamma : \mathfrak{X}^g = F\} \ne \emptyset$, and F depends only on $(\Omega, \Gamma, \mathfrak{P}, \vec{\mathfrak{B}})$ (with Γ taken as a set) and on the Γ-orbit of $\mathfrak{X}$.*

Proof. Well-foundedness. We show that every recursive call of CanHyp on an instance I, made directly or inside Chain, Luks, Agg or CanLC, has strictly smaller measure than I.

1. *Windows of Chain.* Chain is entered only for an intransitive group, so every window V is a proper subset of the current domain (at stage 0 because the group is intransitive, later because $U_i \ne \emptyset$ and $V \cap U_i = \emptyset$). Since $\mathfrak{X}$ is completely occupied and transported copies have the same virtual size, [Neu20, Obs. 4.12] gives that s strictly decreases. (Chain is entered in Step 1, inside Luks, and in step (d) of CanLC; in the last case the objects are completely occupied by (I5).)

2. *Calls inside CanLC.* They are the windows of Chain in step (d), on subsets of $W' \subseteq \Omega_i$ of $(Z^\pi)_*$, which has virtual size at most s by (I4). Each is a proper subset of Ω_i, because every N'-orbit in a Λ'-orbit O has size at most $|O|/t$ (Cost paragraph of Lemma LC). So by (I5) and [Neu20, Obs. 4.12] their virtual size is strictly smaller than s.

3. *Same-domain children* in Step 4 are on proper subgroups: $R_{\mathfrak{A}}$ and L have index at least $(4/3)^k > 1$ in $G' \leq \Gamma$ (Proposition 5.7, Lemma 5.9, [Neu20, Cor. 4.28]), and the virtual size is unchanged, since the object is only transported.

Moreover Chain has at most $|\Omega|$ stages, Lemma B at most $n \log n$ rounds, and the loop of CanLC at most s rounds (by (I4) of Lemma LC). Calls of Theorem A, Corollary D and BC are calls of other algorithms, which terminate. So CanHyp terminates, and we may prove the rest by induction on μ, assuming that the recursive calls of CanHyp return canonizers (Lemma LC and Lemma AGG were stated and proved under this assumption).

Each step returns a canonizer.

- Step 0 is trivial.
- *Chain.* By induction, Lemma F2^{set} applies at every stage, so each C_i is an A-ECM with prefix (U_i, F_i, R_i). At the end $U_r = \Omega$, so C_r is an A-ECM with right group R_r and $Z^\pi = F_r$ for all $\pi \in C_r(Z)$; thus $R_r \leq \mathrm{Aut}(F_r)$. Lemma E4, with the trivial R_r-canonizer on $\{F_r\}$, shows that $(F_r, C_r(Z))$ is an A-canonizer.
- *Luks* and Steps 2, 3(a): Lemma B, Lemma E4, Lemma F3 and the previous item.
- *Step 3(b).* The objects $\mathfrak{B}, N, \psi, \mathcal{C}', G', \varphi$ are (P1) functions of Γ, and Lemma F3 is applied to the G'-canonizers returned by Agg.
- *Step 4.* If $k \leq 10t$ this is Luks. If $k > 10t$, Lemma AGG gives the equivariant data needed: an equivariant family of structures of relative symmetry defect at least $1/4$, or an equivariant pair $(M(y), \Delta(y))$ with $\Delta(y) \leq \mathrm{Aut}_{G'_{M(y)}}(y)$ and $\varphi(\Delta(y))[M(y)] \geq \mathrm{Alt}(M(y))$. By induction the children return canonizers, and Proposition 5.7 (through Lemma S and Lemma F4), respectively Lemma 5.9, give a G'-canonizer. Step 5 keeps the instance valid.

Canonicity of auxiliary data. Every auxiliary object is a (P1) function of the current group, or of a prefix of an ECM (Lemma E2), or an equivariant function of the current object (Lemmas W, B, SIMP, LC, AGG). Hence, for fixed $(\Omega, \Gamma, \mathfrak{P}, \vec{\mathfrak{B}})$, the output depends only on the Γ-orbit of $\mathfrak{X}$. □

10.10 Theorem H

Proof of Theorem H (correctness). For a coset $\sigma\Gamma$, the coset rule gives $\Lambda_{\sigma\Gamma}(\mathfrak{X}) = \sigma\Lambda_\Gamma(\mathfrak{X}^\sigma)$, and $\mathfrak{X}^\sigma$ is a set of $\mathfrak{P}^\sigma$-strings, which is why $\mathfrak{P}^\sigma$ must be Γ-invariant; so let $\sigma = 1$. Let $\mathfrak{X}^+$ be $\mathfrak{X}$ together with the constant string $\perp$ (a new letter) on every block of $\mathfrak{P}$ that carries no string of $\mathfrak{X}$; the map $\mathfrak{X} \mapsto \mathfrak{X}^+$ is injective and equivariant and $\mathfrak{X}^+$ is completely occupied. Compute (P1), from $(\Gamma, \mathfrak{P})$ alone, the normalization of [Neu20, Thm. 4.8]: a set Ω^*, a monomorphism $\iota\colon \Gamma \to \mathrm{Sym}(\Omega^*)$, an almost d-ary sequence $\vec{\mathfrak{B}}^*$ of $\iota(\Gamma)$-invariant partitions, and a level $\mathfrak{P}^*$ of it. The map $\mathfrak{X}^+ \mapsto (\mathfrak{X}^+)^*$ of that theorem is injective, since f maps each P^* onto a block of $\mathfrak{P}$, every block is hit, and $x^* = x \circ f$; it is equivariant by the proof there, and $(\mathfrak{X}^+)^*$ is completely occupied. Run CanHyp on $(\Omega^*, \iota(\Gamma), \mathfrak{P}^*, \vec{\mathfrak{B}}^*, (\mathfrak{X}^+)^*)$; by Proposition 10.2 it returns an $\iota(\Gamma)$-canonizer. Lemma R (with ι and the map $\mathfrak{X} \mapsto (\mathfrak{X}^+)^*$) gives a Γ-canonizer for $\mathfrak{X}$. □

Proof sketch (running time of Theorem H). The recursion tree has the same node types as Neuen's, and the canonical version adds no branch type: the descent to G' ($\leq d$), Luks steps ($\leq k!$, or b cosets for a semi-regular top), the $t!$ cosets in LC, at most k^6 individualizations and at most k^6 structures in Option 1, and two cosets in Option 2 all occur in Neuen's test with the same counts. LC runs $\binom{k}{t}$ single-object loops instead of $4k^{2t}$ pair loops. Isomorphism calls of GNS become calls

of Theorem A on domains of size at most $s \le (n+m)^{O(\log d)}$, costing $(n+m)^{O((\log d)^{c+1})}$, which is absorbed into the exponent. Babai's test on t-ary structures over at most k points becomes BC, at cost $\exp(\mathrm{polylog}\, d)$. Neuen's orbit-by-orbit processing is sequential too [Neu20, src. l. 1677–1686], so the chain rule has the same recursion, and the potential argument [Neu20, src. l. 2710–2725] only needs upper bounds on $|\Lambda[\mathcal{C}']|$, which pass to subgroups. With the choice of Remark 7.2, [Neu20, Lemma 4.36] should give $\sum s_i \le 2^{O(\log^3 d)} s$ at a node with $s_i \le s/2$, and [Neu20, Lemma 2.6] then gives $s^{O(\log^3 d)}(n+m)^{\mathrm{polylog}\, d}$. With $s \le (n+m)^{O(\log d)}$ this is $(n+m)^{O((\log d)^{c'})}$. What is missing is a written transfer of Neuen's accounting [Neu20, Lemma 4.36, Lemma 2.6] to the canonical recursion, including the three points of Remark 10.3. □

Status. Correctness (algorithm CanHyp, Proposition 10.2): proof modulo Theorem A; [Neu20, Thms. 4.8, 4.9, 4.17] (constructions and the listed properties, except (G) of Thm. 4.17, which is not used, Remark 10.3(3)), [Neu20, Lemma 4.16], the virtual-size facts [Neu20, Obs. 4.12, 4.13, Lemma 4.14]; [GNS18, Thm. 5.8]; [Bab16b, Thm. 6(b)]; the CFSG facts of the aggregation [Neu20, Thm. 4.31, Lemma 4.32]; [GNS18, Lemma 7.1]; BC; canonical generating sets; and requirement D3, which includes the structure-graph lemma of Neuen's dissertation [Neu19, Lemma 6.1.17], which we have not seen. Running time: **proof sketch**.

Remark 10.3 (Three points about Neu20 as written).

1. The potential argument in the proof of [Neu20, Lemma 4.36] contains the step discussed in Remark 7.2 [Neu20, src. l. 2716], and the choice of block system [Neu20, src. l. 2683] is equally unconstrained. The same choice applies: $\mathcal{C}' \succeq \mathfrak{B} \succeq \mathfrak{B}_1$, and inserting $\mathcal{C}'$ at the top keeps the sequence almost d-ary for Γ', since $|\mathcal{C}'| \le |\mathfrak{B}| \le d$.

2. *Order of the giant test (cost accounting only).* In the local certificates algorithm the recursive call on W' [Neu20, src. l. 2222] precedes the test "$(\Lambda')^\psi \not\ge \mathrm{Alt}(T)$" [Neu20, src. l. 2224]. The cost proof bounds kernel orbits using the kernel in the *pre-balancing* group Λ, which is a giant [Neu20, src. l. 2442], but bounds virtual size using balance, which holds for orbits of the *post-balancing* group Λ'. If $\psi(\Lambda')$ is not a giant, kernel orbits of Λ' need not be small, and the bound $s_i \le s/2$ is not justified as written. Testing $\psi(\Lambda')$ immediately after balancing, as in step (c) of CanLC, fixes this; correctness of Neu20 is unaffected. Whether the journal version handles this we do not know.

3. *Virtual size outside the window (bound on the iterations and the cost accounting): a question for the author.* We could not verify property (G) of [Neu20, Thm. 4.17] [Neu20, src. l. 1961] for the objects constructed in its proof, which treats it in one word ("analogously", [Neu20, src. l. 2141]); perhaps we misread the definitions. We read $\mathfrak{X}[A]$ as a *set* of restricted strings [Neu20, src. l. 1042], so that strings agreeing on A coincide in $\mathfrak{X}[A]$. Take $\Gamma = 1$, $\mathfrak{P} = \{\{a,b\},\{c,d\}\}$, $W = \{a,c\}$ and let $\mathfrak{X}$ consist of the strings $(a,b) \mapsto (0,0),(1,0)$ on the first block and $(c,d) \mapsto (0,0)$ on the second. Then $\mathfrak{X}[\Omega \setminus W]$ has virtual size 2, but the two copies of $\{a,b\}$ made in Simplify each carry one string on the copy of b, so $\mathfrak{X}'[\Omega' \setminus W']$ has virtual size 3, and Renormalize does not lower this, since f' is onto, $|(f')^{-1}(P')| \le |A|^{f(d)} = 1$ and $|P^*| = |f'(P^*)|$. The computation for (F) does not seem to carry over, because strings on one block with different window parts may agree outside the window, so the multiplicities of the copies may add up to more than the multiplicity of the block's outside part. The same happens at the call of Simplify in the proof of [Neu20, Thm. 4.24]: for $\Gamma = \mathrm{Sym}(T)$ on $T \sqcup \{u,v\}$, $|T| = 9$, $d = 10$, with blocks $T \cup \{u\}$, $\{v\}$ and two strings on $T \cup \{u\}$ that are constant on T, differ there and agree at u, one gets $W' = T$ and virtual size 2 before and 3 after. This concerns the construction only. Property (G) is part of an existence statement, which this example does not refute: for $\Gamma^* := \mathrm{Aut}_\Gamma(\mathfrak{X}_r)$, $\Omega^* = W^* := \Omega$ and one constant string per

block, properties (A)–(G) hold for every input. But that witness is the solution of the problem itself, so the proof of [Neu20, Thm. 4.24] needs (G) for the constructed objects. Is the virtual size in (G) meant with the multiplicities of $\mathfrak{X}$, that is $\sum_P |P \setminus W|\, m_{\mathfrak{X}}(P)^{f(d)+1} = \widehat{s}(\mathfrak{X}, W)$? For that quantity the computation for (F) does carry over (property (S4) of Lemma SIMP), and it equals the virtual size of the outside part whenever the window part is simple. The loop bound and the size sum [Neu20, src. l. 2400–2434] then hold with t_i read as $\widehat{s}$, and the step $t_{p+2} \le t'$ becomes $t_{p+2} \le \widehat{s}(\mathfrak{Z}'_1, W \cup W')$, with $s_{p+1} + \widehat{s}(\mathfrak{Z}'_1, W \cup W') \le t_{p+1}$; the statements of [Neu20, Thm. 4.24] and of the main theorem are unaffected. We use $\widehat{s}$ and not (G).

There is also a harmless typo: "$\ker \psi = N_{(\mathfrak{B})}$" [Neu20, src. l. 2687] should read $N_{(\mathfrak{C})}$.

11 Treewidth: Theorem TW (proof sketch)

This section is a **proof sketch**. It rewrites Wiebking's $n^{\operatorname{polylog} k}$ isomorphism test for graphs of treewidth k [Wie20, Thm. 35] canonically, within the Schweitzer–Wiebking framework. The key observation is that the canonical version needs only Theorem A (through Corollary D), and not the canonical version of Neu20.

11.1 What is known

result	statement	canonical?
[Wie21, Thm. 6.13] (also [GNSW18])	treewidth k, $2^{O(k^2 \log k)} n^{O(1)}$	yes
[LPPS17]	treewidth k, $2^{O(k^5 \log k)} n^5$	yes
[ES16]	bounded treewidth, logspace for each fixed k	yes
[GNSW18] (journal version)	treewidth k, $2^{k \operatorname{polylog} k} n^{O(1)}$	no [Wie21, pp. 9–10]
[Wie20, Thm. 35] = [Wie21, Cor. 6.30]	treewidth k, $n^{\operatorname{polylog} k}$	no
[Wie21, Thm. 6.29], [GNW23]	excluded K_h minor, $n^{\operatorname{polylog} h}$ (uses [Neu20])	no [Wie21, p. 179]
BC [Bab19]	all graphs, $n^{\operatorname{polylog} n}$	yes

GNS conjectured an $n^{O((\log k)^c)}$ isomorphism test for treewidth k [GNS, src. l. 2270–2274], which Wiebking proved. Wiebking's outlook predicts that canonization should follow once GNS is canonical (§1.3).

11.2 What Wiebking's algorithm imports

Wiebking's algorithm [Wie20, proof of Thm. 35] approximates the treewidth, k-improves the graph [Wie20, Lemma 29] (from [LPPS17]), computes the clique-separator decomposition [Wie20, Thm. 30] (from [Lei93, ES16]), and then runs a bottom-up procedure IsoTree.

component	non-canonical element	canonical replacement
treewidth approximation [Ami01]	output need not be invariant	invariant doubling search with invariant checks (fix TW-3)
k-improvement G^k	—	invariant [Wie20, Lemma 29]
clique-separator decomposition	—	output is isomorphism-invariant [Wie20, Thm. 30]
root of the decomposition tree	arbitrary root	centre of the tree (subdivide a central edge)
children: isomorphisms for all pairs	pairwise	canonical labelings of children (recursion)
"ρ_1 arbitrary labeling", "representative t^*"	choice	colour = canonical form of the child, label = its canonical labeling coset restricted to the adhesion set [Wie21, §6.1.3]
equal adhesion sets: `canObj` on the bag	none (bag $\leq k+1$)	[Wie20, Cor. 27], time $n^{\operatorname{polylog} k}$
distinct adhesion sets: IsoBasicClique	GNS calls, arbitrary choices	rewrite (b) below, with fix TW-1
distinct adhesion sets: `cosetHyper`	Neuen-type isomorphism	Lemma CH: Corollary D only
distinct adhesion sets: GNS on the bag	GNS isomorphism	Corollary D
extension to subtrees	arbitrary extensions of generators	canonical extension [Wie21, pp. 145–152]
IsoBasic	"some S_1, S_2", "some bijection", GNS calls	rewrite (a) below
parent adhesion set	not used in IsoTree	marked in the bag object (fix TW-2)

11.3 Coset-labeled hyperedges under a hyperedge-respecting coset

Wiebking's [Wie20, Lemma 34] (`cosetHyper`; Wiebking cites it as Lemma 9 of the ICALP version of [GNSW18] and refers to [Neu19]; that number does not match the arXiv v1 numbering we use for [GNSW18]) assumes that the given coset "maps the hyperedges in H_1 to hyperedges in H_2". Its only use, in the proof of [Wie20, Thm. 35], comes right after the isomorphisms of the bag with its hyperedges have been computed by [Wie20, Lemma 33], so the hypothesis holds there. Only the coset labels remain. (An earlier version of our notes listed `cosetHyper` as an obstacle requiring canonical Neu20; that was wrong for this application.)

Lemma CH. *Let $\Delta\rho$ be a labeling coset of V with $\Delta \leq \mathrm{Sym}(V)$, $\Delta \in \widehat{\Gamma}_d$. Let $\mathcal{S} \subseteq 2^V$ and $\chi\colon \mathcal{S} \to \mathbb{N}$ be preserved by Δ. Let each $S \in \mathcal{S}$ carry a labeling coset $\tau_S\Theta_S$ of S, with $\tau_S\colon S \to [|S|]$, $\Theta_S \leq \mathrm{Sym}([|S|])$, and $\Theta_S = \Theta_{S'}$ whenever $\chi(S) = \chi(S')$. Let $d' := \max_S \mathrm{cw}(\Theta_S) \leq \max_S |S|$. Then a canonical labeling coset of $X = (V, \mathcal{S}, \chi, \{\tau_S\Theta_S\})$ relative to $\Delta\rho$ can be computed in time $(|V| + \sum_S |S|)^{O((\log \max(d,d'))^c)}$, with c as in Theorem A.*

Note the two conventions in this statement: in $\Delta\rho$ the group sits on the side of V (as in Schweitzer and Wiebking), whereas in $\tau_S\Theta_S$ it sits on the label side (as in [GNSW18]); both are consistent with their sources.

Proof. This is the second case of [GNSW18, Lemma 23] (arXiv v1 numbering; "we can assume that Δ preserves $\mathcal{S}$ and χ"). Let $U := \bigsqcup_S \{S\} \times S$ and let $\hat{\Lambda}$ be the labeling coset of $V \cup U$ that orders V by $\delta\rho \in \Delta\rho$, orders the blocks $\{S\} \times S$ by the order of $S^{\delta\rho}$, and orders inside each block

by some element of $\tau_S\Theta_S$. Since Δ preserves χ, the block of S moves to the block of S^δ with inner permutation in Θ_S, so $\hat{\Lambda}$ is a coset; its group is an extension of $\prod_S \Theta_S$ by Δ, of composition width at most $\max(\operatorname{cw}\Delta, d')$. Let Y be the incidence graph v—(S, v), and return $\mathrm{CL}(Y; \hat{\Lambda})|_V$. GNSW prove the required properties of this output; they canonize Y with a Luks-type algorithm, and we use Corollary D instead. □

Status. Proof modulo Corollary D and [GNSW18, Lemma 23].

11.4 Canonical rewrites of Wiebking's isomorphism-only steps

Everything below takes place in the Schweitzer–Wiebking framework: an isomorphism-invariant labeling coset is refined step by step, and $\mathrm{CL}(X_2; \mathrm{CL}(X_1; \Lambda))$ is a canonical labeling of (X_1, X_2) relative to Λ. Every step uses only Corollary D, (P1) constructions, Theorem A through a homomorphism, or Wiebking's quasipolynomial canonizers on at most $k+1$ points. Each item is a sketch.

(a) isoBasic [Wie20, Lemma 32]. Input: a clique-separator-free graph G (here a bag $G^k[D]$, with separability at most k and degeneracy at most k by the checks of (c)).

- *Branching.* Branch over $S \in \mathcal{S} := \{N(v) : \deg v = \min\deg(G)\}$, an invariant family, by Lemma F4. In a branch, start from the invariant labeling coset $\mathrm{Lab}(S)$.
- *Separator sizes.* Canonize the complete directed graph on S, coloured by the sizes of the leftmost minimal separators $|S_{v,w}|$, relative to the current coset (a string on S^2; Corollary D).
- *Wreath product.* Form $\hat{S} := S \sqcup \bigsqcup_{v,w} \hat{S}_{v,w}$ and the coset $\hat{\Lambda}$ that orders S by the current labeling, the blocks by (v^λ, w^λ), and the inside of each block arbitrarily. This is an invariant construction, with composition width at most $\max(|S|, \mathrm{sep}) \le k$.
- *Identification.* Canonize the identification graph relative to $\hat{\Lambda}$ (Corollary D), and pass to the induced labeling coset of the quotient, ordering classes by least label.
- Repeat until $S = V$ (at most n rounds), and finally canonize G relative to the coset (Corollary D).

Each $\hat{\Lambda}$ orders S first, so its group stabilizes S; the final group is $\mathrm{Aut}(G, S)$, and Lemma F4 over the invariant family $\mathcal{S}$ gives $\mathrm{Aut}(G)$.

(b) isoBasicClique [Wie20, Lemma 33], with fix TW-1. Input: G as in (a) and a set $\mathcal{H}$ of cliques of G, possibly coloured by $\chi\colon \mathcal{H} \to \mathbb{N}$.

- Keep the branch S of (a), so that all cosets have composition width at most k.
- The clique cover α of [Wie20, src. l. 185–207] (peel the vertices of minimum degree; $\alpha(v) = N(v) \cup \{v\}$ at the moment v is peeled) is invariant, and $|\alpha(v)| \le \mathrm{degeneracy}(G) + 1 \le k + 1$.
- For each v, canonize the hypergraph $\mathcal{H}_v := \{S' \in \mathcal{H} : S' \subseteq \alpha(v)\}$ on $\alpha(v)$, with the colours χ, by Wiebking's hypergraph canonization [Wie20, Thm. 9] in time $(k+1+|\mathcal{H}_v|)^{\mathrm{polylog}(k+1)}$.
- Colour v by the canonical form of $\mathcal{H}_v$, and canonize relative to the current coset (Corollary D).

- Form the wreath product over $U := V \sqcup \bigsqcup_v \alpha(v) \times \{v\}$ from the canonical cosets of the $\mathcal{H}_v$, canonize the identification graph (Corollary D), and induce the coset on V.

The output group preserves $\mathcal{H}$: identification forces the restriction to $\alpha(v)$ into the canonical coset of $\mathcal{H}_v \to \mathcal{H}_{v^\varphi}$, and every hyperedge lies in some $\alpha(v)$. *Fix TW-1.* Wiebking's version never needs χ, because in the distinct-adhesion case the label image encodes the class. With compressed labels it does not, and Lemma CH requires χ to be preserved. So χ is passed into the hypergraph canonizations as above or, equivalently, after (b) the string $\chi\colon \mathcal{H} \to \mathbb{N}$ is canonized under the induced action of the output group on $\mathcal{H}$ (Theorem A through a homomorphism; $\widehat{\Gamma}_d$ is closed under quotients). The resulting group preserves $\mathcal{H}$ and χ.

(c) isoTree, with fixes TW-2 and TW-3. An instance is $\hat{X} = (\hat{G}, T, \beta, r, S)$, where S is the adhesion set to the parent ($S = \emptyset$ at the global root). Let $D := \beta(r)$, and let the children t have adhesion sets $S_t := D \cap \beta(t)$.

- *Children.* Canonical labelings by recursion. A child's colour is its canonical form (with its adhesion set marked); its label is its canonical labeling coset restricted to S_t and compressed to $\tau_t \Theta_t$. Equal colours give equal Θ_t, as Lemma CH needs.

- *Fix TW-2 (parent adhesion set).* The root-bag object must contain S; otherwise the canonical labeling of $\hat{X}$ need not map S consistently and the parent's step fails. In the distinct-adhesion case, add S as a hyperedge with a reserved colour. (S is a clique in G^k, so the clique cover covers it, and by [Wie20, Thm. 30(3)] it differs from the children's adhesion sets.) In the equal-adhesion case S equals every child's adhesion set and is marked as such.

- *Equal adhesion sets.* The bag has at most $k+1$ vertices (check (i) below). Apply Wiebking's canonization of objects with multiple labeling cosets [Wie20, Cor. 27] to the bag graph, the child labels and colours, and the marked S, in time $n^{\operatorname{polylog} k}$. (The $|D|^{O(\mathrm{cw})}$ route of [Wie21, Thm. 6.11] costs $2^{O(k \log k)}$ and must not be used when $k \gg \log n$.) Extend to the subtrees as in the equal-adhesion cases of the proof of [Wie21, Thm. 6.11] (pp. 148–151), bundling children with identical colour and label.

- *Distinct adhesion sets.* Branch over the minimum-degree neighbourhoods of $Y := G^k[D]$ (Lemma F4). Let Λ_1 be the output of (b) on $(Y, \{S_t\} \cup \{S\})$ with colours, including the χ-step of TW-1. Let Λ_2 be the output of Lemma CH relative to Λ_1, with colours and labels from the children. Let Λ_3 be the canonical labeling coset of $G[D]$ relative to Λ_2 (Corollary D). Take the minimum over the branches, and extend canonically to the subtrees by the "partition into singletons" case of the proof of [Wie21, Thm. 6.11] (pp. 151–152), which accepts any canonical coset on D whose group respects the labeled hyperedges; Λ_3 is such a coset.

- *Root.* Use the centre of the decomposition tree. A subdivided central edge gives a bag that is a clique in G^k with two equal adhesion sets of size at most $k+1$. Components are sorted by canonical form.

- *Fix TW-3 (the parameter).* For $k = 1, 2, 4, \ldots$ compute G^k and its clique-separator decomposition, and accept the first k for which: (i) every equal-adhesion bag has at most $k+1$ vertices; (ii) every basic bag $G^k[D]$ has degeneracy at most k; (iii) every adhesion set has at most $k+1$ vertices. These checks are invariant and polynomial. Canonicity holds for every k; the checks only certify the time bound. They hold for every $k \geq \mathrm{tw}(G)$: by [Wie20, Lemma 29(3)], $\mathrm{tw}(G^k) = \mathrm{tw}(G) \leq k$, so $\mathrm{degeneracy}(G^k[D]) \leq \mathrm{tw}(G^k[D]) \leq k$, equal-adhesion bags have at

most $\mathrm{tw}(G^k)+1 \le k+1$ vertices by [Wie20, Thm. 30(3)], and adhesion sets are cliques of G^k. So the accepted k is at most $\max(1, 2\,\mathrm{tw}(G))$. Check (ii) uses degeneracy rather than minimum degree: the time analysis needs $|\alpha(v)| \le k+1$ for the clique cover in (b), which Wiebking derives from "minimum degree $\le$ tw" for every peeled subgraph, i.e. from degeneracy. For $k < \mathrm{tw}(G)$, $\mathrm{tw}(G^k)$ can be much larger than $\mathrm{tw}(G)$, so a minimum-degree check would not certify that bound. Separability at most k needs no check, since local connectivity is monotone under induced subgraphs and $\mathrm{sep}(G^k) \le k$.

11.5 Theorem TW

Proof sketch (Theorem TW). Assemble (a)–(c). Each node of the clique-separator decomposition makes polynomially many calls of Corollary D with composition width at most $k+1$ on domains of polynomial size, polynomially many calls of Theorem A through homomorphisms (TW-1), and calls of [Wie20, Thm. 9, Cor. 27] on at most $k+1$ points. There are polynomially many nodes and at most n branches per bag. The total is $n^{O((\log k)^{c''})}$ with $k \le \max(1, 2\,\mathrm{tw}(G))$. □

Status. Proof sketch, conditional on Theorem A (via Corollary D); [Wie20, Lemma 29, Thm. 30, Thm. 9, Cor. 27]; the canonical-extension construction in the proof of [Wie21, Thm. 6.11]; and [GNSW18, Lemma 23]. Fixes TW-1, TW-2, TW-3 come from an internal review. The rewrites (a)–(c) have been checked against Wiebking's text step by step, but they are not written as full proofs.

Theorem TW is incomparable with [Wie21, Thm. 6.13] ($2^{O(k^2 \log k)} n^{O(1)}$); it is faster once $k \ge (\log n)^{1/2+\varepsilon}$. For $k = \mathrm{polylog}(n)$ it gives $n^{(\log\log n)^{O(1)}}$, against BC's $n^{\mathrm{polylog}\, n}$. A second route, through the minor result (as in [Wie21, Cor. 6.30]) and Corollary H4 below, would need Theorem H; the route above needs less.

12 Further corollaries

Every statement in this section is conditional or a sketch, as marked.

Corollary H1 (coset-labeled hypergraphs). *Given a coset-labeled hypergraph $(V, \mathcal{E}, \mathfrak{p})$ in the sense of [GNW23] (labeling cosets of the hyperedges, and colours) with $|E| \le d$ for all $E \in \mathcal{E}$, and a labeling coset $\Delta\rho$ of V with $\Delta \in \widehat{\Gamma}_d$, a canonical labeling coset can be computed in time $(n+m)^{O((\log d)^{c'})}$, with c' as in Theorem H.*

Proof. Refine each edge colour c to (c, Θ); this is invariant, since isomorphisms in the sense of [GNW23] require equal Θ. First canonize the coloured hypergraph $(V, \mathcal{E}, \chi)$ relative to $\Delta\rho$ by Theorem H; its right group then preserves $\mathcal{E}$ and χ. Then apply Lemma CH with $d' \le d$. □

Status. Proof modulo Theorem H and Corollary D; hence its running time is only as good as the sketch for Theorem H. The isomorphism version in [GNW23] cites [Neu20] and [Neu19, Thm. 6.6.7].

Corollary H2 (t-CR-bounded graphs; sketch). *Canonical labelings of t-CR-bounded graphs and pairs ([Neu20, Def. 5.1, Thm. 5.3, Cor. 5.4]) can be computed in time $n^{\mathrm{polylog}\, t}$; and for the closure $D = \mathrm{cl}_t(X)$ of [GNW23] one can compute an invariant labeling coset of D of composition width at most t containing all isomorphisms restricted to D.*

Proof sketch. Neuen's proof maintains a coset on the colour classes and refines it, each refinement being an isomorphism of quotient graphs (GNS) or of sets of strings (Neu20). Canonically, one maintains an invariant labeling coset: small classes are split by a wreath product with $\mathrm{Sym}(P)$,

$|P| \le t$, ordered by the parent label, followed by Corollary D on the quotient graph; refinement rounds canonize the set of strings relative to the current coset (Theorem H), induce a labeling coset on the new classes by the lexicographic order of the transported strings, and canonize the quotient graph (Corollary D). The composition width stays at most t, and there are polynomially many steps. The closure output is a weak canonical labeling (an equivariant coset containing the automorphisms), not a canonical one. □

Status. Proof sketch, modulo Theorem H and Corollary D.

Corollary H3 ($K_{3,h}$-minor-free graphs, Euler genus; sketch). *Canonical forms of $K_{3,h}$-minor-free graphs can be computed in time $n^{\operatorname{polylog} h}$, and of graphs of Euler genus g in time $n^{\operatorname{polylog} g}$.*

Proof sketch. Canonical forms of 3-connected (vertex- and arc-coloured) components combine into canonical forms of the graph by the standard canonical decomposition into 2- and 3-connected components (not re-derived here). For a 3-connected component, branch over ordered triples (v_1, v_2, v_3) (n^3 branches, Lemma F4); [Neu20, Cor. 6.3], which holds for any three vertices, makes the result $(h-1)$-CR-bounded, and H2 applies. By [Neu20, Thm. 6.1] (Ringel), graphs of Euler genus g exclude $K_{3,4g+3}$. To obtain a bound in terms of the unknown genus, run an invariant doubling search over h, using the invariant $(h-1)$-CR test. □

Status. Proof sketch, modulo H2.

Corollary H4 (excluded minors and topological subgraphs; conditional). *Canonical forms of graphs excluding K_h as a minor, or as a topological subgraph, can be computed in time $n^{\operatorname{polylog} h}$.*

Proof sketch. The non-canonical elements of the proof of [GNW23, main theorem] are: "fix one representative", "an arbitrary bijection" (twice), "$v_1 \in X_1$ arbitrary", and calls to the t-CR closure theorem, the coset-labeled hypergraph theorem, Wiebking's canonization for multiple labeling cosets (canonical already), and Babai's test on fewer than h vertices. Canonically: colours and labels come from canonical labelings of the parts as in [Wie21, §6.1.3]; equal separators are handled by Wiebking's multiple-labeling-coset canonization on fewer than h points; distinct separators by H1 relative to the coset from H2; branching over $v \in X$ by Lemma F4; and BC on fewer than h vertices. The decomposition steps of [GNW23] are invariant constructions and are used as black boxes. The proof of [Neu24a] has the same structure; its extra call computes a set stabilizer, which Theorem A handles. □

Status. Conditional on Theorem H, H1 and H2; the canonical dynamic programming is only **sketched**. To answer Question 9.7 of [Wie21] (parameter the Hadwiger number of the input), one also needs a parameter-free form: the canonical counterpart of [GNW23, main theorem] must have the form "either a K_h minor is certified, or a canonical form is output", with the failure outcome produced from canonical data so that it is invariant, followed by an invariant doubling over h. We have not written this out.

Prior art for H4. Canonization of graphs excluding a fixed minor is known: in polynomial time for each fixed excluded minor by Grohe [Gro12, Gro17], and in time $f(H) \cdot n^{O(1)}$ by Lokshtanov, Pilipczuk, Pilipczuk and Saurabh [LPPS22]. Neither gives a bound of the form $n^{\operatorname{polylog} h}$, so Question 9.7 remains open as stated, and H4, if completed, would be new only in its dependence on h.

Hypergraphs under $\operatorname{Sym}(V)$**.** Theorem H with $d = n$ gives $(n+m)^{\operatorname{polylog} n}$, which is Wiebking's hypergraph canonization [Wie21, Thm. 5.4(1)] again; nothing new.

FPT treewidth canonization (speculation). The FPT isomorphism test of [GNSW18] in time $2^{k \operatorname{polylog} k} n^{O(1)}$ uses GNS only for strings under subgroups of automorphism groups of trees, which lie in $\widehat{\Gamma}_d$, so Theorem A covers those calls; its other non-canonical step, coset intersection for multiple coloured cosets on $O(k^2)$ points, has a canonical replacement by Wiebking. So the obstacle named in [Wie21, p. 10] may be removed. We have not traced the rest of that algorithm, and we make no claim.

13 Assumptions from the literature

Versions. GNS: arXiv:1802.04659v3. Neu20: arXiv:2002.06997v2. Wie20: arXiv:1911.11257 (the version in our possession). GNSW: arXiv:1803.06858v1. SW19: arXiv:1806.07466. BC: the STOC 2019 paper. Wie21: the dissertation, printed page numbers. Theorem numbers for arXiv papers were computed from their LaTeX sources by counting theorem environments; please treat them as pointers and check them against the labels quoted.

#	result	used for
1	[GNS18, Thm. 2.1]: polynomial-time permutation group algorithms (Seress), in deterministic versions D1	throughout
2	[GNS18, Lemma 2.4]: $\widehat{\Gamma}_d$ closed under subgroups and quotients; closure under extensions (Jordan–Hölder)	§2, §8
3	[GNS18, Thm. 3.13]: structure of large primitive $\widehat{\Gamma}_d$-groups, via Thm. 4.3	change of action
4	[GNS18, Thms. 4.3, 4.6], *including the constructions in their proofs*: ι a homomorphism; $x \mapsto x^*$ injective and equivariant; the bound on $\lvert\Omega^*\rvert$; the almost d-ary sequence; computability; all choices deterministic from canonical generators D3. Also [GNS18, Lemma 4.4] (from [BLS87]) and [GNS18, Lemma 4.5]	Lemma R
5	[GNS18, Obs. 4.2]	inheritance of sequences
6	[GNS18, Thm. 5.8] (= [Neu20, Thm. 4.6])	L3, Lemma 5.1, Lemma LC
7	[GNS18, Thm. 5.9] = [Bab16b, Thm. 6(b)] (= [Neu20, Lemma 4.7])	Lemmas 5.1, 5.6, LC
8	[GNS18, Lemma 6.2] and the termination argument of [GNS18, Lemma 6.3]	L1–L5
9	[GNS18, Cor. 6.5] (from [DM96, Thm. 5.2A,B])	index $\geq (4/3)^k$
10	[GNS18, Thm. A.2] (CFSG: non-giants are at most 5-transitive); [GNS18, Lemma A.3] = [Bab16a, Cor. 2.4.13]	case (iv)
11	[GNS18, Lemma 7.1] = [Bab16a, Thm. 3.2.1], deterministic D2; with $\lvert\mathcal{C}\rvert \geq 2$, $\mathcal{C}$ an equipartition, and $k \leq d$	Step 4, §7.3
12	[GNS18, Lemma 2.6] (an inequality)	§7
13	[Bab19, Thm. 2.2] (canonical form of strings under any group, quasipolynomial; a preliminary report that sketches the proof), called only for S_k, $k \leq d$, on relational structures of arity $O(\log d)$ on $[k]$ (domains of size $\leq 2k^t = d^{O(\log d)}$; D4, Remark 5.8); and Babai's string isomorphism test [Bab19, Thm. 2.1]	Prop. 5.7
14	Canonical generating sets: [AGvM+18, Lemma 6.2] = [GNSW18, Lemma 21] = [SW19, Lemma 15] (arXiv numbering)	(P1)

#	result	used for
15	Lexicographically least element of a coset in polynomial time (standard; point-stabilizer chain)	(P1)
Additionally for Theorem H		
16	[Neu20, Thm. 4.8] (normalize), [Neu20, Thm. 4.9] (renormalize), [Neu20, Thm. 4.17] (simplify on window), with their constructions and listed properties, and the claims in their proofs; deterministic D3	Lemma SIMP, Theorem H
17	[Neu19, Lemma 6.1.17] (structure graphs), as used by [Neu20]; **not seen by us**	normalization
18	[Neu20, Lemma 4.16] (extending window automorphisms)	Lemma LC
19	[Neu20, Obs. 4.12, 4.13, Lemma 4.14] (virtual size); [Neu20, Lemma 4.36, Lemma 2.6] for the time sketch	Lemma LC, Theorem H
20	[Neu20, Thm. 4.31, Lemma 4.32, Cor. 4.28] (as items 9, 10)	Lemma AGG
Additionally for Theorem TW and the corollaries		
21	[Wie20, Lemma 29] (from [LPPS17]), [Wie20, Thm. 30] (from [Lei93, ES16])	decomposition
22	[Wie20, Thm. 9] (hypergraph canonization), [Wie20, Cor. 27] (multiple labeling cosets)	Theorem TW
23	[GNSW18, Lemma 23] (arXiv v1), second case	Lemma CH
24	The canonical-extension construction in the proof of [Wie21, Thm. 6.11] (pp. 145–152)	Theorem TW
25	[Neu20, Def. 5.1, Thm. 5.3, Cor. 5.4, Thm. 6.1, Cor. 6.3]; the structure theorems of [GNW23, Neu24a] as invariant constructions	H2–H4

Not assumed; reproved here. The Babai–Luks framework lemmas F1–F4; the canonical bounded-degree reduction of [BL83] (Section 8), including [BL83, Lemma 4.3]; the recursion lemma (Lemma REC); the potential argument (Lemma 7.4); the canonical versions of [Neu20, Lemmas 4.3, 4.15] (Lemmas B and W, which we prove directly).

New here, as far as we know. Lemma T and its use in Definition 5.3 and Lemmas 5.4, 5.6; Lemmas $\mathrm{T}^{\mathrm{set}}$, LC, SIMP and AGG (the single-object local certificates loop for Neu20); the observation that Wiebking's treewidth algorithm needs only canonical GNS; Lemma 5.9 and Lemma R as stated (routine); the choice of Remark 7.2.

14 Related work

- *Bounded degree.* Luks [Luk82]; Babai, Kantor and Luks [BKL83] ($n^{O(d/\log d)}$); canonical forms by Babai and Luks [BL83] and by Fürer, Schnyder and Specker [FSS83]. [BL83] introduces canonical labeling cosets, canonical forms relative to a coset, and the half-edge trick.
- *Quasipolynomial time.* Babai's isomorphism test [Bab16b, Bab16a] and canonical form [Bab19]. BC makes the local-certificate structure strongly canonical [Bab19, Thm. 12.1] by different means (a structure built from the certificates, and a selection in asymmetric test sets, Claim 12.4), which need $t \geq 2 \log n$.

- n^{polylog} *in a parameter.* GNS [GNS18] (maximum degree); Neuen [Neu20] (hypergraphs under $\widehat{\Gamma}_d$, t-CR-bounded graphs, genus); Wiebking [Wie20] (treewidth); Grohe, Neuen and Wiebking [GNW23] (excluded minors); Neuen [Neu24a] (excluded topological subgraphs). All are isomorphism tests. Neuen's survey [Neu26] states that canonization "is possible for many of the algorithms discussed", without a specific statement for these results. See also the survey [GN21].

- *Canonization frameworks.* Schweitzer and Wiebking [SW19], Wiebking [Wie20, Wie21]: labeling cosets, hypergraphs, multiple labeling cosets, canonical FPT for treewidth [Wie21, Thm. 6.13], and DeepWL [GSW21] (which turns polynomial-time DeepWL isomorphism tests into canonization; GNS is group-theoretic and not of this form). Wiebking's dissertation lists our problem as Question 9.6 and contains the canonical test set device [Wie21, Lemma 5.34].

- *Treewidth.* Canonical FPT algorithms by Lokshtanov, Pilipczuk, Pilipczuk and Saurabh [LPPS17] and by Grohe, Neuen, Schweitzer and Wiebking [GNSW18, Wie21]; logspace canonization for bounded treewidth by Elberfeld and Schweitzer [ES16].

- *Excluded minors and rank width.* Canonization for each fixed excluded minor in polynomial time by Grohe [Gro12, Gro17], and in time $f(H)\,n^{O(1)}$ by Lokshtanov, Pilipczuk, Pilipczuk and Saurabh [LPPS22]. Canonization for bounded rank width by Grohe and Neuen [GN23], and in parallel by Levet, Rombach and Sieger [LRS23]. Neuen also gave an FPT isomorphism test parameterized by genus [Neu24b].

- *Practical canonizers.* Individualization–refinement tools (e.g. [MP14]) routinely relabel individualized vertices to reference labels. Lemma T is the group-theoretic analogue at a fixed ordered tuple.

- *Prior-art searches.* Searches of the arXiv API, Semantic Scholar, OpenAlex and the web through 5 November 2026 found no $n^{\mathrm{polylog}\,d}$ canonization for maximum degree d, no canonical version of [Neu20], and no $n^{\mathrm{polylog}\,k}$ canonization for treewidth k. We did not have access to the SIAM journal version of GNS or to Neuen's dissertation [Neu19]. Since the dissertation predates Wiebking's 2021 statement of Question 9.6, it is unlikely to contain a solution.

15 Status and open points

15.1 Status of every statement

statement	status
Lemmas F1–F4, R, S, T, 5.9; Prop. 5.2, 5.7; Lemmas 5.1, 5.4, 5.6	proofs given
Proposition 6.1 (correctness of CanAD)	proof modulo §13
Lemma REC, Lemma 7.4	proofs given
Theorem A	proof modulo §13
Corollary B	proof modulo Theorem A; small computer check of the construction
Corollary D	proof modulo Theorem A
Lemmas E1–E4, T^{set}	proofs given
Lemmas W, $F2^{set}$, B, SIMP, LC, AGG	proofs modulo Theorem A and the results of Neu20 listed in §13
Proposition 10.2 (correctness of CanHyp)	proof modulo §13, including the unseen [Neu19, Lemma 6.1.17]
Theorem H	correctness: proof modulo §13 (Proposition 10.2); running time: **proof sketch**
Lemma CH	proof modulo Corollary D and [GNSW18, Lemma 23]
Theorem TW	**proof sketch**, conditional on Theorem A, with fixes TW-1–TW-3
H1	proof modulo Theorem H
H2, H3	**proof sketches**
H4	**conditional** on Theorem H, H1, H2; **sketch**; parameter-free form not written
FPT treewidth canonization	speculation; no claim
Remark 7.2	question about the written proof of [GNS18, Lemma 7.2]; the proposed choice is proved to work (Lemma 7.4)
Remark 7.1	presentation point in [GNS18, Thm. 7.3]; covered by Lemma REC
Remark 10.3	three points in the cost analysis of [Neu20]; correctness of Neu20 unaffected

15.2 How this draft was checked

Three internal adversarial reviews were carried out (all AI-assisted, none of them refereeing).

- The first attacked the canonical colour construction (coset versus group transport, reference tuples, canonical generating sets), the exactness of colour classes, the completeness of the list of choice points (Table 2), the running time, the Luks reduction, prior art, and the question why the problem was considered open. It found no break; its corrections (explicit membership tags; credit to [Wie21, Lemma 5.34]; writing out the running time and Corollary B) are incorporated.
- The second checked the running time and the canonical Luks reduction line by line, with a computer check of Section 8 on small graphs, and found four undocumented assumptions (determinism and canonical generating sets for all subroutines feeding (P1) choices, including BC; closure of $\widehat{\Gamma}_d$ under extensions; $|\mathcal{C}| \geq 2$; $k \leq d$), a slip in the description of the right subgroup in Lemma F1 and Lemma 5.1, and imprecise wording for the semi-regular step and the chain rule. All are corrected here.

- The third checked Theorem H, Theorem TW and the corollaries. It found the repairs TW-1–TW-3, the descent group in step (d) of CanLC, the coset convention for $\mathfrak{P}$ in Theorem H, the need for a parameter-free form of H4, and the missing prior art [Gro12, Gro17, LPPS22]. All are incorporated.

A separate check of Remark 7.2 reproduced the example by computer for small m. The written-out algorithm CanHyp of §10.9, Lemma E4, Lemma F2^{set} and the proofs of Lemmas W, B and SIMP were added after these reviews, in response to an internal referee report, and have not been independently checked.

15.3 Questions for experts

Question 15.1. (Lemmas 5.4, 5.6.) Is the colour of an ordered tuple, computed under $J_0 = H_{(\vec{T}_0)}$ after transport, a single well-defined J_0-canonizer on the layered certificate strings? Does "all orderings of T have the same colour" force $\psi_T(G(T,x)) = \mathrm{Sym}(T)$? Do the colours cost at most $k^t n$ in recursive sizes, with no relation between t and n?

Question 15.2. (Lemma R; rows 1–5 of Table 2.) Can every auxiliary choice in the proofs of [GNS18, Thms. 4.3, 4.6] be made as a deterministic function of the group, in particular the $\mathcal{P}$-sequence of [GNS18, Thm. 3.13] and the Johnson identifications? Is Ω^* ordered canonically, and is the restriction to one G^*-orbit still injective on strings? We checked the relevant claims in the proofs but did not re-verify [GNS18, Thm. 3.13] or the group theory behind [GNS18, Lemma 4.5].

Question 15.3. Is [Bab19, Thm. 2.2] accepted as proved? BC is a preliminary report that sketches its proof; Theorem A uses it only for S_k, $k \le d$, on relational structures of arity $O(\log d)$ (Remark 5.8).

Question 7.3 (GNS Lemma 7.2) should be checked first.

Question 15.4. (Neuen.) Is the "all pairs in parallel" device of [Neu20] needed for any reason other than making renormalization consistent? Our single-object loop at the setwise stabilizer of T_0 claims to make it unnecessary. Is the order of the giant test (Remark 10.3(2)) handled in the journal version? Is the virtual size in property (G) of [Neu20, Thm. 4.17] meant with the multiplicities of the whole instance (Remark 10.3(3))?

Question 15.5. (Wiebking.) Is it right that the treewidth algorithm of [Wie20] needs only GNS, with `cosetHyper` applied only to hyperedge-respecting cosets, so that its canonical version follows from canonical GNS plus [Wie20, Thm. 9, Cor. 27] on at most $k+1$ points?

Question 15.6. Do the experts know of unpublished work on Question 9.6 of [Wie21]? The device is theirs.

15.4 Known limitations

- We have not seen the SIAM journal version of GNS, Neuen's dissertation [Neu19], or the journal versions of [Neu20] and [GNSW18].
- BC is a preliminary report that sketches its proof. Theorem A depends on it only for structures of arity $O(\log d)$ on at most d points (Remark 5.8); Wiebking's work uses it in the same way.
- We did not re-verify the group theory in [GNS18, Thm. 3.13, Lemma 4.5], or the internals of [Neu20, Thms. 4.8, 4.9, 4.17] beyond the facts used.

- The running time of Theorem H, Theorem TW and H2–H4 are sketches. The correctness proof of Theorem H (Proposition 10.2) is written out, but it relies on the structure-graph lemma of [Neu19], which we have not seen, and on the constructions of [Neu20] as listed in §13.
- No formal verification was attempted, and the computer checks cover only small instances of Section 8 and of the example in Remark 7.2.

Acknowledgements and disclosure

[To be completed by the author(s).] This draft was prepared with substantial assistance from an AI system (Claude, Anthropic), including the derivations, the internal reviews and the small computer checks. It has not been refereed. The canonical test set device is due to Daniel Wiebking, and the overall approach follows Grohe, Neuen and Schweitzer, Babai and Luks, and Babai.

References

[AGvM+18] Eric Allender, Joshua A. Grochow, Dieter van Melkebeek, Cristopher Moore, and Andrew Morgan. Minimum circuit size, graph isomorphism, and related problems. *SIAM Journal on Computing*, 47(4):1339–1372, 2018.

[Ami01] Eyal Amir. Efficient approximation for triangulation of minimum treewidth. In *Proceedings of the 17th Conference on Uncertainty in Artificial Intelligence (UAI)*, 2001.

[Bab16a] László Babai. Graph isomorphism in quasipolynomial time. arXiv:1512.03547v2, 2016.

[Bab16b] László Babai. Graph isomorphism in quasipolynomial time [extended abstract]. In *Proceedings of the 48th Annual ACM Symposium on Theory of Computing (STOC)*, pages 684–697, 2016.

[Bab19] László Babai. Canonical form for graphs in quasipolynomial time: preliminary report. In *Proceedings of the 51st Annual ACM Symposium on Theory of Computing (STOC)*, pages 1237–1246, 2019.

[BKL83] László Babai, William M. Kantor, and Eugene M. Luks. Computational complexity and the classification of finite simple groups. In *Proceedings of the 24th Annual Symposium on Foundations of Computer Science (FOCS)*, pages 162–171, 1983.

[BL83] László Babai and Eugene M. Luks. Canonical labeling of graphs. In *Proceedings of the 15th Annual ACM Symposium on Theory of Computing (STOC)*, pages 171–183, 1983.

[BLS87] László Babai, Eugene M. Luks, and Ákos Seress. Permutation groups in NC. In *Proceedings of the 19th Annual ACM Symposium on Theory of Computing (STOC)*, pages 409–420, 1987.

[DM96] John D. Dixon and Brian Mortimer. *Permutation Groups*, volume 163 of *Graduate Texts in Mathematics*. Springer, 1996.

[ES16] Michael Elberfeld and Pascal Schweitzer. Canonizing graphs of bounded tree width in logspace. In *Proceedings of the 33rd Symposium on Theoretical Aspects of Computer Science (STACS)*, 2016.

[FSS83] Martin Fürer, Walter Schnyder, and Ernst Specker. Normal forms for trivalent graphs and graphs of bounded valence. In *Proceedings of the 15th Annual ACM Symposium on Theory of Computing (STOC)*, pages 161–170, 1983. Not consulted for this draft.

[GN21] Martin Grohe and Daniel Neuen. Recent advances on the graph isomorphism problem. In *Surveys in Combinatorics 2021*, volume 470 of *London Mathematical Society Lecture Note Series*. Cambridge University Press, 2021. arXiv:2011.01366.

[GN23] Martin Grohe and Daniel Neuen. Canonisation and definability for graphs of bounded rank width. *ACM Transactions on Computational Logic*, 24(1):6:1–6:31, 2023. Conference version LICS 2019; arXiv:1901.10330.

[GNS18] Martin Grohe, Daniel Neuen, and Pascal Schweitzer. A faster isomorphism test for graphs of small degree. arXiv:1802.04659v3 (30 Nov 2019). Conference version FOCS 2018; journal version SIAM J. Comput. 52(6):FOCS18-1–FOCS18-36, 2023, which was not accessible to us. All result numbers refer to arXiv v3, 2018.

[GNSW18] Martin Grohe, Daniel Neuen, Pascal Schweitzer, and Daniel Wiebking. An improved isomorphism test for bounded-tree-width graphs. In *Proceedings of the 45th International Colloquium on Automata, Languages, and Programming (ICALP)*, 2018. Journal version ACM Trans. Algorithms 16(3), 2020. We use arXiv:1803.06858v1.

[GNW23] Martin Grohe, Daniel Neuen, and Daniel Wiebking. Isomorphism testing for graphs excluding small minors. *SIAM Journal on Computing*, 52(1):238–272, 2023. Conference version FOCS 2020. We use arXiv:2004.07671v2.

[Gro12] Martin Grohe. Fixed-point definability and polynomial time on graphs with excluded minors. *Journal of the ACM*, 59(5), 2012. Conference version LICS 2010.

[Gro17] Martin Grohe. *Descriptive Complexity, Canonisation, and Definable Graph Structure Theory*, volume 47 of *Lecture Notes in Logic*. Cambridge University Press, 2017.

[GSW21] Martin Grohe, Pascal Schweitzer, and Daniel Wiebking. Deep Weisfeiler Leman. In *Proceedings of the 2021 ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2021. arXiv:2003.10935.

[Lei93] Hanns-Georg Leimer. Optimal decomposition by clique separators. *Discrete Mathematics*, 113:99–123, 1993.

[LPPS17] Daniel Lokshtanov, Marcin Pilipczuk, Michał Pilipczuk, and Saket Saurabh. Fixed-parameter tractable canonization and isomorphism test for graphs of bounded treewidth. *SIAM Journal on Computing*, 46(1):161–189, 2017.

[LPPS22] Daniel Lokshtanov, Marcin Pilipczuk, Michał Pilipczuk, and Saket Saurabh. Fixed-parameter tractability of graph isomorphism in graphs with an excluded minor. In *Proceedings of the 54th Annual ACM Symposium on Theory of Computing (STOC)*, 2022. arXiv:2210.14638.

[LRS23] Michael Levet, Puck Rombach, and Nicholas Sieger. Canonizing graphs of bounded rank-width in parallel via Weisfeiler–Leman. arXiv:2306.17777, 2023.

[Luk82] Eugene M. Luks. Isomorphism of graphs of bounded valence can be tested in polynomial time. *Journal of Computer and System Sciences*, 25(1):42–65, 1982.

[MP14] Brendan D. McKay and Adolfo Piperno. Practical graph isomorphism, II. *Journal of Symbolic Computation*, 60:94–112, 2014.

[Neu19] Daniel Neuen. *The Power of Algorithmic Approaches to the Graph Isomorphism Problem*. PhD thesis, RWTH Aachen University, 2019. DOI 10.18154/RWTH-2020-00160. Not obtained for this draft.

[Neu20] Daniel Neuen. Hypergraph isomorphism for groups with restricted composition factors. In *Proceedings of the 47th International Colloquium on Automata, Languages, and Programming (ICALP)*, 2020. Journal version ACM Trans. Algorithms 18(3), 2022. We use arXiv:2002.06997v2.

[Neu24a] Daniel Neuen. Isomorphism testing for graphs excluding small topological subgraphs. *ACM Transactions on Algorithms*, 20(3):25, 2024. Conference version SODA 2022. We use arXiv:2011.14730v3.

[Neu24b] Daniel Neuen. Isomorphism testing parameterized by genus and beyond. *SIAM Journal on Discrete Mathematics*, 38(1):453–484, 2024. arXiv:2106.14869.

[Neu26] Daniel Neuen. Parameterized complexity of graph isomorphism testing. *Computer Science Review*, page 100918, 2026. DOI 10.1016/j.cosrev.2026.100918.

[Ser03] Ákos Seress. *Permutation Group Algorithms*, volume 152 of *Cambridge Tracts in Mathematics*. Cambridge University Press, 2003.

[SW19] Pascal Schweitzer and Daniel Wiebking. A unifying method for the design of algorithms canonizing combinatorial objects. In *Proceedings of the 51st Annual ACM Symposium on Theory of Computing (STOC)*, 2019. arXiv:1806.07466.

[Wie20] Daniel Wiebking. Graph isomorphism in quasipolynomial time parameterized by treewidth. In *Proceedings of the 47th International Colloquium on Automata, Languages, and Programming (ICALP)*, 2020. arXiv:1911.11257.

[Wie21] Daniel Wiebking. *A Decomposition-Compatible Canonization Framework for the Graph Isomorphism Problem*. PhD thesis, RWTH Aachen University, 2021. DOI 10.18154/RWTH-2021-06242. Defended 14 June 2021. Page numbers refer to the printed pagination.